Extremal dimensions and critical isomorphism growth of metabelian three-Engel Lie algebras

Baohua Li

Abstract

We determine sharp dimensions and near-minimum isomorphism populations for metabelian three-Engel Lie algebras in characteristic two and their binary quadratic orbit realizations. Two minimum dimension formulas and their complete equality classifications measure the cost of compatibility with the original polar bracket. At the unrestricted metabelian minimum the type count grows linearly with class, while the first admissible dimension surplus already supports doubly exponential growth. We prove fixed-surplus laws, recover exact principal orbits, and identify a sharp normalized-surplus threshold $\sqrt{2}$ for concentration at minimum derived dimension. An explicit logarithmic population profile describes surplus allocation. In a growing critical window we determine the full finite-product partition and the joint law of derived defect and diagonal rank, with exponentially small relative error. Complete isomorphism fibers realize the classical rectangular-matrix rank-deficiency law in actual integer dimension layers. In a fixed macroscopic window a common head normalization controls all module and action defects and gives a complete multiplicative partition. A natural compatibility constraint selects an augmentation module and has an explicit rarity exponent while preserving the leading allocation. As an application, relaxing a restricted-square constraint changes the leading population law and typical allocation. The proofs combine a multigraded commutator module, intrinsic recovery of coordinate actions, and uniform cocycle bounds. The metabelian correspondence is proved independently on all linear morphisms. The population measure is uniform on unlabelled isomorphism types; ordinary counts are stated over every finite characteristic-two field with at least four elements.

Keywords. Metabelian Lie algebra; three-Engel identity; extremal dimension; isomorphism growth; critical threshold; quadratic map.

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1 Introduction

How does an isomorphism population change immediately above its smallest possible dimension? We answer this question for metabelian three-Engel Lie algebras in characteristic two and for their binary quadratic orbit realizations. The minimum is rigid, but the first admissible surplus already supports a doubly exponential population. A larger surplus produces an explicit transition in the typical derived dimension. The same structural estimates also determine how a natural compatibility constraint changes the typical module and the size of its unlabelled population.

The binary equation is

$$f \circ f = f + \mathrm{id}\,. \tag{1}$$

For quadratic f, its polar is a Lie bracket. We impose metabelianity and prove a complete normal form and an ordinary $\mathbb{F}_4$-Lie correspondence, retaining every linear intertwiner. This proof uses only the elementary polar identities and the metabelian assumption; the global structural theory of [21] is not an input. The ordinary formulation also permits counting over every finite characteristic-two field $\mathbb{F}_q$ with $q \geq 4$.

1.1 From exact minima to the first surplus

Within the metabelian class, write $\mu(c)$ for the least binary dimension at polar class c, and $\nu(c)$ for the corresponding compatible metabelian minimum. For $c \geq 5$ the sharp formulas are

$$\mu(c) = 2^c + 2c - 6, \qquad \nu(c) = 2^c + 2c - 2.$$

We classify all equality objects, their automorphisms, and the small class endpoints. The four-dimension difference measures the extra cost of extending the scalar structure to the original polar bracket. The auxiliary ordinary Lie structure exists throughout; compatibility is an additional condition on the original polar algebra.

At the minimum, the metabelian type count grows linearly in c. At dimension $\mu(c)+2$, if A_c counts all binary-linear conjugacy classes, then

$$\log_4 A_c = 2^{c-1} + O(c^2).$$

Theorem 5.1 constructs indecomposable families and bounds the entire layer. The fixed-surplus law, exact principal orbits and full coordinate recovery determine how the additional dimensions enter the population.

1.2 A sharp population transition

For ordinary objects over $\mathbb{F}_q$, put $d = c - 1$ and $n = 2^d$; d is the class-dependent core scale, not the intrinsic generator rank of every object. Let r be total dimension surplus over the minimum, and let $\pi_{d,r}(q)$ be the proportion of unlabelled isomorphism types with minimum possible derived dimension. Theorem 6.2 gives

$$\limsup r/\sqrt{n} < \sqrt{2} \implies \pi_{d,r}(q) \to 1, \qquad \liminf r/\sqrt{n} > \sqrt{2} \implies \pi_{d,r}(q) \to 0.$$

We determine the macroscopic logarithmic population profile and dimension allocation, then resolve the competing integer defects through a discrete staircase. Theorem 6.6 then determines the entire multiplicative partition in the growing window $|r - (\sqrt{2n} - d + 1/2)| \leq \lfloor \sqrt{n}/d^2 \rfloor/4$. It includes the joint law of defect and diagonal rank and gives classical matrix rank-deficiency limits on true integer layers. The measure throughout these population theorems is uniform on isomorphism types, as specified in the corresponding statements; it is not a uniform ideal measure.

The proof mechanism joins three tasks. A multigraded commutator module gives sharp terminal quotients. Intrinsic recovery then identifies every allowed change of coordinates in the near-minimum layers. Finally action normalization and cocycle estimates compare competing allocations of the same dimension surplus. In particular the complete residual module group couples central two-cocycles to quotient actions. Its weights, together with the core parabolic, recover the full matrix rank census even as the defect crosses $d-1$. An all-module tail estimate then passes from conditional branches to the entire type population.

Theorem 4.1 supplies a common head normalization for every liftable extra action, including those visible on the core socle. It controls nonsplit module defects uniformly. Theorem 6.11 uses it to extend the complete finite-product population description to the macroscopic integer range $3/2 \le r/\sqrt{n} \le 8/5$. The critical and macroscopic multiplicative statements retain their respective domains; the logarithmic profile has the wider bounded-ratio range.

1.3 Constraints and their population costs

The extremal compatibility cost has a population counterpart. Requiring the existence of a square with range in L' and zero on L' excludes the maximum-socle derived modules. At the first permitted socle defect, the square supplies a missing head direction and forces the augmentation ideal $J = \operatorname{rad} R$. Theorem 7.2 determines the complete compatible partition and its precise n-scale rarity exponent $\gamma_\alpha^2/2$, while preserving the ordinary leading profile and scaled allocation.

As a focused application, marking the square gives a different type space and a constant-one n-scale cardinality cost. Relaxing zero on L' to zero on $Z(L) \cap L'$ instead changes the leading exponent to $rn + 2r^3/27$ and the typical derived share to $h/r = 1/3$, in the same macroscopic window. These statements use complete automorphism fibers and weighted all-module bounds. The square-parameter and orbit viewpoints are classical; the population comparisons here require their realization in the entire unlabelled type layer.

1.4 Sources and scope

The commutator-module presentation is classical Koszul theory [17, 12]; partial Engel linearization is used in [20]. Cohn's construction [5] supplies a classical regular compatible model and the zero-parameter slice identified below. The equality classification keeps the additional parameters and all their coordinate changes. Generic algebra enumeration in [2] uses a different fixed-variety regime. Here the minimum depends on class and the populations occupy layers immediately above it.

The companion [21] develops general quadratic orbit structure, and [22] studies ordinary central-depth laws. The ideal-growth companion [15] fixes generator rank and allows unbounded restricted-power height. Its ideal/probability measure and growth limit differ from the type layers studied here. A dated local revision of the ideal-growth companion proves arithmetic locality for fixed homogeneous sources of effective dimension at least three; that marked ideal measure does not imply the unlabelled type probabilities here. This comparison uses no companion theorem as a proof input. These papers develop separate parts of the common research manuscript [23].

The independent correspondence is proved in 2.1. The sharp minima and equality cases are grouped in Section 3. Section 4 describes the common structure behind the population estimates. Fixed surplus, the sharp transition and exact finite partitions then lead to the constraint results of Sections 7–8. The appendices provide the complete orbit and uniform-bound arguments. All brackets are alternating and left-normed. Dimensions are binary for orbit solutions and explicitly field-valued in the ordinary counts.

2 The polar bracket and the metabelian correspondence

Proposition 2.1. *Every solution of* (1) *satisfies*

$$f^3 = \mathrm{id}, \qquad f(0) = 0, \qquad f(x + f(x)) = x.$$

Its only fixed point is zero. Every nonzero orbit spans a binary plane, and these planes partition $V \setminus \{0\}$*. In particular* $\dim V$ *is even.*

Proof. Evaluate (1) at $f(x)$. It gives $f^3(x) = f^2(x) + f(x) = x$, so f is bijective. If $a = f(0)$, then $f(a) = a$; applying f^3 at zero gives $a = 0$. A fixed point in (1) must be zero. The remaining identity follows from $x + f(x) = f^2(x)$. The nonzero orbits have length three and sum zero, and $3 \mid 2^{\dim V} - 1$ implies the parity assertion. □

Theorem 2.2. *Let* f *be a quadratic solution. Its polar form* $[x, y] = f(x + y) + f(x) + f(y)$ *is a Lie bracket. It satisfies*

$$[x, f(x)] = 0, \tag{2}$$

$$f([x, y]) = [f(x), f(y)] + [x, y] + [f(x) + f(y), [x, y]], \tag{3}$$

$$[[a, b], [c, d]] + [[a, c], [b, d]] + [[a, d], [b, c]] = 0. \tag{4}$$

Proof. The polar is alternating and bilinear because f is quadratic and $f(0) = 0$. Expand $f(x + f(x)) = x$ by polarization. The terms $f(x) + f^2(x)$ sum to x, leaving (2). Take the third additive difference at zero of $x \mapsto [x, f(x)]$. The terms involving individual values of f cancel, leaving

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

This is Jacobi.

Next polarize $f^2 = f + \mathrm{id}$. Since $f(x + y) = f(x) + f(y) + [x, y]$, the expansion of the left side gives (3).

For (4), introduce Boolean variables $t_1, \ldots, t_4$ and vectors $x_1, \ldots, x_4$. The exact expansion is

$$f\Big(\sum_i t_i x_i\Big) = \sum_i t_i f(x_i) + \sum_{i<j} t_i t_j [x_i, x_j].$$

In applying f again, the coefficient of $t_1 t_2 t_3 t_4$ can only come from a polar cross term between two disjoint pair summands. There are three such pairings, giving the left side of (4). The composition has degree at most two, so this coefficient vanishes. All coefficient comparisons take place in the Boolean function algebra, with $t_i^2 = t_i$. □

The two polarization steps have distinct roles. Binary degree two gives an alternating bilinear form; the commuting identity (2) forces its Jacobi identity. The fourth polarization then supplies (4), an additional restriction on the resulting Lie algebra. The polynomial in (4) is the associative standard polynomial of degree four expressed as a Lie element in characteristic two [8, Theorem 9]. It supplies an additional identity beyond Jacobi.

Proposition 2.3. *All terms of the derived, lower central, and upper central series of the polar Lie algebra* $L = (V, [\ ,\ ])$ *are* f*-invariant. The center is* f*-invariant and* f *is linear on it. Every nonzero factor of any of these series has even binary dimension.*

Proof. Let I be an f-invariant ideal. In (3), take $x \in I$. Each term on the right belongs to $[I, L]$, which is an ideal. Quadratic expansion extends invariance from bracket generators to their span: the extra polar terms also lie in that ideal. The same argument with $x, y \in I$ proves invariance of $[I, I]$, using that it is an ideal of L. Induction starts with $I = L$.

Polarization of (2) gives

$$[x, f(y)] + [y, f(x)] + [x, [x, y]] + [y, [x, y]] = 0. \tag{5}$$

If $x \in Z(L)$, this says $[y, f(x)] = 0$ for every y. Thus $Z(L)$ is invariant; its polar is zero, so f is linear there. Pass successively to quotients by invariant central ideals to obtain the upper central assertion. Every series factor inherits a solution of (1), so Proposition 2.1 proves its parity. □

Lemma 2.4. *A nonabelian nilpotent polar Lie algebra has binary abelianization dimension at least four.*

Proof. The abelianization is a nonzero even-dimensional space with the linear action induced by f. If its dimension were two, any nonzero coset $x+L'$ and its image $f(x)+L'$ would form a binary basis. The subalgebra H generated by $x, f(x)$ is abelian by (2). Moreover $L = H + L'$. Inductively, $L = H + \gamma_j(L)$ implies $L = H + \gamma_{j+1}(L)$, because $[H + \gamma_j, H + \gamma_j] \subseteq H + \gamma_{j+1}$. Nilpotency then gives $L = H$, a contradiction. □

2.1 A complete metabelian correspondence

Assume in this section that the polar Lie algebra is metabelian. Then $W = L'$ is abelian, $f|_W$ is linear, and the induced map on $U = V/W$ is linear. Both operators satisfy $T^2 + T = \mathrm{id}$. We give U and W the $\mathbb{F}_4$-structures in which these operators are multiplication by ω.

Lemma 2.5 (The operator pencil)**.** *In any binary-linear splitting of $V \to U$, write*

$$f(x, z) = (\omega x, \omega z + L_x z + q(x)). \tag{6}$$

The binary-linear pencil $x \mapsto L_x$ satisfies

$$L_x\omega = \omega L_x, \quad L_x^3 = 0, \quad L_{\omega x} = \omega L_x + L_x^2, \quad L_x L_y = L_y L_x, \quad L_x^2 L_y = 0.$$

Proof. Put $J = \omega\,\mathrm{id}_W$ and $M_x = J + L_x$. Both M_x and $\mathrm{id} + M_x$ are invertible: they are the restrictions on coset differences of the permutations f and $f + \mathrm{id} = f^2$. Substitution in (1) gives

$$M_{\omega x} M_x = \mathrm{id} + M_x.$$

For $A = M_x$, the three matrices along the ω-orbit are $A, \mathrm{id} + A^{-1}, (\mathrm{id} + A)^{-1}$. Their sum is J, because the three L-terms sum to zero. Thus J is a rational function of A, and in particular commutes with it. Clearing the denominators gives

$$JA(\mathrm{id} + A) = A^3 + A + \mathrm{id}\,.$$

Since $J^2 + J = \mathrm{id}$ and $J^3 = \mathrm{id}$, this is $(A + J)^3 = 0$. It follows that L_x commutes with J and is cube zero. Expanding $(J + L_x)^{-1}$ now gives $L_{\omega x} = JL_x + L_x^2$. The additivity of the left side forces the L_x's to commute.

For $P = L_x, Q = L_y$, the identity $(P + Q)^3 = 0$ yields $P^2Q = PQ^2 =: T$. Apply the same identity to $P' = JP + P^2 = L_{\omega x}$ and Q. It gives $(J^2 + J)T = P^2Q^2$, hence $T = P^2Q^2 = PT$. As P is nilpotent, this implies $T = 0$. □

Theorem 2.6 (Normal form and equivalence)**.** *Every metabelian quadratic solution is binary-linearly conjugate to*

$$f(x, z) = \big(\omega x, \omega z + P(x)z + Q(x) + P(x)Q(x)\big) \quad \text{on } \mathbb{F}_4^d \oplus W, \tag{7}$$

where

$$P(x) = \sum_i x_i P_i, \qquad Q(x) = \sum_{i<j} x_i x_j a_{ij}, \tag{8}$$

$$P_i P_j = P_j P_i, \qquad P_i^2 = 0, \tag{9}$$

$$P_i a_{jk} + P_j a_{ik} + P_k a_{ij} = 0 \quad (i < j < k), \tag{10}$$

$$W = \sum_i P_i W + \operatorname{span}_{\mathbb{F}_4}\{a_{ij} : i < j\}. \tag{11}$$

Conversely these conditions define a quadratic solution with intrinsic polar image W.

The category of these maps with all binary-linear intertwiners is equivalent to the category of finite-dimensional metabelian $\mathbb{F}_4$-Lie algebras $\mathfrak{l}$ satisfying

$$(\operatorname{ad} u)^2|_{\mathfrak{l}'} = 0 \qquad (u \in \mathfrak{l}). \tag{12}$$

with all $\mathbb{F}_4$-Lie homomorphisms as morphisms. For corresponding objects, $C_{\mathrm{GL}(V,2)}(f) \cong \mathrm{Aut}_{\mathbb{F}_4}(\mathfrak{l})$. Their lower central dimensions agree after multiplication by two.

Proof. Lemma 2.5 provides the preliminary actions. We first normalize a section and separate the scalar weights of the cocycle (Steps 1–2). The decisive step is to combine binary degree two with intrinsic generation to force square zero (Step 3). We then verify sufficiency, construct the auxiliary Lie algebra and compare lower central series (Steps 4–5). Finally, tracking changes of coordinates upgrades the normal form to an equivalence on morphisms (Step 6).

Step 1: a normalized section. Choose an $\mathbb{F}_4$-basis e_i of U, with arbitrary lifts v_i in V. Define a binary-linear section by $e_i \mapsto v_i$, $\omega e_i \mapsto f(v_i)$. The images form a basis modulo W. Since $f(v_i + f(v_i)) = v_i$, the cocycle q in (6) vanishes on each coordinate $\mathbb{F}_4$-line.

Step 2: scalar weights. Initially put $P_i = L_{e_i}$, without yet asserting $P_i^2 = 0$. Lemma 2.5 gives

$$L_{\alpha e_i} = \alpha P_i + (\alpha + \alpha^2) P_i^2 \quad (\alpha \in \mathbb{F}_4).$$

Set $N(x) = \sum_i x_i(P_i + P_i^2)$. Then $L_x = N + N^2$, $N^3 = 0$, and $L_{\omega x} = \omega N + \omega^2 N^2$. Decompose $q = q_0 + q_1 + q_2$ into the three scalar weights $q_j(\omega x) = \omega^j q_j(x)$. Projection to each weight is a linear combination of $q(x), q(\omega x), q(\omega^2 x)$, so preserves binary degree and the vanishing on coordinate lines.

The cocycle equation is

$$q(\omega x) = \omega^2 q(x) + (\omega N + \omega^2 N^2) q(x).$$

Its weight components give

$$q_0 = N q_2 + \omega N^2 q_1, \qquad q_1 = \omega N q_0 + \omega^2 N^2 q_2.$$

Using $N^3 = 0$ yields

$$q_0 = N q_2, \qquad q_1 = N^2 q_2. \tag{13}$$

Step 3: binary degree forces square zero. The reduced $\mathbb{F}_4$-polynomial of a weight-two binary-quadratic map has the form $\sum_i b_i x_i^2 + \sum_{i<j} a_{ij} x_i x_j$. Line normalization removes the first sum, so $q_2 = Q$. Consequently

$$q_1 = N^2 Q = \sum_{k, i<j} x_k^2 x_i x_j P_k^2 a_{ij}.$$

This is where the degree bound becomes decisive. For $k = i$ or j, the exponent pattern is $(3, 1)$; otherwise it is $(2, 1, 1)$. Both patterns have binary degree three, whereas the weight projection

q_1 has degree at most two. The cubed or squared coordinate identifies k, and the remaining coordinates identify $\{i, j\}$. Thus each displayed term has its own reduced monomial, forcing

$$P_k^2 a_{ij} = 0 \qquad \text{for all } k, i, j. \tag{14}$$

For completeness, the one-coordinate function basis $1, x, x^2, x^3$ has binary degrees $0, 1, 1, 2$. Its leading classes form a basis in the associated graded degree filtration. Tensoring over the coordinates proves that distinct reduced monomials of the displayed degree three cannot cancel to degree two. This justifies the comparison in the binary degree filtration.

We now extend this vanishing from the cocycle coefficients to the whole intrinsic derived ideal. Let $\mathfrak{m}W = \sum_i P_i W$. This is an $\mathbb{F}_4$-subspace because the P_i are $\mathbb{F}_4$-linear. By the preliminary identities, every P_k^2 annihilates $\mathfrak{m}W$. Modulo $\mathfrak{m}W$, the polar values in (6) are exactly the values of the polar of Q. For $i < j$ and $\alpha \in \mathbb{F}_4$,

$$\mathrm{pol}(Q)(e_i, \alpha e_j) = \alpha a_{ij}.$$

All polar values of Q lie in the $\mathbb{F}_4$-span of the a_{ij}, and the displayed values contain all their scalar multiples. Their binary span is therefore exactly that $\mathbb{F}_4$-span. Since W is the binary span of all polar values, this proves (11). Together with (14), it gives $P_k^2 = 0$ on all of W. Now $L_x = P(x)$ and $q = Q + PQ$. The degree-three terms of PQ with three distinct indices vanish precisely when (10) holds. The other terms have exponent patterns $(2, 1)$, of binary degree two. This proves necessity.

Step 4: sufficiency. Conditions (9) and (10) make $P(x)^2 = 0$ and make PQ binary-quadratic. Under $x \mapsto \omega x$, the functions Q and PQ have weights two and zero. Thus

$$(Q + PQ)(\omega x) = \omega^2 Q + PQ = \omega^2 (Q + PQ) + \omega P(Q + PQ).$$

Together with $(\omega \,\mathrm{id} + \omega P)(\omega \,\mathrm{id} + P) = \omega^2 \,\mathrm{id} + P$, this verifies $f^2 = f + \mathrm{id}$ by substitution. Mixed polar values generate $\mathfrak{m}W$; modulo that space, the values of the polar of Q generate the a_{ij}'s over $\mathbb{F}_4$. Equation (11) therefore makes the polar image exactly W.

Step 5: the associated Lie algebra. Put $\beta(x, y) = Q(x + y) + Q(x) + Q(y)$ and define

$$[(x, z), (y, t)]_0 = (0, P(x)t + P(y)z + \beta(x, y)). \tag{15}$$

Jacobi follows from commuting operators and (10). This is a metabelian $\mathbb{F}_4$-Lie algebra with derived ideal W, and it satisfies (12). Conversely, splitting the abelianization of any algebra with these properties gives exactly such operators and an alternating cocycle. Choose its cross-term refinement Q to recover (7).

For both the original polar bracket and (15), the action on W is $P(x)$. Hence, with $\mathfrak{m}$ the ideal generated by these actions,

$$\gamma_2 = W, \qquad \gamma_k = \mathfrak{m}^{k-2} W \quad (k \geq 3). \tag{16}$$

This proves the lower central comparison.

Step 6: morphisms. A binary-linear intertwiner maps the intrinsic space W into W'. Its diagonal blocks $A : U \to U'$ and $C : W \to W'$ commute with the intrinsic scalar actions, hence are $\mathbb{F}_4$-linear. Write its off-diagonal block as $t_s + t_a$, where $t_s(\omega x) = \omega t_s(x)$ and $t_a(\omega x) = \omega^2 t_a(x)$. Direct substitution in $Sf = f'S$ gives the two conditions

$$CP(x) = P'(Ax)C, \tag{17}$$

$$Q'(Ax) = CQ(x) + t_a(x) + P'(Ax)t_s(x). \tag{18}$$

Indeed the unseparated cocycle equation reduces to the second one by multiplication with $\mathrm{id} + P'(Ax)$, whose inverse is itself. Polarization over $\mathbb{F}_4$ eliminates t_a and gives

$$\beta'(Ax, Ay) = C\beta(x, y) + P'(Ax)t_s(y) + P'(Ay)t_s(x). \tag{19}$$

Equations (17) and (19) say exactly that $(x, z) \mapsto (Ax, Cz+t_s(x))$ is an $\mathbb{F}_4$-Lie homomorphism.

Conversely, for any such Lie homomorphism the function $Q'(Ax) + CQ(x) + P'(Ax)t_s(x)$ has zero polar and is an ordinary homogeneous quadratic. It is therefore a unique sum of squares $\sum b_i x_i^2$. Taking this as t_a gives a unique intertwiner. Scalar parts respect composition because A and C are $\mathbb{F}_4$-linear. Thus the correspondence is an equivalence on both objects and all morphisms, proving also the centralizer assertion. No inverse of A, C, or the intertwiner was used. Thus the argument includes maps with nontrivial kernel or cokernel. □

Corollary 2.7. *Every finite-dimensional metabelian polar algebra is nilpotent and satisfies* $(\operatorname{ad} v)^3 = 0$ *for every* v. *If* $w = \dim_{\mathbb{F}_4} W$, *its class is at most* $w + 1$.

Proof. The commuting square-zero operators generate a nilpotent associative ideal. They have a common nonzero kernel whenever $W \neq 0$; repeating on the quotient gives a strictly triangular flag, so all products of w operators vanish. Apply (16). Finally $\operatorname{ad} v$ maps V into W, and its square on W is zero. □

2.2 An intrinsic change of addition

Let $\mathcal{Q}$ have finite binary-quadratic solutions $f^2 = f + I$ with metabelian polar algebra as objects, and all binary-linear intertwiners $Tf = gT$ as morphisms. Let $\mathcal{E}$ have finite-dimensional metabelian $\mathbb{F}_4$-Lie algebras satisfying $(\operatorname{ad} u)^2 \mathfrak{l}' = 0$, with all $\mathbb{F}_4$-Lie homomorphisms.

Proposition 2.8 (An involutive twist of binary Lie structures)**.** *Let* $(E, +, B)$ *be a metabelian binary Lie algebra with derived ideal* W, *satisfying* $(\operatorname{ad} u)^2 W = 0$ *for every* u. *Set* $P_u = \operatorname{ad} u|_W$ *and*

$$C(u, v) = (I + P_u + P_v)B(u, v), \qquad u \star v = u + v + C(u, v).$$

Then $(E, \star, C)$ *is another binary Lie algebra in the same category. Applying this operation twice recovers* $(+, B)$. *It is a canonical involutive functor on this precise Lie category, with the same underlying morphism functions.*

Proof. The operators P_u commute because W is abelian, and square to zero by hypothesis. For $a = P_x, b = P_y, c = P_z$, put $w = C(x, y) = (I + a + b)B(x, y)$. Direct expansion gives

$$C(x \star y, z) + C(x, z) + C(y, z) = bB(x, z) + aB(y, z) + (I + a + b + c)c(I + a + b)B(x, y).$$

The last operator is c, so this is zero by the original Jacobi identity. Thus C is additive for $\star$ in both arguments, before associativity is assumed. Both associations of three inputs now equal $x + y + z + C(x, y) + C(x, z) + C(y, z)$. The operation is therefore an abelian group operation, with identity zero and exponent two. On W, it is the old addition, and $C(x, w) = B(x, w)$ for $w \in W$. The Jacobi sum for C is the old Jacobi sum plus $\sum_{\text{cyc}} P_x(P_y + P_z)B(y, z)$; the latter vanishes by multiplying the old Jacobi identity by P_x, P_y, P_z and summing.

The new derived ideal is W. One inclusion follows from $C(E, E) \subseteq W$; for the other use

$$B(u, v) = C(u, v) + C(u \star v, C(u, v)).$$

The actions on W are unchanged, so the new square condition holds. If $S = u \star v$, $w = C(u, v)$, then $S \star w \star C(S, w) = u + v$: the last extra term is $P^2_{u+v} w = 0$. The second twisted bracket is $(I + P_u + P_v)^2 B(u, v) = B(u, v)$. This proves involutivity. Every original Lie homomorphism preserves the defining formulas; involutivity gives the converse as well. □

There is a choice-free realization of that equivalence. In the next formulas $+$ and B denote the original binary addition and polar bracket of a source object. Define on its underlying set

$$C(u, v) = B(u, v) + B(u + v, B(u, v)), \qquad u \oplus v = u + v + C(u, v), \qquad \omega \cdot u = f(u). \quad (20)$$

The other two $\mathbb{F}_4$ scalars are zero and the identity, and $\omega^2 u = f^2(u)$.

Theorem 2.9 (Canonical all-morphism equivalence)**.** *The operations* (20) *make the underlying set a vector space over* $\mathbb{F}_4$*, and* C *makes it an object of* $\mathcal{E}$*. On morphisms the correspondence uses the same set map. It is fully faithful and has a canonical inverse: for an object* $(L, \oplus, C)$ *of* $\mathcal{E}$*, define*

$$u + v = S \oplus C(u, v) \oplus C(S, C(u, v)), \qquad S = u \oplus v, \qquad f(u) = \omega u. \tag{21}$$

These are the binary vector addition and map of an object of $\mathcal{Q}$*. The two constructions are inverse on objects and on their underlying morphism functions.*

Proof. It suffices to establish the formulas in any normal-form coordinates; their right sides are already intrinsic. Put

$$H_f(x, z) = (x, (I + P(x))(z + Q(x))), \qquad H_f^{-1}(x, z) = (x, (I + P(x))z + Q(x)).$$

For the morphisms in Step 6 of Theorem 2.6, direct use of its cocycle equation gives the naturality identity

$$H_g T_f = T_L H_f. \tag{22}$$

Indeed the second coordinate inside $I + P'(Ax)$ becomes $C(z + Q(x)) + (I + P'(Ax))t_s(x)$, whose image is $C(I + P(x))(z + Q(x)) + t_s(x)$.

For completeness, write $u = (x, z)$, $v = (y, t)$, $a = P(x)$, $b = P(y)$, and $\beta = Q(x + y) + Q(x) + Q(y)$. A calculation using $a^2 = b^2 = 0$, $ab = ba$, gives both

$$H_f^{-1}(H_f u + H_f v) + u + v = [H_f u, H_f v]_0 = B(u, v) + (a + b)B(u, v).$$

Here H_f^{-1} is the identity on W, and $(a + b)B(u, v) = B(u + v, B(u, v))$. Thus H_f transports $\oplus, C, f$ to ordinary vector addition, the auxiliary Lie bracket, and multiplication by ω. This proves that (20) defines the claimed structure in every class, not merely in a low-class truncation.

To recover the old addition put $w = C(u, v) \in W$ and $S = u \oplus v$. On $V \times W$, the two bracket formulas agree and their actions square to zero. Hence $S \oplus w = S + w + C(S, w)$, and subsequently adding $C(S, w)$ by $\oplus$ cancels that last term. The result is $u + v$, proving (21).

For an arbitrary object of $\mathcal{E}$, choose a vector-space splitting only to verify the construction: its usual action and cocycle data give the existing normal-form solution, and the preceding calculation transports that solution to exactly (21). Thus this formula is associative, has exponent two, and gives the required quadratic solution; no splitting enters its definition. Applying the two formulas successively recovers the original operations.

Every source intertwiner preserves $+, f, B$, hence $\oplus, C$ and the $\mathbb{F}_4$ scalars. Conversely, a target Lie homomorphism preserves the right side of (21) and $f = \omega$, so it is an original binary-linear intertwiner. This proves full faithfulness and the asserted inverse functor. □

Exactness and central extensions. The canonical functor changes operations, not the underlying sets or morphism functions. Kernels and images therefore have the same point sets. Short exact sequences are preserved and reflected, with $\dim_{\mathbb{F}_2} = 2 \dim_{\mathbb{F}_4}$ on corresponding objects. Even ranks are also immediate from the original orbit-parity argument and are not a separate new invariant.

The original derived ideal is the same set W as the new derived ideal. On W the additions coincide, and on $V \times W$ the brackets coincide; hence all lower central terms agree as sets. The original polar bracket can be recovered as

$$B(u, v) = C(u, v) \oplus C(u \oplus v, C(u, v)).$$

Thus the two centers coincide as sets. An original f-stable binary subspace is the same set as a new $\mathbb{F}_4$-Lie subalgebra; original f-stable polar Lie ideals correspond to new $\mathbb{F}_4$-Lie ideals. No claim is made about original ideals lacking f-invariance. These assertions follow in both directions directly from the two operation formulas. In particular, central short exact sequences whose middle object remains metabelian correspond to central Lie extensions. This does not cover central extensions that leave the metabelian category.

Scope. The formula uses the metabelian normal-form identities, especially commuting square-zero actions on the derived ideal. The general construction in [21] treats non-metabelian objects with an additional quartic term. The canonical change of addition preserves the original classification through its full morphism correspondence.

3 Sharp metabelian bounds and equality cases

The metabelian correspondence turns the dimension problem into a quotient problem for one multigraded commutator module. We first use that module to determine the least dimension at each polar class, then classify every equality case. Section 5 examines the much larger populations immediately above this minimum.

3.1 A multigraded commutator module

For $d \geq 2$, put

$$R_d = \mathbb{F}_4[t_1, \dots, t_d]/(t_1^2, \dots, t_d^2), \qquad \mathfrak{m} = (t_1, \dots, t_d).$$

Let W_d be the R_d-module generated by v_{ij}, $i < j$, subject to

$$t_i v_{jk} + t_j v_{ik} + t_k v_{ij} = 0 \qquad (i < j < k). \tag{23}$$

We set $v_{ji} = v_{ij}$ and $v_{ii} = 0$. This is the classical free metabelian commutator-module presentation, base-changed to R_d; compare [17, Theorem 6.2] and [12, Sections 2–3].

Every derived module in Theorem 2.6 is a quotient of W_d: send t_i to P_i and v_{ij} to a_{ij}. The intrinsic-generation condition makes this homomorphism surjective. Conversely, every quotient of W_d, with these images, gives the normal form (7).

Grade R_d by coefficient degree, and put the generators v_{ij} in coefficient degree zero. We also use the finer grading $\deg t_i = e_i$, $\deg v_{ij} = e_i + e_j$.

Proposition 3.1. *The coefficient-degree Hilbert polynomial of W_d is*

$$H_d(T) = \sum_{k=2}^{d} \binom{d}{k} \big((k-1)T^{k-2} + kT^{k-1}\big) + \binom{d}{2} T^2. \tag{24}$$

Consequently

$$\dim_{\mathbb{F}_4} W_d = (d-1)2^d + 1 - d + \binom{d}{2}.$$

The universal solution on $\mathbb{F}_4^d \oplus W_d$ has class $\max(4, d+1)$.

Proof. A raw multihomogeneous monomial is $t_S v_{ij}$, where S is square-free. Its multidegree has entries zero, one, or two, and at most two entries are doubled. Fix its support of size $k \geq 2$.

If no index is doubled, the raw coordinates are the edges of the complete graph on the support. Relations (23) are its triangle sums. The quotient edge space has dimension $k-1$, in coefficient degree $k-2$.

If exactly one index j is doubled, only edges incident with j occur. Triangle relations through j identify every pair of these edges, leaving dimension one. There are k choices of j, in coefficient degree $k-1$.

If exactly two indices i, j are doubled, there is one raw coordinate. A third index in the support kills it by a triangle relation, since the other two terms contain a squared variable. The coordinate survives exactly when the support is $\{i, j\}$, giving the final term of (24). These cases exhaust the multigrading.

The maximum coefficient degree is two for $d = 2, 3$, and $d-1$ for $d \geq 4$. Equation (16) gives the class. □

In particular, binary degree two permits unbounded polar nilpotency class. For the exact minimum dimension we must retain a terminal commutator in as small a quotient of W_d as possible.

3.2 Terminal tensors and contraction modules

Lemma 3.2. *Let $\mathfrak{l}$ be a metabelian $\mathbb{F}_4$-Lie algebra satisfying* (12)*, of class $c \geq 3$. Some nonzero terminal commutator has a repeated input between its first bracket pair and an operator position. If $c \geq 5$, there is a subalgebra of the same class generated by exactly $c-1$ elements modulo its derived ideal.*

Proof. The last lower central ideal is spanned by

$$T(x_1, x_2; x_3, \ldots, x_c) = P(x_3) \cdots P(x_c)\beta(x_1, x_2).$$

Indeed $W = \operatorname{span}\beta + \mathfrak{m}W$, and multiplication by $\mathfrak{m}^{c-2}$ kills the second summand. The operator positions are symmetric, and a repeated input within them gives zero.

Suppose every value $T(x, y; x, z_4, \ldots, z_c)$ were zero. Polarizing the repeated input gives

$$T(x, y; z, \ldots) = T(z, y; x, \ldots).$$

Together with symmetry of the first pair and the operator positions, this makes T fully symmetric. Jacobi on the first three inputs then gives $3T = T = 0$, a contradiction.

Choose a nonzero repeated-input value. Its inputs span a subspace U_0 of the abelianization of dimension at most $c-1$. Choose a linear section on a basis of U_0 and use its linear lifts for all the inputs. Changing a lift inside W does not alter the terminal commutator: the additional term belongs to $\mathfrak{m}^{c-1}W = 0$. These basis lifts generate a subalgebra H of class c, with abelianization exactly U_0. Indeed, any linear combination of the chosen generators whose image in U_0 vanishes is zero, and all remaining Lie words lie in H'. Its module is a quotient of W_{d_0}, where $d_0 = \dim U_0$. By Proposition 3.1, class $c \geq 5$ requires $d_0 \geq c-1$. Thus $d_0 = c-1$. □

For $d \geq 4$, the top coefficient degree of W_d is $d-1$. It has basis $T_1, \ldots, T_d$, where

$$T_j = \Big(\prod_{h \neq i} t_h\Big) v_{ij} \quad (i \neq j). \tag{25}$$

Triangle relations identify the choices of i.

Lemma 3.3. *The top component is the* $\mathrm{GL}(d, 4)$*-module* $\det(U) \otimes U$*. For every nonzero functional λ on that component, extended by zero to the other degrees,*

$$\dim_{\mathbb{F}_4} R_d\lambda = 2^d - 2.$$

If a quotient M of W_d retains a nonzero top component, then $\dim_{\mathbb{F}_4} M \geq 2^d - 2$*, whether or not the quotient is graded.*

Proof. Under a diagonal basis change A, T_j scales by $\det(A)A_{jj}$. A transvection $e_j \mapsto e_j + \alpha e_k$ sends T_j to $T_j + \alpha T_k$, and fixes the other top basis vectors. To check the formula choose $i \notin \{j,k\}$ in (25); any extra product has a squared variable. For the remaining vectors an index outside the affected entries can be chosen since $d \geq 4$. Together with permutations these transformations prove the module assertion. They act transitively on nonzero dual vectors, including scalar multiples.

Normalize $\lambda(T_j) = 1$, $\lambda(T_k) = 0$ for $k \neq j$. We claim

$$\operatorname{Ann}_{R_d}(\lambda) = \Big(\prod_{i \neq j} t_i\Big). \tag{26}$$

The annihilator is multigraded: contractions by distinct monomials have distinct resulting dual multidegrees. For a square-free monomial t_S not containing j, the remaining multidegree is of the one-doubled-index type. It survives unless S contains every other index. If $j \in S$, the remaining multidegree has no doubled index. It survives whenever an index is omitted, as witnessed by an edge joining that index to j; it vanishes when all indices are used. These are exactly the two monomials in the ideal on the right of (26). This proves the claim and the dimension formula.

For an arbitrary quotient M, pull back a functional nonzero on the top image to $\phi \in W_d^*$. Its highest homogeneous component ϕ_{d-1} is nonzero. Independent homogeneous contractions of ϕ_{d-1} lift to independent contractions of ϕ, by comparison of their leading dual degrees. Hence

$$\dim M \geq \dim R_d\phi \geq \dim R_d\phi_{d-1} = 2^d - 2.$$

The filtered contraction argument applies to every quotient $W_d \to M$, including those with nongraded kernel. □

3.3 Exact minima

Theorem 3.4. *Among metabelian quadratic solutions of polar class $c \geq 2$, the minimum binary dimension is*

$$\mu(2) = 6, \quad \mu(3) = 8, \quad \mu(4) = 12, \qquad \mu(c) = 2^c + 2c - 6 \quad (c \geq 5).$$

Proof. Write $d = \dim_{\mathbb{F}_4}(V/W)$, $w = \dim_{\mathbb{F}_4} W$. Nonabelian objects have $d \geq 2$ by Lemma 2.4. A nonzero product $P_1 \cdots P_k v$, for commuting square-zero operators, yields 2^k independent vectors

$$v_S = \Big(\prod_{i \in S} P_i\Big)v \qquad (S \subseteq \{1, \ldots, k\}).$$

In a relation, choose a term with nonzero coefficient of smallest support and multiply by its complementary operators. Every other surviving term repeats an operator and vanishes, while the chosen term does not. Taking $k = c - 2$ gives $w \geq 2^{c-2}$. For $c = 2, 3, 4$ this gives the respective lower bounds $6, 8, 12$.

They are attained by cyclic quotients of $W_2 = R_2 v_{12}$: use $R_2/\mathfrak{m}$, $R_2/(t_2)$, and R_2, respectively, as the derived module in (7). The first two representatives are

$$\begin{aligned}
&(a, b, z) \longmapsto (\omega a, \omega b, \omega z + ab),\\
&(a, b, y, z) \longmapsto (\omega a, \omega b, \omega y + az + a^2 b, \omega z + ab).
\end{aligned}$$

A class-four attaining model on $\mathbb{F}_4^6$ is

$$\begin{aligned}
(a, b, c, d, e, h) \longmapsto \big(&\omega a, \omega b, \omega c + ab, \omega d + ac + a^2 b,\\
&\omega e + bc + ab^2, \omega h + ae + bd\big).
\end{aligned} \tag{27}$$

It is the normal form with $a_{12} = c$ and commuting square-zero actions $P_1 c = d$, $P_2 c = e$, $P_1 e = P_2 d = h$, all other basis images zero. Thus its derived module has basis c, d, e, h and its lower-central dimensions are $(12, 8, 6, 2, 0)$.

Let $c \geq 5$. Proposition 3.1 gives $d \geq c - 1$. Lemma 3.2 supplies a same-class subalgebra with $d_0 = c-1$. Its derived module retains the top of W_{c-1}, so Lemma 3.3 gives $w \geq 2^{c-1} - 2$. Therefore

$$\dim_{\mathbb{F}_2} V = 2(d + w) \geq 2^c + 2c - 6.$$

To attain the bound, set $d = c - 1 \geq 4$, choose a top functional λ as in Lemma 3.3, and put

$$M_d = W_d / \{w : \lambda(rw) = 0 \text{ for every } r \in R_d\}. \tag{28}$$

The denominator is an R_d-submodule. The quotient has dimension $2^d - 2$ and retains the terminal value, so its solution has class $d + 1 = c$ and binary dimension $2(d + 2^d - 2)$. □

The change of formula at class five comes from the top degree of the universal module. For two and three quotient generators,

$$H_2(T) = 1 + 2T + T^2, \qquad H_3(T) = 3 + 8T + 6T^2.$$

In these cases the degree-two double-endpoint terms contribute at the top, allowing class four even with two quotient generators. For $d \geq 4$, they lie below the top degree $d - 1$; the terminal component $\det(U) \otimes U$ then controls class $d + 1$ uniformly. This is why the contraction argument yields one formula from class five onward, while the cyclic quotients of W_2 give the three smaller values.

The extremal quotient M_d in (28) has coefficient Hilbert polynomial

$$(d - 1) + \sum_{k=1}^{d-1} \binom{d}{d - 1 - k} T^k.$$

For example, the class-five extremizer has derived Hilbert vector $(3, 6, 4, 1)$ and binary lower central dimensions $(36, 28, 22, 10, 2, 0)$. The contraction construction supplies an explicit finite module description at every class, without an orbit search.

3.4 All minimum-dimensional metabelian solutions

The bound $\mu(c)$ also admits an equality classification. Let E_c denote the number of binary-linear conjugacy classes of metabelian quadratic solutions of class c and dimension $\mu(c)$. The three small classes close directly in the cyclic module W_2.

Lemma 3.5 (The three small minima)**.** *There is exactly one metabelian extremizer in each of classes two, three, and four. Thus $E_2 = E_3 = E_4 = 1$.*

Proof. Equality in the separate bounds $d \geq 2$ and $w \geq 2^{c-2}$ used in Theorem 3.4 forces $d = 2$ and $w = 1, 2, 4$, respectively. Intrinsic generation makes W cyclic over $R_2 = k[t_1, t_2]/(t_1^2, t_2^2)$, $k = \mathbb{F}_4$, generated by a_{12}: iterate $W = ka_{12} + \mathfrak{m}W$ and use $\mathfrak{m}^3 = 0$. Hence $W = R_2/I$, with the cocycle represented by its cyclic generator. For $w = 1$ the ideal is $I = \mathfrak{m}$, and for $w = 4$ it is zero.

If $w = 2$, then $I \subseteq \mathfrak{m}$ has dimension two. Every nonzero ideal of R_2 contains its socle kt_1t_2: multiply a nonzero linear part by a suitable generator, or start with a socle element. Thus I also contains a nonzero linear form u, after subtracting a multiple of t_1t_2, and $I = (u)$. A change of basis in $\mathrm{GL}_2(k)$ takes (u) to (t_2). The accompanying nonzero determinant factor on a_{12} is absorbed by a rescaling of the cyclic module generator. These are precisely the three quotients displayed above, up to the isomorphisms of Theorem 2.6. □

For the uniform range, equality first fixes the derived module. We then classify its surviving cocycles under all changes of quotient basis and splitting; this last step distinguishes equality objects with identical lower central dimensions. Fix $c \geq 5$, and put $k = \mathbb{F}_4$, $d = c - 1$, $m = d - 1$, and

$$U = ke_0 \oplus H, \qquad H = \operatorname{span}_k\{e_1, \dots, e_m\}, \qquad R = k[t_0, \dots, t_m]/(t_0^2, \dots, t_m^2).$$

Write $K = (t_1, \dots, t_m)$, $\tau = t_0 \cdots t_m$, and $\tau_{ij} = \prod_{\ell \notin \{i,j\}} t_\ell$. For $\kappa = (\kappa_i) \in H^*$ and $\lambda = (\lambda_{ij})_{0 \leq i < j \leq m} \in k^{\binom{d}{2}}$, define an alternating cocycle $\beta_{\kappa,\lambda} : \Lambda^2 U \to K$ by

$$\begin{aligned} \beta_{\kappa,\lambda}(e_0, e_i) &= t_i + \lambda_{0i}\tau_{0i} && (1 \leq i \leq m), \\ \beta_{\kappa,\lambda}(e_i, e_j) &= t_0(\kappa_i t_j + \kappa_j t_i) + \lambda_{ij}\tau_{ij} && (1 \leq i < j \leq m). \end{aligned} \tag{29}$$

Let $F_{\kappa,\lambda}$ be the map in (7) with

$$W = K, \qquad P(x)w = \Big(\sum_{i=0}^{m} x_i t_i\Big)w, \qquad Q(x) = \sum_{i<j} x_i x_j \beta_{\kappa,\lambda}(e_i, e_j).$$

Thus these representatives require only multiplication in a square-zero polynomial ring.

Theorem 3.6 (Unrestricted metabelian extremizers)**.** *Every metabelian quadratic solution of class $c \geq 5$ and binary dimension $\mu(c)$ is binary-linearly conjugate to some $F_{\kappa,\lambda}$. Its derived module is isomorphic to K.*

The exact equivalence relation on the displayed parameters is the following ordinary Lie-cohomology orbit relation:

$$[\beta_{\kappa',\lambda'}] = \big[a\,\sigma_B \beta_{\kappa,\lambda}(B^{-1}\cdot, B^{-1}\cdot)\big], \qquad B = \begin{pmatrix} a & 0 \\ v & M \end{pmatrix}, \tag{30}$$

where $a \in k^\times$, $M \in \mathrm{GL}(H)$, $v \in H$, $\sigma_B(t_j) = \sum_i B_{ij} t_i$, and brackets denote classes in $H^2(U, K)$ for the action $e_i w = t_i w$. In particular, no nonlinear substitutions or Galois identifications are required.

Their number E_c is

$$E_c = \begin{cases} (7c - 13)/2, & c \geq 5 \text{ odd}, \\ 7(c-2)/2 + 2\mathbf{1}_{6|c}, & c \geq 6 \text{ even}. \end{cases} \tag{31}$$

Proof. Equality in the contraction bound forces the derived module to be the dual of $R/(t_1 \cdots t_m)$, hence to be K. An ordinary cohomology calculation gives a terminal star together with the κ- and λ-classes in (29); intrinsic generation is equivalent to the star being nonzero, and it has been normalized to one. Module intertwining maps give exactly (30). Theorem 2.6 then transfers the auxiliary-Lie classification to binary conjugacy. Appendix B proves the common-module assertion, the cohomology and morphism statements, and the complete orbit count. □

For $d \geq 5$, the orbit relation is especially explicit. If $A^{[2]}$ means entrywise squaring, its action is

$$\kappa' = a^2 M^{-T}\kappa, \qquad \lambda' = a \det(B)(\Lambda^2 B^{-T})^{[2]}\lambda. \tag{32}$$

After setting $\Omega = \lambda^{[2]}$ and writing $\Omega = e_0^* \wedge \ell + b$, the classification reduces to an alternating form b on H, two covectors on its radical, and one zero/nonzero flag. In the corank-two stratum an additional three-valued scalar invariant survives exactly when $6 \mid c$; this accounts for the correction in (31). At $c = 5$, two cohomological degrees coincide and the parameter representation is nonsplit. A symmetric-matrix description nevertheless gives all eleven classes, including this endpoint in the same count.

Models inside Cohn's construction. The regular Lie model and the zero-λ slice have a concrete classical source. Take characteristic two and the finite ground field $k = \mathbb{F}_4$ in Cohn's construction [5, Section 2, pp. 402–403]. Reindex its transcendental elements as $\xi_0, \xi_1, \ldots$, put $u = u(0)$, and let $D_i = \operatorname{ad} \xi_i|_M$. For $S \subseteq \{0, \ldots, d-1\}$, set

$$v_S = \Big(\prod_{i \in S} D_i\Big) u = \Big(\prod_{i \in S} \xi_i\Big) \sum_{T \subseteq S} u(\mathbf{1}_T).$$

Here $\mathbf{1}_T$ is the indicator sequence indexing Cohn's module basis. The displayed expansion is triangular in those basis vectors, so the v_S's are independent even over Cohn's larger coefficient field. The operators commute and square to zero, and $D_i v_S = v_{S \cup \{i\}}$ if $i \notin S$, and zero otherwise. Consequently the k-span of the ξ_i's and v_S's is the regular-module semidirect product $k^d \ltimes R_d$, of dimension $d + 2^d$ and class $d + 1$. Its polar realization is (38), or the compatible model F_0 after writing a ring element as its constant term plus an element of J.

The zero-λ models in (29) also occur as finite subalgebras. Choose

$$e_0 = \xi_0 + v_\varnothing, \qquad e_i = \xi_i + \kappa_i v_{\{0\}} \quad (1 \le i < d).$$

Their derived module is spanned by the v_S's with $S \cap \{1, \ldots, d-1\} \neq \varnothing$, and is therefore K. Their action and brackets are multiplication by t_i, $\beta(e_0, e_i) = t_i$, and $\beta(e_i, e_j) = t_0(\kappa_i t_j + \kappa_j t_i)$. Thus this finite subalgebra has dimension $d + 2^d - 2$, class $d + 1$, and gives $F_{\kappa,0}$ through Theorem 2.6. These finite identifications are derived here from Cohn's construction; his paper establishes the infinite-dimensional example rather than an extremal dimension formula.

Cohn's full characteristic-two algebra satisfies

$$[[[x, y], x], y] = 0.$$

Indeed, for $x = a + u'$, $y = b + v'$, its bracket is $D_b u' + D_a v'$; the commuting operators D_a, D_b square to zero, so applying $D_a D_b$ kills it. In the present family,

$$[[[e_i, e_j], e_i], e_j] = t_i t_j \beta(e_i, e_j) = \lambda_{ij} \tau.$$

Hence the nonzero-λ models lie outside every Lie subquotient of Cohn's characteristic-two example. The sharp bounds and full orbit classifications determine how the classical slice and these additional models populate the extremal layers. The same finite-subalgebra identifications hold over any finite field of characteristic two; the binary-map interpretation uses $k = \mathbb{F}_4$.

3.5 When the original polar bracket is $\mathbb{F}_4$-bilinear

Return to the normal form (7) and keep the intrinsic scalar actions on $U = V/W$ and W. Its auxiliary bracket (15) is always $\mathbb{F}_4$-bilinear. The following definition concerns the original polar bracket of f.

Definition 3.7. A metabelian solution is *compatible* if V admits an $\mathbb{F}_4$-vector space structure that restricts to the intrinsic structure on W, induces the intrinsic structure on U, and makes the original polar bracket $\mathbb{F}_4$-bilinear.

In the criterion below we use the symmetric convention $a_{ji} = a_{ij}$ and $a_{ii} = 0$.

Theorem 3.8. *For a normal form with data P_i, a_{ij}, the following are equivalent:*

(i) *the solution is compatible;*

(ii) *there are* $b_1, \dots, b_d \in W$ *satisfying*

$$P_i b_i = 0, \qquad P_i b_j = P_j a_{ij} \quad (i \neq j); \tag{33}$$

(iii) *the auxiliary* $\mathbb{F}_4$*-Lie algebra has a restricted square* r *with* $r(W) = 0$ *and* $r(\mathfrak{l}) \subseteq W$.

If a compatible scalar lift exists, the number of such scalar actions on the fixed binary space, with the prescribed actions on U, W*, is* 4^{dk}*, where* $k = \dim_{\mathbb{F}_4} \bigcap_i \ker P_i$.

Proof. Every binary-linear lift of multiplication by ω has the form

$$T(x, z) = (\omega x, \omega z + t(x)).$$

The relation $T^2 + T = \mathrm{id}$ is exactly $t(\omega x) = \omega^2 t(x)$. Thus $t(x) = b(x) = \sum_i x_i^2 b_i$. The involutory binary shear $S(x, z) = (x, z + b(x))$ conjugates T to standard scalar multiplication. It changes the cocycle of f to

$$Q + b + P(Q + b).$$

The first two terms have $\mathbb{F}_4$-bilinear polars. The last term has scalar weight zero. Its polar could be $\mathbb{F}_4$-bilinear only if that polar were zero, since an $\mathbb{F}_4$-bilinear polar has weight two. An additive zero-constant function of weight zero is zero: additivity and invariance under ω give its value at $\omega^2 x = x + \omega x$ equal to zero, and then invariance gives its value at x equal to zero. Compatibility is therefore equivalent to

$$P(x)(Q(x) + b(x)) = 0. \tag{34}$$

After the triangle cancellations, comparison of x_i^3 and $x_i x_j^2$ gives precisely (33).

We recall explicitly the restricted-square axioms in characteristic two: $r(\alpha u) = \alpha^2 r(u)$, $r(u+v) = r(u) + r(v) + [u, v]_0$, and $\operatorname{ad} r(u) = (\operatorname{ad} u)^2$. Any square with the range and vanishing conditions in (iii) must have the form

$$r(x, z) = (0, P(x)z + q(x)), \qquad q(x) = Q(x) + b(x). \tag{35}$$

Its first two axioms are automatic. For $u = (x, z), v = (y, t)$, the two adjoint expressions have final coordinates

$$P(x)P(y)z + P(x)\beta(x, y), \qquad P(x)P(y)z + P(y)q(x),$$

respectively. Condition (34) implies $P(x)\beta(x, y) = P(y)q(x)$: take the coefficient of s in $P(x + sy)q(x + sy) = 0$. Its degree in s is at most three, below $|\mathbb{F}_4|$, so the comparison is valid for functions over $\mathbb{F}_4$. This proves the third restricted axiom. Conversely that axiom, applied to u, gives $[u, r(u)]_0 = 0$, hence (34).

Finally two solutions of (33) differ by independent choices of d vectors in $\bigcap_i \ker P_i$. The displayed count follows. It counts scalar actions, not their conjugacy classes. □

Example 3.9. Write the eight-dimensional class-three model over $\mathbb{F}_4$ as

$$N(a, b, y, z) = (\omega a, \omega b, \omega y + az + a^2 b, \omega z + ab).$$

On $W = \operatorname{span}_{\mathbb{F}_4}\{y, z\}$, its data are $P_1 z = y$, $P_1 y = 0$, $P_2 = 0$, and $a_{12} = z$. The equation with $(i, j) = (2, 1)$ in (33) would require

$$0 = P_2 b_1 = P_1 a_{21} = y,$$

which is impossible. Its auxiliary bracket is $\mathbb{F}_4$-bilinear, but the original polar bracket admits no compatible scalar action. Thus the class-three metabelian minimum $\mu(3) = 8$ lies outside the compatible subclass, whose minimum is $\nu(3) = 12$. The general dimension cost is proved in the next section.

Remark 3.10. The restricted square in Theorem 3.8 is a square on the auxiliary algebra. The theorem imposes both its range and its restriction to the derived ideal. General correspondences between quadratic maps and restricted Lie structures, such as [16, Section 9], carry their own additional identities. Here the exact additional condition is the solvable linear system (33).

3.6 Sharp compatible bounds and extremizers

3.6.1 The abelianization bound

Lemma 3.11. *A nonabelian compatible solution of class c, with $d = \dim_{\mathbb{F}_4}(V/W)$, satisfies $c \le d$.*

Proof. Use a restricted square from Theorem 3.8. The identity $P(x)\beta(x,y) = P(y)r(x)$, with a lift of x understood, implies

$$P(x)P(y)\beta(x,y) = 0. \tag{36}$$

In the universal module this kills the submodule K spanned by $t_it_jv_{ij}$, $i<j$. It has dimension $\binom{d}{2}$ and lies in coefficient degree two. Every multiplication by an endpoint kills it by a square, and multiplication by a third index kills it by a triangle relation.

The space K is $\mathrm{GL}(U)$-invariant: the expression $P(x)P(y)\beta(x,y)$ is additive in each variable and homogeneous of degree two in each. For additivity in x, the cross term is

$$P(y)\big(P(x)\beta(z,y) + P(z)\beta(x,y)\big) = P(y)^2\beta(x,z) = 0$$

by Jacobi. Its basis values therefore span precisely K.

Put $\overline{W}_d = W_d/K$. The multigrading in Proposition 3.1 shows that its highest coefficient degree is $d-1$, for every $d \ge 2$, and that its top has the basis T_j of (25). This top is $\det(U)\otimes U$ also for $d = 2, 3$: the same diagonal and transvection calculations apply, with any additional double-endpoint term now in K. For $d = 2$ this is immediate from $T_1 = t_1v_{12}, T_2 = t_2v_{12}$.

Every compatible derived module M is a quotient of $\overline{W}_d$, so $c \le d+1$. Suppose $c = d+1$. Choose a functional on M nonzero on its terminal component. A change of quotient basis normalizes its top pullback to $\lambda(T_j) = 1$, $\lambda(T_k) = 0$ for $k \ne j$. Using $P_ja_{ij} = P_ib_j$, we have, for $i \ne j$,

$$\mathrm{im}(T_j) = \Big(\prod_{h\ne j} P_h\Big)b_j. \tag{37}$$

Write $b_j = \sum c_{kl}a_{kl} \pmod{\mathfrak{m}M}$, using intrinsic generation. The product on the right kills $\mathfrak{m}M$. If j is an endpoint of v_{kl}, then $(\prod_{h\ne j} t_h)v_{kl}$ is a top vector whose doubled endpoint is different from j. If j is not an endpoint, the vector has both endpoints doubled and is in K. Thus the normalized top functional is zero on the right side of (37), but one on the left. This contradiction proves $c \le d$. □

3.6.2 Counting preimages of an operator cube

Lemma 3.12. *A compatible solution of class $c \ge 2$ satisfies*

$$\dim_{\mathbb{F}_4} W \ge 2^{c-1} - 1.$$

Proof. The case $c = 2$ says $W \ne 0$. Let $c \ge 3$. By Lemma 3.2, choose a nonzero terminal value

$$P(x)P(z_1)\cdots P(z_{c-3})\beta(x,y).$$

Restrictedness gives $P(x)\beta(x,y) = P(y)b$, where $b = r(x) \in W$. Put $m = c-2$, $u_1 = y, u_2 = z_1, \ldots, u_m = z_{c-3}$. Then $P(u_1)\cdots P(u_m)b \neq 0$. The 2^m vectors

$$b_S = \Big(\prod_{i\in S} P(u_i)\Big)b \qquad (S \subseteq \{1,\ldots,m\})$$

are independent by the complementary-operator argument used in Theorem 3.4. Let C be their span. Since $[x, r(x)]_0 = 0$, $P(x)$ kills b, hence all of C.

For each nonempty S, choose $i \in S$ and set

$$v_S = \Big(\prod_{j\in S\setminus\{i\}} P(u_j)\Big)\beta(x,u_i).$$

These vectors lie in the intrinsic derived ideal, and restrictedness gives $P(x)v_S = b_S$. Thus

$$\dim\ker(P(x)|_W) \geq 2^m, \qquad \operatorname{rank}(P(x)|_W) \geq 2^m - 1.$$

Rank-nullity yields $\dim W \geq 2^{m+1} - 1 = 2^{c-1} - 1$. □

Theorem 3.13. *For every $c \geq 2$, the minimum binary dimension among compatible solutions of polar class c is*

$$\nu(c) = 2^c + 2c - 2.$$

Proof. Lemmas 3.11 and 3.12 give the lower bound by addition. For attainment put $s = c-1$, $R_s = \mathbb{F}_4[t_1,\ldots,t_s]/(t_i^2)$, and use

$$f(x,z) = \left(\omega x, \omega z + \Big(\sum_i x_i t_i\Big)z\right) \quad \text{on } \mathbb{F}_4^s \oplus R_s. \tag{38}$$

The square of $\sum x_i t_i$ is zero, so direct substitution gives (1). The polar bracket is $\mathbb{F}_4$-bilinear and has intrinsic derived ideal $\mathfrak{m}R_s$. The scalar actions induced by f on that ideal and on the abelianization are the displayed ones, so the map is compatible. Its lower central ideals are $\gamma_k = \mathfrak{m}^{k-1}R_s$ for $k \geq 2$, giving class $s+1 = c$. The derived and quotient dimensions are $2^s - 1$ and $s+1$. The binary dimension is therefore $2(s + 2^s) = 2^c + 2c - 2$. □

The regular Lie model in (38) is a finite subalgebra of Cohn's characteristic-two construction [5]; Section 3.4 gives the identification. The preceding bounds establish its precise extremal dimension.

For $c \geq 5$, the extra four binary dimensions have an exact structural meaning. The sharp bounds in the two subclasses are

	d	w
metabelian	$c-1$	$2^{c-1}-2$
compatible metabelian	c	$2^{c-1}-1$

where each entry is the corresponding lower bound, attained simultaneously. Compatibility therefore requires one additional $\mathbb{F}_4$-dimension in the quotient and one in the derived ideal, giving four additional binary dimensions. The restricted square explains the strengthened bounds: it constrains terminal commutators and supplies independent preimages for the nonempty faces of the operator cube. The elementary semidirect construction attains this cost.

3.6.3 A common module and a full parameter family

The equality cases of the two bounds have a rigid module structure, although the resulting maps admit many inequivalent parameters. Throughout this part put $k = \mathbb{F}_4$, $s = c - 1 \geq 2$, and

$$R = k[t_1, \dots, t_s]/(t_1^2, \dots, t_s^2), \quad J = (t_1, \dots, t_s), \quad \tau = t_1 \cdots t_s, \quad Z = k\tau. \tag{39}$$

The socle $Z = \operatorname{Ann}_R(J)$ is one-dimensional. We use the parameter space

$$\mathcal{V}_s = J^2/Z = \bigoplus_{j=2}^{s-1} \Lambda^j(k^s), \qquad D_s = \dim_k \mathcal{V}_s = 2^s - s - 2. \tag{40}$$

In characteristic two the exterior algebra is the displayed commutative square-zero algebra. Each parameter is represented by its unique sum of terms of degrees two through $s-1$. These are degrees of fixed ring coefficients; the maps below still have binary degree at most two.

Proposition 3.14 (Canonical extremal models)**.** *For $p \in \mathcal{V}_s$, put*

$$a_p(X, V) = \sum_{i=1}^{s} X_i t_i + Vp \quad (X \in k^s,\ V \in k).$$

Then

$$F_p(X, V, z) = \big(\omega X, \omega V, \omega z + a_p(X, V)(z + V)\big), \qquad z \in J, \tag{41}$$

is a compatible solution of class $c = s + 1$ and binary dimension $\nu(c)$. Its intrinsic derived ideal is J, and

$$\gamma_j = J^{j-1} \quad (j \geq 2), \qquad \dim_k \gamma_j = \sum_{h=j-1}^{s} \binom{s}{h}.$$

Proof. Every element of J squares to zero. Write $r_p = F_p - \omega\,\mathrm{id}$. At the image under F_p, the expressions a_p and $z + V$ become ωa_p and $(\omega + a_p)(z + V)$, respectively. Hence $r_p(F_p(y)) = \omega^2 r_p(y)$, proving $F_p^2 = F_p + \mathrm{id}$. The displayed expression is quadratic over k, and thus has binary degree at most two. Its original polar bracket is k-bilinear:

$$[(X, V, z), (Y, T, w)] = (0, 0, a_p(X, V)(w + T) + a_p(Y, T)(z + V)).$$

With quotient basis $e_1, \dots, e_s, v$, the action and cocycle are

$$P(e_i) = t_i \cdot, \quad P(v) = p \cdot, \quad \beta(e_i, e_j) = 0, \quad \beta(e_i, v) = t_i. \tag{42}$$

The square values of r_p are $b_i = 0$, $b_v = p$. The cocycle values generate J, and the core actions generate its augmentation filtration, proving all assertions about the derived ideal and lower central series. The dimension is $2((s + 1) + (2^s - 1)) = 2^c + 2c - 2$. □

Lemma 3.15 (The common extremal module)**.** *Every compatible solution of class $c \geq 3$ and dimension $\nu(c)$ has derived module isomorphic to J under $s = c - 1$ independent quotient directions acting by $t_1, \dots, t_s$. There is one remaining quotient direction, whose action, after a change of basis, is multiplication by a parameter in $\mathcal{V}_s$. In particular all these extremizers have the lower central dimensions in Proposition 3.14.*

Proof. Equality in the two separate bounds forces $d = c$ and $\dim_k W = 2^{c-1} - 1$. In the proof of Lemma 3.12, put $m = c - 2$, $T = P(x)$, $A = k[t_1, \dots, t_m]/(t_i^2)$, and $C = Ab$. The complementary-operator argument identifies C with the regular module A. Every nonempty face lies in $\operatorname{im} T$, whereas $C \subseteq \ker T$. Equality of dimensions now gives

$$\ker T = C \cong A, \qquad \operatorname{im} T = I_A b, \quad I_A = (t_1, \dots, t_m).$$

Here and below kernels and images are taken on W.

For completeness, the required splitting follows directly from the Frobenius pairing. Let ℓ_A take the top coefficient in A, and extend $ab \mapsto \ell_A(a)$ from C to a linear functional ℓ on W. Nondegeneracy of $(a, a') \mapsto \ell_A(aa')$ defines an A-linear retraction $\rho : W \to A$ by $\ell_A(a\rho(w)) = \ell(aw)$ for every $a \in A$. Thus $W = C \oplus N$, and T identifies N with $I_A b$. In these coordinates $W = A \oplus I_A$, $T(a, q) = (q, 0)$. The map $(a, q) \mapsto t_0 a + q$ identifies this with the augmentation ideal of $A[t_0]/t_0^2$, giving the claimed J.

The directions $x, u_1, \dots, u_m$ are independent: applying an action relation to b first kills all coefficients of the u_i, and $T \neq 0$ kills the last one. There is therefore exactly one remaining direction v. The top-coefficient pairing makes R self-injective: $R \cong \mathrm{Hom}_k(R, k)$, and $\mathrm{Hom}_R(-, \mathrm{Hom}_k(R, k)) \cong \mathrm{Hom}_k(-, k)$ is exact. Every R-endomorphism of J extends to multiplication on R, so $\mathrm{End}_R(J) = R/Z$. Accordingly $P(v)$ is multiplication by some $p \in R/Z$. Its square is zero, forcing its constant coefficient to vanish. Subtracting its linear part from v leaves $p \in J^2/Z$, as required. □

Theorem 3.16 (All compatible extremizers)**.** *Every compatible solution of class $c \geq 3$ and dimension $\nu(c)$ is binary-linearly conjugate to a model F_p in* (41)*.*

Proof. Theorem 3.8 allows us first to write the solution as $\omega\,\mathrm{id} + r$, with a compatible restricted square r. Use the common module from Lemma 3.15. The remaining data are core cocycle values $a_{ij} = \beta(e_i, e_j)$, core square values $b_i = r(e_i)$, mixed values $A_i = \beta(e_i, v)$, and $b_v = r(v)$. We normalize them in three steps. All changes of section take values in the intrinsic ideal J.

Core cohomology. Lemma A.1 computes the relevant low-degree complex explicitly. A section change puts the core data in the form

$$b_i = \kappa_i t_i, \qquad a_{ij} = \kappa_i t_j + \kappa_j t_i \quad (\kappa_i \in k). \tag{43}$$

The restricted identities give $t_i A_i = p b_i$, so $A_i = \kappa_i p + t_i z_i$, with $z_i \in R$. Mixed Jacobi gives $t_i t_j (z_i + z_j) = 0$; hence all z_i have the same constant η. If $\eta = 0$, the mixed values have no linear terms, and the linear core values span at most the proper hyperplane $\ker(\sum \kappa_i t_i^*)$ of J/J^2 (or zero if $\kappa = 0$). This contradicts intrinsic generation of J. Replace v by $\eta^{-1} v$, scaling p, A_i by η^{-1} and b_v by η^{-2}, to make $\eta = 1$. Also $t_i b_v = p A_i \in J^3$ for all i; since $s \geq 2$, b_v has no linear part.

Removing the core class. Replace e_i by $e_i + \kappa_i v$. The new square values $b_i + \kappa_i^2 b_v + \kappa_i A_i$ have no linear terms. The new actions are multiplication by $t_i + \kappa_i p$. The substitution $\phi(t_i) = t_i + \kappa_i p$ is an automorphism of R, because its linear part is the identity. Reidentify J by ϕ^{-1}; in particular the parameter must be changed simultaneously to

$$p_{\mathrm{new}} = \phi^{-1}(p) \pmod{Z}. \tag{44}$$

The core square values still have no linear terms. By the same cohomology calculation their class is zero, so a J-valued section change kills every b_i and a_{ij}.

Removing the mixed freedom. Now (A_i) is a one-cocycle. Its linear part is still (t_i), so Lemma A.1 gives $A_i = t_i + t_i w$ for a common $w \in J$. Replacing the section of v by $v + w$ makes $A_i = t_i$. Restrictedness then gives $t_i(b_v + p_{\mathrm{new}}) = 0$ for every i, whence $b_v = p_{\mathrm{new}} + \zeta\tau$. The last term is removed by the binary-linear central shear $z \mapsto z + \zeta\tau V^2$. We have reached precisely (42), with $b_i = 0$, $b_v = p_{\mathrm{new}}$, proving the theorem.

All ring substitutions in this argument reidentify the finite-dimensional k-space J; they are linear maps on that space. No nonlinear change of the input vector is used. □

3.6.4 The exact parameter orbits

Theorem 3.17 (Binary conjugacy of extremizers)**.** *Let $s \geq 2$. The maps F_p, F_q are binary-linearly conjugate if and only if*

$$\sigma_B(p) = \delta q \pmod{Z} \quad \text{for some } B \in \mathrm{GL}_s(k),\ \delta \in k^\times, \qquad \sigma_B(t_i) = \sum_j B_{ji} t_j. \tag{45}$$

Thus the compatible extremizers of class $c = s + 1$ are in bijection with the orbits of $\mathcal{G}_s = \mathrm{GL}_s(k) \times k^\times$ on $\mathcal{V}_s$, acting by $p \mapsto \delta^{-1}\sigma_B(p)$.

Proof. Let $SF_p = F_q S$. The intrinsic ideal J is preserved, so $S(u, z) = (Au, Cz + h(u))$. The diagonal blocks commute with the intrinsic action of ω, and are therefore k-linear; h is initially only binary-linear. The mixed polar relations show that C conjugates the unital action algebras, both equal to $H = R/Z$. The kernel of $U \to \mathrm{rad}(H)/\mathrm{rad}(H)^2$ is kv. Hence

$$Av = \delta v', \qquad Ae_i = \sum_j B_{ji} e'_j + \alpha_i v', \quad B \in \mathrm{GL}_s(k),\ \delta \neq 0.$$

Compare the zero core brackets modulo J^2. All terms from h lie in J^2, and the remaining equation is $\alpha_j B_i + \alpha_i B_j = 0$ for $i \neq j$. Independence of the columns of B and $s \geq 2$ force $\alpha_i = 0$ for every i. The induced algebra automorphism therefore sends each t_i to its linear substitution $B(t_i)$. The action of v gives (45). This proves necessity without assuming the off-diagonal block is k-linear.

For the chosen representatives, the condition is equality in R, because linear substitutions preserve degree. The map

$$S(X, V, z) = (BX, \delta V, \delta\sigma_B(z))$$

then intertwines the two models directly. For arbitrary representatives with $\theta = \sigma_B(p) + \delta q \in Z$, the same conclusion holds after adding $\delta\theta V^2$ to the last coordinate. Thus passing modulo the socle loses no invariant. □

The final equivalence group consists of linear variable substitutions and one common scalar. Arbitrary nonlinear automorphisms of R do not preserve this normal form with its mixed cocycle; nor is an extra Galois action allowed, since the diagonal blocks commute with the fixed intrinsic ω.

Corollary 3.18 (Extremal diversity)**.** *Let N_c be the number of compatible extremal conjugacy classes of polar class $c \geq 3$, and put $s = c - 1$. Then*

$$N_c = \frac{1}{3|\,\mathrm{GL}_s(4)|} \sum_{B \in \mathrm{GL}_s(4)} \sum_{\delta \in k^\times} 4^{\sum_{j=2}^{s-1} \dim_k \ker(\Lambda^j B - \delta I)}, \tag{46}$$

$$N_c \geq \left\lceil \frac{4^{D_s}}{3|\,\mathrm{GL}_s(4)|} \right\rceil > \frac{1}{3} 4^{D_s - s^2}, \qquad N_c \leq 4^{D_s}. \tag{47}$$

In particular $\log_4 N_c = 2^{c-1} + O(c^2)$.

Proof. Apply Burnside's lemma to Theorem 3.17. The parameter set has 4^{D_s} points and every orbit has size at most $3|\,\mathrm{GL}_s(4)|$. Finally $|\,\mathrm{GL}_s(4)| = \prod_{j=0}^{s-1}(4^s - 4^j) < 4^{s^2}$. □

Corollary 3.19. *For every $c \geq 4$, at least $c - 2$ compatible extremizers are given by $p = 0$ and the monomials $p = t_1 \cdots t_j$, $2 \leq j \leq s - 1 = c - 2$.*

Proof. The parameter action preserves degree and distinguishes zero from nonzero parameters. The listed models are therefore inequivalent. □

The growth in Corollary 3.18 is double exponential in the polar class c. In the extremal binary dimension $n = \nu(c) = 2^{s+1}+2s$, the bound is instead exponential, $N_c > 2^{n-2s^2-4s-4}/3$. For example, class seven and dimension 140 give at least 532249921743 classes by the displayed formula; no enumeration of that many models is involved.

Classical tools and the extremal interface. The algebra R is the group algebra of an elementary abelian two-group, and J is its augmentation ideal, or first syzygy of the trivial module. These are classical objects; see Dade's treatment of the syzygy modules and their endo-trivial properties [6, Part II, Section 7 and Theorem 10.1]. The low-degree complex used above is the ordinary periodic tensor resolution for this algebra. Its interpretation as restricted cohomology agrees with the Ext convention in [10, Section 2]. The extremal conclusions here require the equality-forced common module, elimination of every remaining cocycle and square parameter, and the necessity of the precise binary conjugacy criterion. They retain the mixed interactions that are absent from a classification of the module J alone.

3.6.5 Automorphisms of every extremal model

The general lifting problem for automorphisms of an abelian Lie extension is governed by its module action and extension class; see [3, Theorems 1.1–1.2]. For the extremal models these conditions can be solved explicitly, including the unique binary correction needed to commute with the map.

Theorem 3.20 (Automorphisms of compatible extremizers)**.** *Put $k = \mathbb{F}_4$, and let $c \geq 3$, $s = c - 1$,*

$$R_s = k[t_1, \dots, t_s]/(t_1^2, \dots, t_s^2), \qquad J_s = (t_1, \dots, t_s), \qquad \tau_s = t_1 \cdots t_s.$$

Choose $p \in J_s^2/k\tau_s$, represented by its unique sum of terms of degrees $2, \dots, s-1$. For the canonical extremizer

$$F_p(X, V, w) = \big(\omega X, \omega V, \omega w + P_p(X, V)w + VP_p(X, V)\big), \qquad P_p(X, V) = \sum_i X_i t_i + Vp,$$

where $X \in k^s$, $V \in k$, $w \in J_s$, and $\omega^2 + \omega = 1$, define

$$\mathcal{S}_p = \{(B, \delta) \in \mathrm{GL}_s(k) \times k^\times : \sigma_B(p) = \delta p \text{ in } J_s^2/k\tau_s\}, \qquad \sigma_B(t_i) = \sum_j B_{ji} t_j.$$

The binary-linear centralizer fits into a split exact sequence

$$1 \longrightarrow K_s \longrightarrow C_{\mathrm{GL}(2(s+2^s),2)}(F_p) \longrightarrow \mathcal{S}_p \longrightarrow 1.$$

The kernel K_s is independent of p up to group isomorphism. It has $2^{s+1} + s - 2$ free k-parameters, with the group law described in Appendix A.1; in particular,

$$\big|C_{\mathrm{GL}(2(s+2^s),2)}(F_p)\big| = 4^{\,2^{s+1}+s-2}|\mathcal{S}_p| = 4^{\,2^c+c-3}|\mathcal{S}_p|.$$

The splitting sends (B, δ) to $(X, V, w) \mapsto (BX, \delta V, \delta\sigma_B(w))$.

Proof. The intrinsic derived ideal identifies the action algebra with $R_s/k\tau_s$. Its radical modulo its square identifies the core quotient directions and the remaining line. Comparing core brackets modulo J_s^2 forces the induced ring automorphism to be the linear substitution σ_B, giving the displayed map to $\mathcal{S}_p$. The stated splitting is verified by substitution. Appendix A.1 solves its kernel explicitly and proves that each auxiliary k-Lie automorphism has exactly one binary-linear lift commuting with F_p. No additional choices of the antiscalar block enter the order. □

For $p = 0$, one has $\mathcal{S}_0 = \mathrm{GL}_s(4) \times \mathbb{F}_4^\times$. For $p \neq 0$, projection onto B identifies $\mathcal{S}_p$ with the projective stabilizer of kp in $\mathrm{GL}_s(4)$; there is no further factor of three in this latter description.

Remark 3.21 (The class-two boundary). For the six-dimensional central model $F(X, V, z) = (\omega X, \omega V, \omega z + XV)$, the action algebra has no radical directions. The preceding sequence is replaced by

$$C_{\mathrm{GL}(6,2)}(F) \cong (k^2, +) \rtimes \mathrm{GL}_2(k), \qquad |C_{\mathrm{GL}(6,2)}(F)| = 2880.$$

This follows from the automorphisms of the three-dimensional Heisenberg algebra and the same unique binary lifting argument.

3.6.6 Asymptotic enumeration and generic symmetry

Lemma 3.22 (A uniform fixed-space bound)**.** *Let $k = \mathbb{F}_4$, $s \geq 2$, and put*

$$\mathcal{E}_s = \Lambda(k^s), \qquad \mathcal{V}_s = \bigoplus_{j=2}^{s-1} \Lambda^j(k^s), \qquad \mathcal{G}_s = \mathrm{GL}_s(k) \times k^\times.$$

Let (B, δ) act on $\mathcal{V}_s$ by $p \mapsto \delta^{-1}(\Lambda B)p$. For every $(B, \delta) \neq (I, 1)$,

$$\operatorname{codim}_{\mathcal{V}_s} \operatorname{Fix}(B, \delta) \geq r_s := 2^{s-2} - s - 2.$$

Proof. Put $T = \Lambda B$, $A = T - \delta I$, and let L_v denote left exterior multiplication by $v \in k^s$. The identity

$$AL_v - L_{Bv}A = \delta L_{Bv-v}$$

follows from $TL_v = L_{Bv}T$. If $B \neq I$, choose $Bv \neq v$. Multiplication by a nonzero exterior generator has rank 2^{s-1}, so subadditivity of rank gives

$$2^{s-1} \leq \operatorname{rank}(AL_v) + \operatorname{rank}(L_{Bv}A) \leq 2 \operatorname{rank} A.$$

If $B = I$ and $\delta \neq 1$, then A is invertible. Finally, deleting degrees $0, 1, s$ loses at most $s + 2$ dimensions of rank. Since fixed points are the kernel of $A|_{\mathcal{V}_s}$, the claimed bound follows. □

Corollary 3.23 (Asymptotic enumeration and generic automorphisms)**.** *Write*

$$D_s = 2^s - s - 2, \qquad g_s = 3|\,\mathrm{GL}_s(4)|, \qquad \eta_s = 3\,4^{\,s^2+s+2-2^{s-2}},$$

and let $N_{s+1} = |\mathcal{V}_s/\mathcal{G}_s|$. Then

$$N_{s+1} = \frac{4^{D_s}}{g_s}(1 + \epsilon_s), \qquad 0 \leq \epsilon_s \leq (g_s - 1)4^{-r_s} \leq \eta_s.$$

For a uniformly chosen parameter $p \in \mathcal{V}_s$,

$$\Pr\big(\mathrm{Stab}_{\mathcal{G}_s}(p) \neq 1\big) \leq \epsilon_s.$$

The proportion of parameter orbits having nontrivial stabilizer is at most $2\epsilon_s/(1 + \epsilon_s)$. Consequently

$$N_{s+1} \sim C_4\, 4^{\,2^s - s^2 - s - 2}, \qquad C_4 = \frac{1}{3\prod_{j=1}^{\infty}(1 - 4^{-j})}.$$

Under the canonical classification, these are the compatible extremizers of polar class $c = s+1$. Thus asymptotically almost all, whether counted by parameters or by conjugacy classes, have full binary-linear centralizer K_s from Theorem 3.20. For every $s \geq 9$, such extremizers exist, since $\eta_s \leq 3\,4^{-36} < 1$.

Proof. Burnside's lemma gives the stated formula with

$$\epsilon_s = \sum_{g \in \mathcal{G}_s \setminus \{1\}} 4^{-\operatorname{codim} \operatorname{Fix}(g)}.$$

The fixed-space bound and $g_s < 3\,4^{s^2}$ give the error estimate. The union bound gives the assertion for parameters. Let β_s be the proportion of parameters with nontrivial stabilizer. Counting stabilizer orders shows that the proportion of nonfree orbits is $(\beta_s + \epsilon_s)/(1 + \epsilon_s)$, which is at most the claimed bound because $\beta_s \le \epsilon_s$. Finally,

$$|\operatorname{GL}_s(4)| = 4^{s^2} \prod_{j=1}^{s} (1 - 4^{-j}),$$

the infinite product has positive limit, and $\eta_s \to 0$. At $s = 9$ its exponent is -36, and it decreases thereafter. The centralizer assertion now follows from Theorem 3.20. □

It follows in particular that, for every $c \ge 10$, the minimum binary-linear centralizer order among compatible extremizers is exactly 4^{2^c+c-3}. It is attained by asymptotically almost all such classes. This is a minimum of the symmetry group size within the already minimal-dimensional family.

3.6.7 The first three extremal classifications

The canonical parameters give complete classifications at three low-class compatible minima.

Corollary 3.24. *The numbers of binary-linear conjugacy classes of compatible extremizers in polar classes* $3, 4, 5$ *are* $1, 2, 7$*, respectively. Their binary dimensions are* $12, 22, 40$*. For class five, write* $p = \alpha + \beta$*, with* $\alpha \in \Lambda^2 k^4$*,* $\beta \in \Lambda^3 k^4$*, and use the notation* $t_{ij} = t_i t_j$*,* $t_{ijk} = t_i t_j t_k$*. The seven parameter representatives and their distinguishing conditions are listed in Table 1.*

Table 1: The seven compatible class-five extremizers. The conditions refer to the quadratic and cubic parts of the parameter, not to the binary degree of the map, which remains at most two.

Parameter representative	Distinguishing condition
0	$\alpha = \beta = 0$
t_{123}	$\alpha = 0,\ \beta \neq 0$
t_{12}	$\operatorname{rank} \alpha = 2,\ \beta = 0$
$t_{12} + t_{123}$	$\operatorname{rank} \alpha = 2,\ 0 \neq \beta \in \alpha \wedge k^4$
$t_{12} + t_{134}$	$\operatorname{rank} \alpha = 2,\ \beta \notin \alpha \wedge k^4$
$t_{12} + t_{34}$	$\operatorname{rank} \alpha = 4,\ \beta = 0$
$t_{12} + t_{34} + t_{134}$	$\operatorname{rank} \alpha = 4,\ \beta \neq 0$

Proof. Put $s = c - 1$. The canonical orbit correspondence reduces the assertion to the action of $\operatorname{GL}_s(k) \times k^*$ on J^2/Z, where the second factor scales the whole parameter. For $s = 2$ this parameter space is zero. For $s = 3$ it is $\Lambda^2 k^3$; the usual alternating-bivector normal form gives the two ranks zero and two.

For $s = 4$, the homogeneous decomposition is $J^2/Z = \Lambda^2 k^4 \oplus \Lambda^3 k^4$. If $\alpha = 0$, the cubic part is either zero or a nonzero decomposable three-vector, giving the first two rows. Suppose $\operatorname{rank} \alpha = 2$, and normalize $\alpha = t_{12}$. Write $k^4 = A \oplus B$, with $A = \langle t_1, t_2 \rangle$, $B = \langle t_3, t_4 \rangle$. The

stabilizer of the line $k\alpha$ consists of block matrices $\left(\begin{smallmatrix} M & N \\ 0 & P \end{smallmatrix}\right)$; the common parameter scale is then $\det M$. There is a decomposition

$$\Lambda^3 k^4 = (\Lambda^2 A \wedge B) \oplus (A \wedge \Lambda^2 B).$$

If the second component is zero, the first is either zero or nonzero, and $\mathrm{GL}(B)$ is transitive in the latter case. If the second component is nonzero, the block N removes the first component, and $\mathrm{SL}(A)$ is transitive on the remaining nonzero vector in A. These are exactly the three rank-two rows.

Finally normalize the rank-four part to $\alpha = t_{12}+t_{34}$. The map $u \mapsto \alpha\wedge u$ is an isomorphism $k^4 \to \Lambda^3 k^4$, as is seen directly on the displayed basis. It intertwines the symplectic stabilizer of α with its action on k^4. A nonzero vector extends to a symplectic basis, so that group is transitive on nonzero vectors. Thus the cubic part gives precisely the last two rows. All conditions in the table are invariant, so the listed orbits are pairwise distinct and exhaustive. □

4 The structural mechanism of near-minimum enumeration

For the ordinary counts fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$, and put

$$d = c - 1, \quad n = 2^d, \quad w_0 = n - 2, \quad \dim L' = w_0 + h, \quad \dim(L/L') = d + s, \quad h + s = r.$$

The scale d is determined by class; the quotient dimension is $N = d + s$. The commutator-module contraction produces a class-c core with d quotient directions. Its action ring is $R = k[t_0, \ldots, t_{d-1}]/(t_i^2)$. The full derived module $W = L'$ has a filtration with one factor $K = (t_1, \ldots, t_{d-1})$ and h trivial factors (Lemma C.2). This is a filtration, not a splitting assumption.

Write $S_0 = \mathrm{Soc}_R W$, $\sigma = \dim S_0 = h + 1 - t$, and $S = Z(L) \cap W$. The latter is intrinsic and equals the common kernel on S_0 of all extra quotient actions. Their maximum rank on S_0, taken over the finite base field, is denoted by ρ. Thus $\dim S \leq \sigma - \rho$.

Theorem 4.1 (A common head normalization)**.** *There is an R-module decomposition $W = M \oplus Z$, with Z trivial and $\mathrm{Soc}_R M \subseteq \mathrm{rad}\, M$, such that the core bracket generates all but at most t dimensions of the head $M/\mathrm{rad}\, M$. After one module shear, all extra actions that extend this core bracket lie in a single linear space of endomorphisms. Its restriction kernel on S_0 has dimension at most*

$$w_0 + \big(C_d + (d+1)t\big)(d-1) + t\sigma, \qquad C_j = \binom{j}{2}.$$

The normalization applies simultaneously to the entire extra-action tuple, including actions nonzero on the core socle.

The point is the replacement of the unrestricted head cost $h(d-1)$ by $O(d^3 + d^2 t)$. The normalization comes from an actual module automorphism. It does not discard a visible socle action or count only split modules. Appendix J proves the theorem and the resulting all-module covering.

The second task is to determine which parameters remain distinct under every Lie isomorphism. On the maximum-socle modules E_Q, the action algebra, its radical filtration and the extra quotient subspace are intrinsic. Recovering them identifies every core change and shear. The complete module endomorphisms, semilinear lifts and residual action on the whole cohomology fiber are given in Appendix I. Their orbit weights produce the finite matrix-rank census, rather than a count of labelled extension parameters.

Finally, extra actions consume ordinary two-cohomology. The rank-sensitive bound of Lemma G.2 compares this loss with their remaining freedom. The head normalization makes that comparison uniform over all module defects in the macroscopic window below. These three steps connect sharp dimension constraints to the entire unlabelled type population.

5 Fixed-surplus populations

At the sharp minimum the metabelian types have linear growth in the polar class. The first two added binary dimensions already produce a doubly exponential population. This section identifies the mechanism and gives the fixed-surplus law and exact principal orbits.

For orientation, put $n = 2^{c-1}$, and let r be the surplus in k-dimension, corresponding to $2r$ binary dimensions at $q = 4$. The three population scales are

$r \geq 1$ fixed :	fixed-surplus law (Theorem 5.3);
$r/\sqrt{n} \to \alpha \in (0, \infty)$:	growth and allocation (Theorem 6.5);
$r/\sqrt{n} \to \infty$:	minimum-derived density tends to zero (Theorem 6.1).

The critical staircase then resolves the transition on the integer defect scale in Section 6.4.

Counting convention. Throughout this section, $d = c - 1$ is the class-dependent core parameter. It is distinct from the point rank $d(L)$ of an individual solution: in a surplus branch the quotient can have dimension $d + s$, where s is the quotient surplus. The corresponding proof appendices use the same local convention. At $q = 4$, ordinary $\mathbb{F}_4$-Lie isomorphism counts agree with binary-linear conjugacy counts through the metabelian correspondence. We specify the field whenever the count is stated over a general $\mathbb{F}_q$.

5.1 Two more binary dimensions

The linear class count at $\mu(c)$ suggests a stability question: does a small dimension surplus preserve the rigidity of the minimum? The next theorem gives a sharp answer. Already the first admissible increase in dimension supports a much larger family, while keeping the same derived module as a metabelian extremizer.

Theorem 5.1 (A sharp jump above the minimum)**.** *For each $c \geq 5$, there is a family of indecomposable, noncompatible metabelian quadratic solutions of polar class c and binary dimension*

$$\mu(c) + 2 = 2^c + 2c - 4.$$

Put $d = c-1$ and $D_{\mathrm{par}} = 2^d - d - 3$. If T_c counts the binary-linear conjugacy classes represented by this family, and A_c counts all metabelian classes of polar class c and binary dimension $\mu(c) + 2$, then

$$\frac{4^{D_{\mathrm{par}}} - 1}{3\, 4^d |\operatorname{GL}_d(4)|} \leq T_c \leq 4^{D_{\mathrm{par}}} - 1, \qquad T_c \leq A_c \leq E_c\, 4^{2^d - 2} + 4^{(3d^2 - d)/2}. \tag{48}$$

In particular

$$\log_4 A_c = 2^{c-1} + O(c^2) \qquad (c \longrightarrow \infty).$$

Thus doubly exponential class growth occurs at the first admissible dimension above the metabelian minimum.

Proof. We construct the family, identify an intrinsic line in its quotient, and use that line to bound how many parameters can become conjugate. The key point is parameter recovery under arbitrary binary conjugacy.

Construction. Use

$$R = k[t_0, \dots, t_{d-1}]/(t_i^2), \qquad K = (t_1, \dots, t_{d-1}), \qquad k = \mathbb{F}_4.$$

Write $z = t_1 \cdots t_{d-1}$, $\tau = t_0 z$, and $I = (z) = kz \oplus k\tau = \operatorname{Ann}_R K$. Let $\mathcal{P}_d$ be the space of sums of square-free monomials of degree at least two, excluding z and τ. It is a fixed representative space for $\operatorname{rad}(R/I)^2$, has dimension $D_{\mathrm{par}} = 2^d - d - 3$, and is contained in K.

For $p \in \mathcal{P}_d$, take $U = \operatorname{span}_k\{e_0, \dots, e_{d-1}, v\}$, with actions on K and cocycle

$$\begin{aligned} P(e_i) &= t_i \cdot, & P(v) &= p\cdot, \\ \beta(e_0, e_i) &= t_i \quad (i \geq 1), & \beta(e_0, v) &= p, \end{aligned}$$

all other quotient-basis pair values being zero. The actions commute and square to zero. The only additional triangle equation is $t_i p + p t_i = 0$, and the t_i, $i \geq 1$, generate K. These are therefore data for Theorem 2.6. Explicitly, put $b_p(X, V) = \sum_{i=1}^{d-1} X_i t_i + Vp$. The map is

$$G_p(a, X, V, w) = \big(\omega a, \omega X, \omega V, \omega w + (at_0 + b_p)w + ab_p + a^2 t_0 b_p\big), \quad w \in K. \tag{49}$$

Indeed $Q = ab_p$ and $PQ = a^2 t_0 b_p$, since $b_p^2 = 0$. All displayed terms have binary degree at most two. The quotient has k-dimension $d+1$, and $\dim_k K = 2^d - 2$, giving the stated binary dimension. As $p \in \mathfrak{m}^2$, the radical action ideal is still $\mathfrak{m} = (t_0, \dots, t_{d-1})$, and

$$\mathfrak{m}^{d-1} K = k\tau \neq 0, \qquad \mathfrak{m}^d K = 0.$$

Thus every model has polar class $d + 1 = c$.

Parameter recovery. The unital algebra generated by the actions on the derived ideal is $\mathcal{A} = R/I$, acting faithfully on K. Its radical $\mathfrak{r}$ is intrinsic. The map

$$U \longrightarrow \mathfrak{r}/\mathfrak{r}^2, \qquad u \longmapsto P(u) \pmod{\mathfrak{r}^2},$$

has kernel exactly kv. A binary conjugacy between G_p and G_q induces k-linear maps on U and K, since their intrinsic scalar actions are multiplication by ω. It must therefore induce a quotient map of the form

$$Ae_i = \sum_j B_{ji} e_j + \alpha_i v, \qquad Av = \delta v,$$

where $B \in \mathrm{GL}_d(k)$, $\alpha \in k^d$, and $\delta \in k^\times$. There are $3\,4^d |\,\mathrm{GL}_d(4)|$ possible such maps.

Let C be the map on the derived ideal. Mixed polar brackets give $CP_p(u)C^{-1} = P_q(Au)$. Hence conjugation by C induces an algebra isomorphism $\Phi : \mathcal{A} \to \mathcal{A}$ satisfying

$$\Phi(t_i) = \ell_i + \alpha_i q, \qquad \ell_i = \sum_j B_{ji} t_j, \qquad \Phi(p) = \delta q.$$

In this algebra every radical element squares to zero. Substituting the fixed polynomial representative of p therefore gives the exact identity

$$\delta q = p(\ell + \alpha q) = p(\ell) + qh, \qquad h = \sum_i \alpha_i (\partial_i p)(\ell) \in \mathfrak{r}.$$

The derivatives are taken on that representative in R, before evaluation in R/I; no derivation on the quotient is assumed. Since p has no linear part, h has zero constant term. Consequently $\delta + h$ is a unit, and

$$q = (\delta + h)^{-1} p(\ell) = (\delta^{-1} + \delta^{-2} h) p(\ell) \quad \text{in } R/I. \tag{50}$$

For a fixed quotient map A, this determines the canonical representative q uniquely. Thus a conjugacy class meets $\mathcal{P}_d$ in at most $3\,4^d|\,\mathrm{GL}_d(4)|$ parameters. This necessary condition suffices for the counting bound; it does not require deciding which quotient maps lift to full conjugacies.

Indecomposability and counting. Restrict to $p \neq 0$. The action map $U \to \mathcal{A}$ is then injective: the t_i have independent linear parts and p is a nonzero element of $\mathfrak{r}^2$. A central vector of the original polar algebra must consequently lie in K. There its centralizer condition is $t_i w = 0$ for all i, so its center is exactly $k\tau$, of binary dimension two. In a nontrivial direct sum of solutions, each nilpotent polar summand has a nonzero center of even binary dimension (Proposition 2.3). Such a direct sum would have center dimension at least four. Hence every G_p, $p \neq 0$, is indecomposable. None is compatible, because $\mu(c) + 2 < \nu(c)$.

There are $4^{D_{\mathrm{par}}} - 1$ nonzero parameters, and the fiber bound proves the bounds on T_c in (48).

An upper bound for the entire dimension layer. The separate dimension bounds from Theorem 3.4 leave only two possibilities for the k-dimensions of quotient and derived ideal:

$$(d+1, 2^d - 2) \quad \text{or} \quad (d, 2^d - 1).$$

In the first case, Lemma 3.2 supplies a same-class subalgebra with d quotient generators. Its derived ideal already has dimension at least $2^d - 2$, so equals that of the full algebra. It is therefore one of the E_c metabelian extremizers. After choosing its representative, the extra quotient action belongs to $\mathrm{End}_R K = R/I$. Its constant term is zero, and its linear part can be removed by adding core quotient directions. At most $4^{D_{\mathrm{par}}}$ actions remain. For a fixed action p, the mixed cocycle values $a_i = \beta(e_i, v)$ satisfy

$$t_i a_j + t_j a_i = p\beta(e_i, e_j).$$

Their differences are ordinary one-cocycles. Changing the lift of v by $w \in K$ adds the one-coboundary $(t_i w)_i$. Lemma B.3 gives $\dim_k H^1(U_0, K) = d + 1$, where U_0 is the core quotient. This case therefore contributes at most $E_c 4^{D_{\mathrm{par}}+d+1} = E_c 4^{2^d-2}$ classes.

In the second case the derived module is a quotient of W_d retaining the top. Lemma B.2 gives at most $4^{\binom{d}{2}}$ possible module types and $\dim_k H^2(U, W) \leq d^2$. For each fixed module, changes of section identify cocycles differing by a two-coboundary. Thus at most $4^{\binom{d}{2}+d^2} = 4^{(3d^2-d)/2}$ classes occur in this case. These are upper bounds: no uniqueness of the chosen core or the codimension-one module presentation is required.

Combining the two cases proves the bound on A_c. Since $E_c = O(c)$ and $\log_4 |\,\mathrm{GL}_d(4)| = d^2 + O(1)$, the upper and lower bounds give the asserted logarithmic asymptotic. Finally all solution dimensions are even by Proposition 2.1, so two is the first possible positive surplus above $\mu(c)$. □

The two upper bounds locate the source of the jump. Adding one $\mathbb{F}_4$-dimension to the derived ideal allows at most $4^{(3d^2-d)/2}$ classes. Keeping that ideal at its minimum and adding one quotient direction already produces the doubly exponential contribution. The location of the extra dimension therefore matters as well as its size.

The contrast is already quantitative at class seven. The minimum dimension $\mu(7) = 136$ has exactly $E_7 = 18$ classes, whereas (48) gives at least 32 485 958 indecomposable noncompatible classes in dimension 138. This lower bound comes from the uniform recovery argument, rather than an enumeration of the models.

Lie-algebra interpretation. The associated auxiliary $\mathbb{F}_4$-Lie algebras have the same one-dimensional center $k\tau$, so they too are indecomposable. They admit no restricted square.

Indeed, $(\operatorname{ad} e_0)^2$ is zero on K. If it were $\operatorname{ad} y$, injectivity of the quotient action would force $y \in K$. But

$$(\operatorname{ad} e_0)^2 e_1 = t_0 t_1, \qquad [y, e_1] = t_1 y,$$

and the latter has zero t_0t_1-coefficient because $y \in K$ has no t_0-term. Thus $(\operatorname{ad} e_0)^2$ is not inner, contradicting the restricted-square adjoint axiom. The correspondence therefore also gives the stated number of distinct indecomposable, nonrestrictable Lie algebras of class c and $\mathbb{F}_4$-dimension $2^{c-1} + c - 2$.

The construction uses the classical abelian-extension framework [3]. Its additional feature is that an intrinsic quotient line and (50) control all binary conjugacies despite the large group acting on the derived module. The upper bound also controls the full first-surplus layer. Explicit orbit representatives and an exact finite count for every c remain separate classification questions.

5.2 A law for every fixed surplus

The first-surplus transition belongs to a uniform counting law. It is convenient to formulate it for ordinary Lie algebras over a finite field $k = \mathbb{F}_q$ of characteristic two, with $q \geq 4$. Dimensions in this subsection are over k, unless explicitly called binary. A Lie algebra is *3-Engel* if $(\operatorname{ad} x)^3 = 0$ for every x. The following classical linearization identifies the relevant metabelian category; it is the metabelian consequence of [20, Section 1.2, identity (6)]. We include the short argument to specify the field-size hypothesis.

Lemma 5.2. *Let L be metabelian over a characteristic-two field with at least four elements, and put $W = L'$. Then L is 3-Engel if and only if $(\operatorname{ad} x)^2|_W = 0$ for every $x \in L$.*

Proof. The restricted operators $P_x = \operatorname{ad}(x)|_W$ commute, since $[L', L'] = 0$. If their squares vanish, then $(\operatorname{ad} x)^3 z = P_x^2[x, z] = 0$. Conversely, apply the 3-Engel identity to $x + ty$ and z. Commutativity and characteristic two give

$$0 = P_x^2[x, z] + tP_x^2[y, z] + t^2 P_y^2[x, z] + t^3 P_y^2[y, z].$$

This polynomial has degree at most three, below the field size. Its coefficient of t vanishes, so $P_x^2[y, z] = 0$. Brackets span W, proving the assertion. □

Write $\ell(c)$ for the minimum dimension of a metabelian 3-Engel algebra of class $c \geq 5$. Lemma C.1 gives $\ell(c) = 2^{c-1} + c - 3$. Let $B_{c,r}(q)$ count ordinary k-Lie isomorphism classes of class c and dimension $\ell(c) + r$ in this category.

Theorem 5.3 (The fixed-surplus law)**.** *Put $d = c - 1 \geq 4$ and $D_{\mathrm{par}} = 2^d - d - 3$. For $1 \leq r \leq D_{\mathrm{par}}$,*

$$B_{c,r}(q) \;\geq\; \frac{\displaystyle\prod_{j=0}^{r-1} (q^{D_{\mathrm{par}}} - q^j)}{q^{dr} |\operatorname{GL}_d(q)| \, |\operatorname{GL}_r(q)|}. \tag{51}$$

The lower bound is realized by indecomposable algebras with one-dimensional center and no restricted square. For each fixed q and $r \geq 1$,

$$\log_q B_{c,r}(q) = r\, 2^{c-1} + O_{q,r}(c^2) \qquad (c \longrightarrow \infty). \tag{52}$$

Proof. Put

$$R = k[t_0, \ldots, t_{d-1}]/(t_i^2), \quad K = (t_1, \ldots, t_{d-1}), \quad I = (t_1 \cdots t_{d-1}) = \operatorname{Ann}_R K.$$

Put $\mathfrak{m} = (t_0, \dots, t_{d-1})$, so $\dim K = 2^d - 2$. Take canonical representatives $\mathcal{P}_d$ of $\operatorname{rad}(R/I)^2$, excluding the two monomials in I; their dimension is D_{par}. Choose an ordered independent frame $\mathbf{p} = (p_1, \dots, p_r)$ in $\mathcal{P}_d$. On $U \oplus K$, where $U = \langle e_0, \dots, e_{d-1}, v_1, \dots, v_r \rangle_k$, define commuting actions and an alternating cocycle by

$$P(e_i) = t_i \cdot, \quad P(v_a) = p_a \cdot, \qquad \beta(e_0, e_i) = t_i \ (i \geq 1), \quad \beta(e_0, v_a) = p_a,$$

with all other basis pair values zero. In particular, the Jacobi equation on e_0, v_a, v_b is $p_a p_b + p_b p_a = 0$. The cocycle values generate K, so the derived ideal is exactly K. All radical elements of R square to zero; hence the algebra is 3-Engel by Lemma 5.2. Its lower central ideals are $\gamma_j = \mathfrak{m}^{j-2} K$ for $j \geq 2$, with terminal term $k(t_0 \cdots t_{d-1})$. Its class is $d+1$, and its dimension is $d + r + 2^d - 2 = \ell(c) + r$.

The faithful generated action algebra is $\mathcal{A} = R/I$. The kernel of the intrinsic map $U \to \operatorname{rad}(\mathcal{A})/\operatorname{rad}(\mathcal{A})^2$ is $V_r = \langle v_1, \dots, v_r \rangle_k$. Any isomorphism between two models therefore induces

$$Ae_i = \sum_j B_{ji} e_j + \sum_b \alpha_{bi} v_b, \qquad Av_a = \sum_b \Delta_{ba} v_b,$$

where $B \in \mathrm{GL}_d(k)$, $\Delta \in \mathrm{GL}_r(k)$. There are $q^{dr} |\mathrm{GL}_d(q)||\mathrm{GL}_r(q)|$ possible quotient maps. For target frame $\mathbf{p}'$, conjugation on the derived ideal induces an algebra isomorphism satisfying

$$\Delta^T \mathbf{p}' = \mathbf{p}(\ell + \alpha^T \mathbf{p}'), \qquad \ell_i = \sum_j B_{ji} t_j. \tag{53}$$

All polynomials here are evaluated on their fixed representatives.

For a fixed quotient map this equation has at most one target frame. Indeed, compare two solutions and let e be their difference. Every source polynomial has degree at least two, and all its arguments lie in $\operatorname{rad}(\mathcal{A})$. Exact telescoping of each monomial therefore gives $\Delta^T e = Me$, with every entry of M in that radical. Thus $e = Ne$, where $N = (\Delta^T)^{-1} M$. Since $\operatorname{rad}(\mathcal{A})^d = 0$, also $N^d = 0$, and iteration gives $e = 0$. The exact telescoping identity retains all products involving distinct target parameters. This uniqueness is an instance of the nilpotent implicit-function principle of Bahturin–Olshanskii [1, Sections 1.2–1.3]; the intrinsic quotient subspace and the equation forced by isomorphism specify its application here. Each isomorphism class consequently contains at most $q^{dr} |\mathrm{GL}_d(q)||\mathrm{GL}_r(q)|$ ordered frames from this family, proving (51).

Frame independence makes $U \to \mathcal{A}$ injective. The center is therefore contained in K, where it is the one-dimensional socle $k(t_0 \cdots t_{d-1})$. This implies indecomposability. If $(\operatorname{ad} e_0)^2 = \operatorname{ad} y$, its zero action on K forces $y \in K$. But $(\operatorname{ad} e_0)^2 e_i = t_0 t_i$, whereas $t_i y$ has zero $t_0 t_i$-coefficient. Hence no restricted square exists.

For fixed q, r, the numerator of (51) has logarithm $rD_{\mathrm{par}} + O_{q,r}(1)$, while the denominator has logarithm $d^2 + dr + O_{q,r}(1)$. This proves the required lower estimate. Proposition C.3 gives the matching upper estimate, including all possible derived-module defects, and proves (52). □

Corollary 5.4 (Generic structure)**.** *With isomorphism classes counted uniformly, the proportion in $B_{c,r}(q)$ failing any of the following properties is at most $q^{-2^d + O_{q,r}(d^2)}$: the derived ideal has minimum dimension $2^d - 2$ and is the standard core module K; the action $L/L' \to \mathrm{End}_k(L')$ is injective; the center is one-dimensional; the algebra is indecomposable and has no restricted square.*

Proof. In the upper-bound proof, write the quotient and derived dimensions as $d + s$ and $2^d - 2 + h$, where $s + h = r$. The branches $h \geq 1$ together contribute at most $q^{(r-1)2^d + O_{q,r}(d^2)}$ classes. When $h = 0$, the derived module is K. Removing the linear parts of the extra actions

gives an r-tuple in $\mathcal{P}_d$. Rank-deficient tuples number at most $q^{(r-1)D_{\mathrm{par}}+r(r-1)}$: choose $r-1$ spanning vectors and their coefficient matrix. The remaining cocycle choices cost only $q^{O_{q,r}(d^2)}$. Thus these exceptions have the same bound. Injectivity of the full quotient action is intrinsic, so different choices of core do not affect this division into cases.

For a full-rank tuple, the center and indecomposability follow as above. In a normalized extremal core, (94) gives $\beta(e_0, e_i) \equiv t_i \pmod{\mathfrak{m}K}$. Hence $(\operatorname{ad} e_0)^2 e_i$ still has coefficient one at $t_0 t_i$. The same inner-derivation obstruction excludes a restricted square, independently of the mixed cocycle choices. Dividing the exceptional bound by (51) proves the asserted proportion. □

Corollary 5.5 (Binary fixed-surplus law)**.** *Let $A_{c,r}$ count all metabelian quadratic solutions of polar class c and binary dimension $\mu(c)+2r$, up to binary-linear conjugacy. Then $A_{c,r} = B_{c,r}(4)$, and, for each fixed $r \geq 1$,*

$$\log_4 A_{c,r} = r\, 2^{c-1} + O_r(c^2).$$

Almost all have binary polar center of dimension two and are indecomposable; their auxiliary $\mathbb{F}_4$-Lie algebras admit no restricted square.

Proof. Use Theorem 2.6, Lemma 5.2, and $\mu(c) = 2\ell(c)$. For the generic objects the faithful quotient action forces the original polar center into the derived ideal, where it is the same one-dimensional $\mathbb{F}_4$-socle. Its binary dimension is two, excluding a nontrivial direct sum by Proposition 2.3. The lower-bound family is noncompatible for every r: the compatibility criterion would require $t_i b_0 = t_0 t_i$ with $b_0 \in K$, whose $t_0 t_i$-coefficients are respectively zero and one. □

Corollary 5.6 (Compatibility at fixed surplus)**.** *For every fixed $r \geq 2$, the proportion of compatible objects among the $A_{c,r}$ classes is at most $4^{-2^d+O_r(d^2)}$. At $r=2$, their dimension is precisely $\nu(c)$, and the compatible classes are precisely the extremizers counted by N_c. With $D_{\mathrm{par}} = 2^d - d - 3$,*

$$\frac{N_c}{A_{c,2}} \leq \frac{60(1+\eta_d)\, 4^{-2^d+3d+4}}{(1-4^{-D_{\mathrm{par}}})(1-4^{1-D_{\mathrm{par}}})} = 4^{-2^d+O(d)}. \tag{54}$$

Here η_d is the explicit bound from Corollary 3.23.

Proof. Lemma 3.12 gives $\dim_{\mathbb{F}_4} L' \geq 2^d - 1$, so compatible objects lie in the $h \geq 1$ upper-bound branches. This proves the first assertion. For the sharper $r=2$ estimate, combine

$$N_c \leq \frac{4^{D_{\mathrm{par}}+1}}{3|\operatorname{GL}_d(4)|}(1+\eta_d)$$

with (51). The $\operatorname{GL}_d(4)$ factors cancel, and $|\operatorname{GL}_2(4)| = 180$ gives (54). □

5.3 Exact principal orbits at every fixed surplus

The minimum-derived stratum has a complete orbit model for any number of additional quotient directions. Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$, and put

$$d = c - 1 \geq 4, \quad C_d = \binom{d}{2}, \quad D_{\mathrm{par}} = 2^d - d - 3, \quad w_0 = 2^d - 2.$$

Use R, K, I from (93), and let $\mathcal{P}_d$ be the canonical monomial representatives of $\operatorname{rad}(R/I)^2$. On $U = \langle e_0, \dots, e_{d-1}, v_1, \dots, v_r \rangle_k$, let $P_{\mathbf{p}}(e_i) = t_i \cdot$ and $P_{\mathbf{p}}(v_a) = p_a \cdot$, for $\mathbf{p} \in \mathcal{P}_d^r$. Let η denote the degree-one star coefficient, and set

$$\mathcal{Y}_{d,r} = \coprod_{\mathbf{p} \in \mathcal{P}_d^r} \{[\beta] \in H^2(U, K; P_{\mathbf{p}}) : \eta = 1\}.$$

Each fiber has dimension

$$f_{d,r} = (r+1)d + C_d + r + \binom{r}{2} - 1.$$

Write $\mathcal{Y}_{d,r}^{\mathrm{fr}}$ for its restriction to linearly independent tuples $p_1, \dots, p_r$.

Let P_H be the parabolic preserving $H = \langle e_1, \dots, e_{d-1} \rangle_k$. The effective quotient group is

$$G_{d,r} = \left\{ A_g = \begin{pmatrix} B & 0 \\ \alpha & \Delta \end{pmatrix} : B = \begin{pmatrix} a & 0 \\ b & M \end{pmatrix} \in P_H,\ \alpha \in k^{r \times d},\ \Delta \in \mathrm{GL}_r(k) \right\}, \tag{55}$$

with

$$|G_{d,r}| = (q-1) q^{rd+d-1} |\operatorname{GL}_{d-1}(q)| |\operatorname{GL}_r(q)|.$$

For $\ell_i = \sum_j B_{ji} t_j$, its action is

$$\Delta^T \mathbf{p}' = \mathbf{p}(\ell + \alpha^T \mathbf{p}') \qquad \text{in } R/I, \tag{56}$$

$$[\beta'] = [a \sigma_g \beta(A_g^{-1} \cdot, A_g^{-1} \cdot)], \qquad \sigma_g(t_i) = \ell_i + \sum_a \alpha_{ai} p'_a. \tag{57}$$

The implicit system has a unique solution in $\mathcal{P}_d^r$.

Theorem 5.7 (Complete minimum-derived orbits)**.** *The isomorphism classes of metabelian 3-Engel k-Lie algebras of class c, dimension $\ell(c) + r$, and minimum derived dimension w_0, are exactly $\mathcal{Y}_{d,r}/G_{d,r}$. The faithful-action subfamily is exactly $\mathcal{Y}_{d,r}^{\mathrm{fr}}/G_{d,r}$. Equations* (56)–(57) *are sufficient as well as necessary for equivalence.*

Theorem 5.8 (Multiplicative enumeration at fixed surplus)**.** *For every fixed $q \geq 4$ of characteristic two and fixed $r \geq 1$,*

$$B_{c,r}(q) \sim \frac{\prod_{j=0}^{r-1} (q^{D_{\mathrm{par}}} - q^j)\, q^{C_d + r + \binom{r}{2}}}{(q-1) |\operatorname{GL}_{d-1}(q)| |\operatorname{GL}_r(q)|} \qquad (c \longrightarrow \infty). \tag{58}$$

Almost all isomorphism classes have full automorphism-group order

$$|\operatorname{Aut}_k(L)| = q^{2w_0 + d + r - 1} = q^{2^c + c + r - 6}. \tag{59}$$

For $q = 4$, these are respectively the binary conjugacy count $A_{c,r}$ and its generic binary-linear centralizer order.

The proof retains all cocycle coefficients, including brackets between different added directions. Nontrivial linear parts are controlled by an exterior-rank estimate. For a pure matrix shear, specializing its active variables to zero gives a necessary affine multiplication constraint. This retains the higher cross-products in the implicit substitution and controls the summed Burnside error for every fixed r. Appendix D proves both statements. At $r = 1$, (58) reduces exactly to

$$B_{c,1}(q) \sim \frac{(q^{D_{\mathrm{par}}} - 1) q^{C_d + 1}}{(q-1)^2 |\operatorname{GL}_{d-1}(q)|} \sim \frac{q^{2^d - C_d - 5}}{(1 - q^{-1})^2 \prod_{j=1}^{\infty} (1 - q^{-j})}.$$

6 The critical transition

The fixed-surplus law persists while the surplus is small relative to the square root of the core dimension. At that scale the whole population changes its typical derived dimension. We first locate the transition, then determine its constant, allocation profile, and discrete staircase. Let $B^{\min}_{c,r}(q)$ count the classes in $B_{c,r}(q)$ whose derived ideal has the minimum possible dimension $2^{c-1}-2$, and put

$$\pi_{d,r}(q) = \frac{B^{\min}_{d+1,r}(q)}{B_{d+1,r}(q)}, \qquad d = c-1.$$

Both the numerator and denominator count unlabelled, k-linear Lie isomorphism classes. In particular this is not a parameter-weighted proportion.

Theorem 6.1 (A transition scale for minimum-derived objects)**.** *Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$, and let $d \to \infty$. Uniformly over positive integers r,*

$$\log_q B_{d+1,r}(q) = \Theta\big(r2^d + r^3\big). \tag{60}$$

For integer sequences $r = r(d) \geq 1$,

$$r = o(2^{d/2}) \quad \Longrightarrow \quad \log_q B_{d+1,r}(q) \sim r2^d, \qquad \pi_{d,r}(q) \longrightarrow 1, \tag{61}$$

$$r/2^{d/2} \longrightarrow \infty \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 0. \tag{62}$$

In the second regime, $\log_q \pi_{d,r}(q) \leq -(2/27 - o(1))r^3$.

The small-surplus assertion follows by retaining the derived defect h in Proposition C.3; its loss of $h2^d$ dominates the remaining terms. For the opposite regime, two-step alternating-tensor families supply a cubic population. Their contribution can be counted through the characteristic quotient $L/\gamma_3(L)$, whose dimension discards the large minimum core. Appendix C.1 gives the precise bounds and proof.

Thus $2^{(c-1)/2}$ separates the two off-window behaviors of the minimum-derived population. Theorem 6.2 below sharpens this scale to an explicit critical constant. The finer allocation and finite-partition theorems then resolve the competing defects and actual integer layers at the transition. The large-surplus lower family uses direct sums, but the conclusion concerns only the derived dimension; it makes no assertion that a typical object in that regime is decomposable. At $q = 4$, the same conclusions apply to binary conjugacy classes in dimension $\mu(c) + 2r$, with minimum binary derived dimension $2^c - 4$, by Corollary 5.5.

6.1 The critical constant for minimum derived dimension

The transition scale admits a sharp constant away from its boundary. Two effects determine it. Normalizing the subspace spanned by the extra actions removes a quadratic coordinate redundancy from the minimum-derived count. A single additional central derived direction, coupled to those same quotient directions, supplies enough alternating forms to outweigh that count on the other side. The latter family consists of indecomposable algebras.

Theorem 6.2 (The critical constant)**.** *Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$. For every integer sequence $r = r(d) \geq 1$, as $d \to \infty$,*

$$\limsup_{d\to\infty} \frac{r}{2^{d/2}} < \sqrt{2} \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 1, \tag{63}$$

$$\liminf_{d\to\infty} \frac{r}{2^{d/2}} > \sqrt{2} \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 0. \tag{64}$$

In particular, if $r/2^{d/2} \to \alpha \neq \sqrt{2}$ with $0 \leq \alpha \leq \infty$, the limit is one for $\alpha < \sqrt{2}$ and zero for $\alpha > \sqrt{2}$. These are proportions of unlabelled k-linear isomorphism types.

Proof. Write $n = 2^d$, $D_{\mathrm{par}} = n - d - 3$, and use B_h for the branch with derived dimension $n - 2 + h$, so $\pi_{d,r}(q) = B_0/\sum_h B_h$. For the first implication there is an $\epsilon > 0$ such that eventually $r^2/2 \le (1-\epsilon)n$. In particular $r \le D_{\mathrm{par}}/2$, and

$$Z(d,r) = 2d^2 + 3dr + 2r + 6 = o(n).$$

Proposition E.3 gives, with $a = n - r^2/2 - Z(d,r) \ge (\epsilon - o(1))n$,

$$1 - \pi_{d,r}(q) \le \frac{q^{-a}}{1-q^{-a}} \longrightarrow 0.$$

For the second implication, first consider the indices with $r \le D_{\mathrm{par}}/2$. Here $r \ge 3$ eventually and Proposition E.4 yields

$$\pi_{d,r}(q) \le \frac{B_0}{B_1} \le q^{\,n - r^2/2 + dr + d^2 + r/2 + \log_q(\frac{3}{2}(r+1))}.$$

The strict lower limit assumption gives an $\epsilon > 0$ with $n - r^2/2 \le -\epsilon r^2$ eventually. Since $r \gg d$, every remaining exponent term is $o(r^2)$; the last display tends to zero along these indices. On the remaining indices, $r > D_{\mathrm{par}}/2$, one has

$$\frac{r}{\sqrt{n}} > \frac{n-d-3}{2\sqrt{n}} \longrightarrow \infty.$$

The all-surplus upper-side assertion (62) of Theorem 6.1 therefore applies along this subsequence and also gives $\pi_{d,r}(q) \to 0$. Both subsequences have the same limit. This argument uses no monotonicity of $\pi_{d,r}(q)$ as the dimension changes. □

For $q = 4$, Theorem 2.6 transfers the same critical constant to binary-linear conjugacy classes of metabelian quadratic solutions in binary dimension $\mu(c) + 2r$, where $d = c - 1$. The counted minimum derived dimension is $2^c - 4$. The coupled lower family gives indecomposable quadratic solutions by Remark E.5. Theorems 6.5 and 6.9 refine this constant-scale statement to the macroscopic allocation and the discrete critical profile. Theorem 6.6 resolves the multiplicative partition and adjacent weights in its stated growing window.

The following initial localization is sharpened in Corollary 6.10.

Corollary 6.3 (An initial outer localization of the transition)**.** *Fix a finite characteristic-two field of order $q \ge 4$, put $n = 2^d$, and let $r = r(d) \ge 1$ be integers. Then, as $d \to \infty$,*

$$\begin{aligned} r \le \sqrt{2n} - 3d - 3 \quad &\textit{eventually} \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 1, \\ r \ge \sqrt{2n} - d + 1 \quad &\textit{eventually} \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 0. \end{aligned}$$

The two thresholds are separated by $2d + 4$, giving an outer localization of order $d = c - 1$ around $\sqrt{2n}$. Its proof supplies the outer splice used for the integer selection in Appendix M.

Proof. Set $t = \sqrt{2n}$. The exponent

$$a(d,r) = n - \tfrac{1}{2}r^2 - 2d^2 - 3dr - 2r - 6$$

in Proposition E.3 is decreasing in real $r \ge 0$. Its value at the lower endpoint is

$$a(d, t - 3d - 3) = t + \tfrac{5}{2}d^2 + 6d - \tfrac{9}{2} =: A_d.$$

For all sufficiently large d, the assumed lower-side range is contained in $1 \le r \le D_{\mathrm{par}}/2$. Hence

$$1 - \pi_{d,r}(q) \le \frac{q^{-A_d}}{1-q^{-A_d}} \longrightarrow 0.$$

For the upper side, Corollary E.7 proves the stronger endpoint by combining a uniform count of the minimum-derived stratum with the full-central-cocycle lower family. That comparison is independent of the present lower-side argument. No monotonicity of $\pi_{d,r}(q)$ is assumed. □

6.2 Uniform enumeration through the transition scale

Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$. The minimum-derived stratum admits a uniform multiplicative count even when the surplus grows. Put $d = c - 1$, $n = 2^d$, $D_{\mathrm{par}} = n - d - 3$, $C_d = \binom{d}{2}$, and $w_0 = n - 2$. Write $B_0 = B^{\min}_{d+1,r}(q)$. The range in the following theorem is

$$2d \leq r \leq \frac{D_{\mathrm{par}}}{4d}.$$

It includes every sequence $r/\sqrt{n} \to \alpha$ with $0 < \alpha < \infty$.

Theorem 6.4 (Multiplicative principal enumeration at growing surplus)**.** *Uniformly for the integer range* (121)*, as $d \to \infty$,*

$$B_0 \sim \frac{\prod_{i=0}^{r-1}(q^{D_{\mathrm{par}}} - q^i)\, q^{C_d + r + \binom{r}{2}}}{(q-1)|\,\mathrm{GL}_{d-1}(q)||\,\mathrm{GL}_r(q)|}. \tag{65}$$

Almost all classes in this stratum have full automorphism order $q^{2w_0+d+r-1}$. These statements concern the whole minimum-derived stratum, not the complete dimension layer above the critical constant.

The proof reparametrizes a fixed pure shear by its induced ring automorphism σ and the action subspace Q:

$$\mathrm{span}\{\sigma(t_i) - t_i\} \subseteq Q \subseteq \mathrm{Fix}(\sigma).$$

A displacement-rank inequality and Gaussian incidence counts then control the complete stabilizer sum. This avoids a loss depending on the number of added directions in the fixed-surplus estimate. Appendix E.3 gives the proof and the sharper upper-side localization used in Corollary 6.3. The full dimension layer has the same multiplicative asymptotic on the subrange $2d \leq r \leq \sqrt{2n} - 3d - 3$, by the independent minimum-derived dominance bound. Above the critical constant, (65) counts only the minimum-derived stratum.

6.3 Exact growth and the typical surplus allocation

At a fixed ratio of surplus to the square-root core scale, an explicit profile determines both the growth of the entire population and its typical allocation. Continue to put $n = 2^d$, $d = c - 1$, and count ordinary k-linear Lie isomorphism types uniformly. For $\alpha > 0$, define

$$\gamma_\alpha = \begin{cases} \alpha, & 0 < \alpha \leq \sqrt{2}, \\ (\alpha + \sqrt{\alpha^2 + 6})/3, & \alpha > \sqrt{2}, \end{cases} \qquad \beta_\alpha = \alpha - \gamma_\alpha, \qquad \varphi(\alpha) = \gamma_\alpha + \frac{\beta_\alpha \gamma_\alpha^2}{2}. \tag{66}$$

Theorem 6.5 (Exact logarithmic growth and allocation)**.** *Fix a finite field $k = \mathbb{F}_q$ of characteristic two, $q \geq 4$, and suppose $r/\sqrt{n} \to \alpha \in (0, \infty)$. Then*

$$\log_q B_{d+1,r}(q) = \big(\varphi(\alpha) + o(1)\big) n^{3/2}. \tag{67}$$

For every $\varepsilon > 0$, there is an $\eta > 0$ such that, for all sufficiently large d, all but a proportion $q^{-\eta n^{3/2}}$ of these types satisfy

$$\left| \frac{h}{\sqrt{n}} - \beta_\alpha \right| < \varepsilon, \qquad h = \dim_k L' - (n-2). \tag{68}$$

Thus for $\alpha > \sqrt{2}$, the typical derived share is

$$\frac{h}{r} \longrightarrow \frac{2 - \sqrt{1 + 6/\alpha^2}}{3}, \qquad \frac{s}{r} = 1 - \frac{h}{r}.$$

The core contributes the term γ, while alternating central brackets contribute $\beta\gamma^2/2$. Their competition gives the profile φ and its unique maximizing allocation. The main upper-bound issue is that arbitrary nonsplit derived modules may have additional actions. Restricting those actions to the core socle separates their visible part from a controlled kernel. A common rational flag bounds the visible tuples, and ordinary cohomology charges their rank against the available cocycles. This coupled bound covers every derived module, so the result counts the whole layer. Appendix G proves the theorem using the classical triangular square-zero count and ordinary Chevalley–Eilenberg tools.

At $q = 4$, this also counts metabelian quadratic solutions in binary dimension $\mu(c)+2r$. At the critical ratio the result gives $h = o(\sqrt{n})$ in type-probability. The finer multiplicative partition, the integer staircase, and the diagonal-rank distribution are given next in Section 6.4.

6.4 Multiplicative critical counts and the joint type law

The critical population admits a finite-product partition, including both the allocation of derived surplus and the rank of its diagonal extension. Fix $k = \mathbb{F}_q$, of characteristic two, $q \geq 4$, and put

$$d = c - 1, \quad n = 2^d, \quad m = d - 1, \quad D = n - d - 3, \quad T = \sqrt{2n}, \quad H = \lfloor \sqrt{n}/d^2 \rfloor.$$

Let B_h count all metabelian three-Engel ordinary k-Lie isomorphism types of class $d + 1$, dimension $d + n - 2 + r$, and derived dimension $n - 2 + h$. Thus $B = \sum_h B_h = B_{d+1,r}(q)$. Probabilities are uniform on these unlabelled types. The scale d is determined by class and need not be the object's intrinsic generator rank. Write $C_j = \binom{j}{2}$, $P_j = \prod_{i=1}^{j}(1-q^{-i})$, $P_0 = 1$. For $s = r - h$ put

$$M_h = \frac{q^{sn-3s+(h-1)C_{d+s}+m-h(h+1)}}{(q-1)P_m P_h P_s} \prod_{j<s}(1 - q^{j-D}), \qquad Z_{d,r} = \sum_{h=0}^{H} M_h. \tag{69}$$

Only actual integer surplus layers in the window

$$|x_{d,r}| \leq H/4, \qquad x_{d,r} = r - (T - d + \tfrac{1}{2}) \tag{70}$$

are considered in the following whole-layer theorem.

Theorem 6.6 (The full critical partition and joint law)**.** *There exists $\kappa_q > 0$ such that, as $d \to \infty$, uniformly on* (70)*,*

$$B_{d+1,r}(q) = Z_{d,r}\big(1 + O_q(q^{-\kappa_q n})\big). \tag{71}$$

Except on a set of type-probability $O_q(q^{-\kappa_q n})$, the derived module is a maximum-socle diagonal extension E_Q with faithful polynomial action. Its intrinsic rank $u = \operatorname{rank} Q$ is recovered from its head dimension $m+h-u$. The joint law of (h, u), with an extra symbol for the exceptions, has total variation distance $O_q(q^{-\kappa_q n})$ from

$$\rho_{d,r}(h, u) = \frac{M_h}{Z_{d,r}} \frac{R_{h,m}(u)}{q^{hm}}, \qquad 0 \leq h \leq H, \quad 0 \leq u \leq \min(h, m), \tag{72}$$

where $R_{h,m}(u)$ counts rank-u matrices in $\operatorname{Mat}_{h\times m}(k)$.

The weights $M_h/Z_{d,r}$ are part of the joint law. They follow from complete Lie isomorphism fibers and their automorphism weights. Theorem I.4 also gives the complete branch formula uniformly for $0 \leq h \leq H$, $a\sqrt{n} \leq r \leq b\sqrt{n}$, for fixed $0 < a < b$, including the passage through $h = m$. Replacing P_m, P_s by P_∞ changes the relative error to $O_q(q^{-m})$, rather than the error in (71).

Corollary 6.7 (Adjacent-branch weights)**.** *Let i maximize M_h. Choose its larger neighbor and i (choose $\{0,1\}$ if $i=0$), forming $\{j,j+1\}$. In (70), the probability outside this pair is $O_q(q^{-T/2})$. The pair has respective probabilities $(1,R_j)/(1+R_j)$, each with relative error $O_q(q^{-T/2})$, where, for $s=r-j$,*

$$R_j=\frac{q^{-n+C_{d+s}-j(d+s)+1-j}}{1-q^{-j-1}}\frac{1-q^{-s}}{1-q^{s-1-D}}.$$

These finite-product weights do not assert that arbitrary real displacements or mixing limits are realized by integer layers.

Corollary 6.8 (Rank deficiency on true integer layers)**.** *Fix $\ell\in\mathbb{Z}$ and take*

$$r_d(\ell)=\lfloor T\rfloor+d-2+2\ell,\qquad j_d=m+\ell.$$

Then $\Pr(h=j_d)=1-O_{q,\ell}(q^{-c_qT})$. On this event put $J_d=\min(m,j_d)-u$, assigning any fixed nonnegative value on the exceptional event. Its law converges to

$$p_{|\ell|}(t)=q^{-t(|\ell|+t)}\frac{P_\infty}{P_tP_{|\ell|+t}},\qquad t\ge 0.$$

The total variation error is $O_{q,\ell}(q^{-c_qT})+O_q(q^{-\max(m,j_d)-1})$. For $\ell=0$, the typical defect is m, while $\Pr(u=m)\to P_\infty<1$.

The matrix law is classical [11]; the conclusion realizes it in the entire uniform Lie type population. Appendices I–L prove the complete fibers, uniform errors and whole-layer tail.

Corollary 6.9 (The logarithmic staircase)**.** *Put $\mathcal{S}(x)=\max_{h\ge0}\{hx-h(h-1)\}$. For fixed $A>0$, uniformly on actual integer $|x_{d,r}|\le A$,*

$$\log_q(B/B_0)=T\mathcal{S}(x_{d,r})+O_{q,A}(d).$$

If $x_{d,r}\to x$, all but proportion q^{-cT} have $h=0$ for $x<0$, $h=j$ for $2j-2<x<2j$, and $h\in\{j,j+1\}$ for $x=2j$.

The finite-product ratio supplies the multiplicative information beyond this logarithmic profile. Ridout's theorem also gives the following consequence, proved in Appendix M.

Corollary 6.10 (Integer minimum-derived selection)**.** *Let $\rho_d=\lfloor T+1/2\rfloor+1-d$. Then*

$$\sup_{r\in\mathbb{Z}_{\ge0}}\left|\pi_{d,r}(q)-\mathbf{1}_{\{r<\rho_d\}}\right|\longrightarrow 0.$$

For fixed $A,\epsilon>0$, uniformly on $|x_{d,r}|\le A$, the defect is $\max(0,\lceil x_{d,r}/2\rceil)$ except for proportion at most $q^{-T^{1-\epsilon}}$, eventually.

6.5 The complete population in a macroscopic window

The same finite weights M_h in (69) control a fixed macroscopic range above the threshold. The all-module tail here requires the common head normalization; the earlier critical window alone does not imply it. Put

$$\mathcal{I}_r=\{0,\dots,\lfloor r/5\rfloor\},\qquad Z_{d,r}^{\mathrm{mac}}=\sum_{h\in\mathcal{I}_r}M_h.$$

Theorem 6.11 (Macroscopic partition and typical structure)**.** *Fix $q \geq 4$ of characteristic two. Uniformly over actual integer layers with $3/2 \leq r/\sqrt{n} \leq 8/5$,*

$$B_{d+1,r}(q) = Z^{\mathrm{mac}}_{d,r}\big(1 + O_q(q^{-\kappa_q n})\big)$$

for some $\kappa_q > 0$. Outside a set of that type-probability, the derived module is E_Q, with faithful polynomial quotient action and $\dim(Z(L) \cap L') = h + 1$. The joint law of (h, u), assigning an exception symbol to the other types, has total variation distance $O_q(q^{-\kappa_q n})$ from

$$\frac{M_h}{Z^{\mathrm{mac}}_{d,r}} \frac{R_{h,d-1}(u)}{q^{h(d-1)}}, \qquad h \in \mathcal{I}_r.$$

Moreover $u = d - 1$ with probability $1 - O_q(q^{-c_q\sqrt{n}})$, for some $c_q > 0$.

Appendix K proves the uniform branch estimate and the whole-layer comparison. The critical theorem and this theorem have distinct stated domains; they do not assert a multiplicative formula at every intervening surplus. The logarithmic profile of Theorem 6.5 retains its wider bounded-ratio domain. The matrix full-column-rank error above is also distinct from the smaller error comparing Lie types with the exact finite partition.

7 Compatibility selects a different typical module

Compatibility has a population cost as well as an extremal dimension cost. For an ordinary algebra put $W = L'$. Call it *square-compatible* if it admits a restricted square p with $p(L) \subseteq W$ and $p(W) = 0$. This is an existence condition: the square is forgotten in counting ordinary types. At $q = 4$ it is precisely the scalar compatibility condition of Theorem 3.8 under the metabelian correspondence.

Theorem 7.1 (The first compatible socle defect)**.** *For every class-$d + 1$ square-compatible algebra, each class-preserving d-direction core satisfies $\dim \mathrm{Soc}_R W \leq h$. If equality holds, then*

$$W \simeq J \oplus k^{h-1}, \qquad J = \operatorname{rad} R.$$

If all extra actions kill this core socle, they are positive polynomial multipliers. Subtracting core linear parts puts them in $\operatorname{rad}^2(R/(\tau))$, of dimension $n - d - 2$, where $\tau = t_0 \cdots t_{d-1}$.

The square supplies the one head direction missing from the minimum core and eliminates its diagonal deformation. This is a necessary module classification, before any asymptotic counting. Its proof, including an independent core compatibility calculation preceding the polynomial rigidity argument, is in Appendix N.

Let C_h count the square-compatible ordinary types in branch h, and put $C = \sum_h C_h$. Retain the actual integer macroscopic window $3/2 \leq r/\sqrt{n} \leq 8/5$. For $N = d + s$, $s = r - h$, $z = h - 1$ and $D' = n - d - 2$, define

$$\widetilde{M}_h = \frac{\displaystyle\prod_{j<s} (q^{D'} - q^j)\, q^d (q^s - 1) q^{C_s + z(C_N - d)}}{(q-1) q^{ds+z} |\operatorname{GL}_d(k)||\operatorname{GL}_z(k)||\operatorname{GL}_s(k)|}, \qquad Z^{\mathrm{comp}}_{d,r} = \sum_{h=1}^{\lfloor r/11 \rfloor} \widetilde{M}_h. \tag{73}$$

Theorem 7.2 (Compatible partition and rarity)**.** *Uniformly in the stated integer window,*

$$C = Z_{d,r}^{\mathrm{comp}}\big(1 + O_q(q^{-\kappa_q n})\big).$$

The law of h under uniform compatible ordinary types differs in total variation by that order from $\widetilde{M}_h/Z_{d,r}^{\mathrm{comp}}$. With probability $1 - O_q(q^{-\kappa_q n})$, the derived module is $J \oplus k^{h-1}$, the quotient action is faithful polynomial, and $\dim(Z(L) \cap L') = h$. If $r/\sqrt{n} \to \alpha \in [3/2, 8/5]$, then

$$-\frac{1}{n}\log_q(C/B) \longrightarrow \frac{\gamma_\alpha^2}{2}, \qquad \gamma_\alpha = \frac{\alpha + \sqrt{\alpha^2 + 6}}{3}.$$

Compatible types retain the logarithmic profile $\varphi(\alpha)$ and scaled allocation $\beta_*(\alpha)$ of the ordinary population.

Thus a natural constraint preserves the leading allocation while selecting a different module and imposing an explicit cost at the n scale. Here C/B is a probability on ordinary types. The full compatible fibers and all conditional tails, rather than the small ordinary probability of a bad module, justify the conditional statements.

8 Marking and relaxing a square

Marking the square changes the type space. Let $T_{d,r}^0$ count restricted k-Lie isomorphism types (L, p) with $p(L) \subseteq L'$ and $p(L') = 0$, retaining the same ordinary class and dimension layer. Define $\widehat{M}_h = \widetilde{M}_h q^{h(d+r-h)-d}$.

Corollary 8.1 (The cost after marking a square)**.** *Uniformly on $3/2 \le r/\sqrt{n} \le 8/5$,*

$$T_{d,r}^0 = \left(\sum_{h=1}^{\lfloor r/11 \rfloor} \widehat{M}_h\right)\big(1 + O_q(q^{-\kappa_q n})\big).$$

Along every sequence $r/\sqrt{n} \to \alpha$ in this window, $-n^{-1}\log_q(T_{d,r}^0/B) \to 1$.

The factor $q^{h(d+s)-d}$ comes from the complete extension fiber after section changes, followed by all residual and base actions. The raw number of squares is larger. The ratio $T_{d,r}^0/B$ compares cardinalities of different type spaces; it is not an event probability on ordinary algebras.

A weaker intrinsic constraint can change the leading allocation itself. Put $S = Z(L) \cap W$, $W = L'$, and now require only $p(L) \subseteq W$ and $p(S) = 0$. Since $\mathrm{ad}(w)^2 = 0$, one has $p(W) \subseteq S$ and hence $p^3(L) = 0$. Denote the number of these restricted types by $A_{\mathrm{rel}}(d, r)$.

Theorem 8.2 (A change of allocation under a relaxed square)**.** *Fix $q \ge 4$ of characteristic two. Uniformly over actual integer layers $3/2 \le r/\sqrt{n} \le 8/5$,*

$$\log_q A_{\mathrm{rel}}(d, r) = rn + \frac{2r^3}{27} + O_q(dn).$$

Under uniform unlabelled restricted types, $h/r \to 1/3$ and $s/r \to 2/3$. For each fixed positive ϵ, the probability of a deviation of at least ϵ from either proportion is at most $q^{-\eta_\epsilon n^{3/2}}$, eventually and uniformly over the stated layers.

The added central Frobenius-linear choices on W enter the leading exponent. On an explicit family their loss under the complete Lie automorphism group is only $q^{O(n)}$, while an all-module weighted bound controls the entire new population. Appendix O proves both assertions. The relaxed theorem is logarithmic; no multiplicative count or local fluctuation law is asserted for this new type space.

9 Further questions

The full-layer tail and multiplicative partition are determined in the growing critical window of Theorem 6.6. The macroscopic window of Theorem 6.11 also has a complete partition, including compatible and square-marked versions. Wider postcritical intervals still require more precise control of module and action strata. Allowing the field size to grow also requires new uniform estimates. These questions retain the uniform unlabelled type measure used here.

A second natural object class would test which part of the mechanism survives beyond these Lie extensions. Such a transfer must recover all coordinate changes and compare actual isomorphism fibers and automorphism weights with the cohomology loss; a relabelled version of the present calculation would not suffice.

Compatibility changes both minimum dimension and extremal population. The relaxed square constraint in Theorem 8.2 changes the leading allocation. A multiplicative count and local fluctuation law for that type space would require a more precise analysis of its semilinear section action and residual stabilizers. Other scalar constraints may impose new dimension costs; an intrinsic module criterion would provide a structural starting point.

A Extremal rigidity and automorphism calculations

This appendix gives the low-degree elimination and full kernel calculations used in the compatible equality cases. Put $k = \mathbb{F}_4$ and retain R, J, Z, τ from (39).

Lemma A.1. *Let $k = \mathbb{F}_4$,*

$$R = k[t_1, \dots, t_s]/(t_1^2, \dots, t_s^2), \qquad J = (t_1, \dots, t_s), \qquad \tau = t_1 \cdots t_s, \quad s \geq 2.$$

Give $k = R/J$ its augmentation-module structure, and write $H^n(M) = \operatorname{Ext}_R^n(k, M)$. Then

$$H^1(J) \cong k, \qquad H^2(J) \cong k^s.$$

These isomorphisms have the following explicit descriptions.

(i) *If $a_i \in J$ satisfy*

$$t_i a_i = 0, \qquad t_i a_j + t_j a_i = 0 \quad (i \neq j), \tag{74}$$

then there are a unique $\eta \in k$ and some $w \in J$ such that

$$a_i = \eta t_i + t_i w \qquad (1 \leq i \leq s).$$

Thus the class of $(t_i)_i$ generates $H^1(J)$.

(ii) *Let $b_i, a_{ij} \in J$, with $a_{ij} = a_{ji}$ and $a_{ii} = 0$, satisfy*

$$t_i b_i = 0, \tag{75}$$

$$t_i a_{ij} = t_j b_i, \qquad t_j a_{ij} = t_i b_j \quad (i \neq j), \tag{76}$$

$$t_i a_{jk} + t_j a_{ik} + t_k a_{ij} = 0 \quad (i < j < k). \tag{77}$$

Then there are unique $\kappa_1, \dots, \kappa_s \in k$ and some $w_1, \dots, w_s \in J$ such that

$$b_i = \kappa_i t_i + t_i w_i, \tag{78}$$

$$a_{ij} = \kappa_i t_j + \kappa_j t_i + t_i w_j + t_j w_i. \tag{79}$$

In particular κ_i is read directly from $b_i \equiv \kappa_i t_i \pmod{J^2}$. The class corresponding to $\kappa = (\kappa_i) \in k^s$ has representative

$$b_i = \kappa_i t_i, \qquad a_{ij} = \kappa_i t_j + \kappa_j t_i.$$

Proof. For $A_i = k[t_i]/(t_i^2)$, the complex

$$\cdots \xrightarrow{t_i} A_i \xrightarrow{t_i} A_i \xrightarrow{t_i} A_i \longrightarrow k \longrightarrow 0$$

is an augmented free resolution: at each positive degree both kernel and image are (t_i). Tensor the s resolutions over k and take the total complex. Exact complexes of vector spaces split, so this tensor complex has homology k in degree zero and zero in positive degrees. It is a free R-resolution $P_\bullet \to k$, explicitly

$$P_n = \bigoplus_{|\alpha|=n} RE_\alpha, \qquad dE_\alpha = \sum_{\alpha_i>0} t_i E_{\alpha-\varepsilon_i}. \tag{80}$$

These are tensor-periodic differentials, with no coefficient α_i. Their square is zero because $t_i^2 = 0$ and the mixed terms occur twice in characteristic two.

Put $C^n(M) = \mathrm{Hom}_R(P_n, M)$. A zero-cochain is $w \in M$, a one-cochain is a tuple (w_i), and a two-cochain has components b_i at $E_{2\varepsilon_i}$ and a_{ij} at $E_{\varepsilon_i+\varepsilon_j}$. The first differentials are

$$(\delta^0 w)_i = t_i w, \qquad (\delta^1(w_i))_{ii} = t_i w_i, \qquad (\delta^1(w_i))_{ij} = t_i w_j + t_j w_i. \tag{81}$$

Consequently the one-cocycle equations are (74). Evaluating the next differential at $E_{3\varepsilon_i}$, $E_{2\varepsilon_i+\varepsilon_j}$, $E_{\varepsilon_i+2\varepsilon_j}$, and $E_{\varepsilon_i+\varepsilon_j+\varepsilon_k}$ gives precisely (75)–(77).

Let $\lambda : R \to k$ take the coefficient of τ. The pairing $(r, a) \mapsto \lambda(ra)$ is nondegenerate: the square-free monomial t_S pairs with t_{S^c}. It identifies R with $\mathrm{Hom}_k(R, k)$ as an R-module, and hence identifies the complexes

$$\mathrm{Hom}_R(P_\bullet, R) \cong \mathrm{Hom}_k(P_\bullet, k).$$

The latter is exact in positive cohomological degrees, since the underlying augmented vector-space resolution is exact. Thus $H^n(R) = 0$ for $n > 0$.

Apply $C^\bullet$ to $0 \to J \to R \to k \to 0$. Each P_n is free, so the result is a short exact sequence of cochain complexes. All differentials of $C^\bullet(k)$ vanish, giving $H^0(k) = k$ and $H^1(k) = k^s$. Moreover $H^0(R) = \mathrm{Ann}_R(J) = k\tau$, and its augmentation map to $H^0(k)$ is zero. The long exact sequence therefore gives the connecting isomorphisms

$$H^0(k) \xrightarrow{\sim} H^1(J), \qquad H^1(k) \xrightarrow{\sim} H^2(J).$$

Lifting $1 \in H^0(k)$ to $1 \in R$ gives the one-cocycle $(t_i)_i$. Lifting $\kappa \in H^1(k)$ to the constant tuple $(\kappa_i) \in C^1(R)$ and applying (81) gives the displayed two-cocycle representative. Subtracting these representatives leaves coboundaries with coefficients in J, proving both descriptions. Such a two-coboundary has $b_i = t_i w_i \in J^2$, so its homogeneous degree-one part vanishes; this also proves the stated identification and uniqueness of κ_i. □

For the extension data used in the extremal normal form, replacing the lift of e_i by $e_i + w_i$, with $w_i \in J$, changes the square value and bracket cocycle by

$$b_i \longmapsto b_i + t_i w_i, \qquad a_{ij} \longmapsto a_{ij} + t_i w_j + t_j w_i.$$

These are exactly the legal section coboundaries in the lemma. The constants $\kappa_i \in R$ in the connecting-map calculation are cochain lifts through $R \to k$; they are not section shifts in $W = J$. Thus the lemma distinguishes the removable section terms from the degree-one class that must be eliminated by the separate change of quotient basis in the normal-form proof.

A.1 The automorphism kernel of a compatible extremizer

Retain the notation of Theorem 3.20, and abbreviate $R = R_s$, $J = J_s$, $\tau = \tau_s$, and $\tau_i = \prod_{j\neq i} t_j$. Thus $\mathrm{Soc}(R) = \mathrm{Ann}_R(J) = k\tau$. An element of $R/k\tau$ is always represented with zero τ-coefficient. The auxiliary algebra $E(p)$, on $U \oplus J$ with $U = \langle e_1, \dots, e_s, v\rangle_k$, has

$$[e_i, e_j] = 0, \quad [e_i, v] = t_i, \quad [e_i, w] = t_i w, \quad [v, w] = pw, \quad [J, J] = 0.$$

This is also the original k-bilinear polar bracket of F_p.

The quotient and its splitting. The derived ideal is J, and its unital action algebra is $H = R/k\tau$. An automorphism induces maps $A : U \to U$, $C : J \to J$, and a section change $h : U \to J$. The kernel of

$$U \longrightarrow \mathrm{rad}(H)/\,\mathrm{rad}(H)^2$$

is precisely kv. Consequently $Av = \delta v$ and $Ae_i = \sum_j B_{ji} e_j + \alpha_i v$, with B invertible. The source core brackets vanish. The section-change terms lie in J^2, so their images modulo J^2 give $\alpha_j B_i + \alpha_i B_j = 0$ for $i \neq j$. Since $s \geq 2$, independence of the columns of B forces all $\alpha_i = 0$. It follows that the induced automorphism of H is σ_B, and that $\sigma_B(p) = \delta p$ modulo $k\tau$. The specified representatives make this an equality in R.

The standard Frobenius pairing of R, given by the highest coefficient of a product, yields $\mathrm{End}_R(J) = R/k\tau$. Thus $C = u\sigma_B$ for a unit $u \in H$. The mixed brackets modulo J^2 give $u(0) = \delta$. The maps $(X, V, w) \mapsto (BX, \delta V, \delta\sigma_B(w))$ therefore provide a homomorphic section of the quotient map.

All kernel elements. In the kernel, A is the identity and C is multiplication by $1 + a$, for a unique $a \in J/k\tau$. Write $h_i = h(e_i)$ and $h_v = h(v)$. Bracket preservation is equivalent to

$$t_i h_j + t_j h_i = 0 \qquad (i \neq j), \tag{82}$$

$$t_i(a + h_v) + p h_i = 0 \qquad (1 \leq i \leq s). \tag{83}$$

There is no additional equation $t_i h_i = 0$: the chosen restricted square is not a marked part of the automorphism problem.

Multiplying (82) by t_i shows that $t_i h_i = \lambda_i \tau$ for a unique $\lambda_i \in k$. Then $g_i = h_i + \lambda_i \tau_i$ satisfies $t_i g_i = 0$ and $t_i g_j + t_j g_i = 0$. These relations imply $g_i = t_i z$ for a common $z \in R/k\tau$. Here is a direct coefficient proof. The first relation allows only monomials containing t_i in g_i. For every proper subset $A \subsetneq \{1, \dots, s\}$, define the coefficient of t_A in z to be the coefficient of $t_{A\cup\{i\}}$ in g_i, choosing any $i \notin A$. The second relation makes the answer independent of the choice. Only the highest coefficient remains free, and it is removed modulo $k\tau$.

Since $p\tau_i = 0$, equation (83) now gives

$$\begin{gathered} h_i = t_i z + \lambda_i \tau_i, \qquad h_v = a + pz + \zeta\tau, \\ a \in J/k\tau,\ z \in R/k\tau,\ \lambda \in k^s,\ \zeta \in k. \end{gathered} \tag{84}$$

These parameters are unique, and every choice satisfies the two kernel equations. Since $a^2 = 0$, multiplication by $1 + a$ is invertible. There are $(2^s - 2) + (2^s - 1) + s + 1 = 2^{s+1} + s - 2$ free parameters.

The group law. Let $\mathrm{pr} : R \to R$ remove the highest term, and let $[t_A]r$ denote a monomial coefficient. For left parameters (a, z, λ, ζ) and right parameters $(a', z', \lambda', \zeta')$, the composite

has coordinates

$$\begin{aligned}
a'' &= \mathrm{pr}(a + a' + aa'),\\
z'' &= \mathrm{pr}\left(z + z' + az' + \sum_i \lambda_i'[t_i]a\,\tau_i\right),\\
\lambda'' &= \lambda + \lambda',\\
\zeta'' &= \zeta + \zeta' + [\tau](aa').
\end{aligned}$$

Indeed block composition is $(u,h)(u',h') = (uu', h + uh')$, and $a\tau_i = [t_i]a\,\tau$. The formula contains no p, proving that the abstract kernel is independent of the parameter. It has exponent four: a square has $a = \lambda = \zeta = 0$, so its square is the identity; taking $a = t_1$, $z = 1$, and $\lambda = \zeta = 0$ gives an element whose square is nonidentity.

For completeness, conjugation by the section corresponding to (B,δ) is explicitly described in block coordinates by

$$u \longmapsto \sigma_B(u), \qquad h \longmapsto \delta\sigma_B \circ h \circ A^{-1}, \quad A = \mathrm{diag}(B,\delta).$$

Thus the kernel and the stabilizer action specify the full semidirect product, not only its order.

The unique binary lift. Any binary-linear map commuting with F_p preserves the intrinsic derived ideal. Its diagonal blocks commute with multiplication by ω, so they are k-linear. Write its off-diagonal block as $h_s + h_a$, with h_s linear over k and $h_a(\alpha\xi) = \alpha^2 h_a(\xi)$. Comparing the polar identity at ξ and $\omega\xi$ shows that h_a takes values in the common action kernel $k\tau$. It therefore contributes nothing to the polar bracket, and the k-linear part is an automorphism of $E(p)$.

Conversely, let S_0 be such an auxiliary automorphism and put $r_p = F_p - \omega\,\mathrm{id}$. The difference

$$D(\xi) = r_p(S_0(\xi,0)) - S_0(r_p(\xi,0))$$

is Frobenius semilinear: its polar vanishes, and the two restricted squares have the same adjoint square. Its values belong both to J and to the Lie center, hence to $k\tau$. It vanishes on the derived ideal. Since $\omega^2 + \omega = 1$, adding the central off-diagonal map D to S_0 gives a map commuting with F_p. The lift is unique; the difference of two lifts would satisfy $(\mathrm{id} + P_p(A\xi))h_a(\xi) = 0$, and the first factor is invertible. Taking the scalar part respects composition, so this is a group isomorphism.

For the kernel parameters above, the actual binary transformation is

$$\begin{aligned}
(X,V,w) \longmapsto \Big(X, V, (1+a)w + \sum_i X_i(t_i z + \lambda_i\tau_i)\\
+ V(a + pz + \zeta\tau) + \tau\sum_i \lambda_i X_i^2\Big).
\end{aligned}$$

The last term is forced by λ; it is not an additional family of choices. Indeed $D(e_i) = \lambda_i\tau$ and $D(v) = 0$.

The class-two boundary. For $s = 1$, the auxiliary algebra is the three-dimensional Heisenberg algebra. Its automorphisms are $(u,z) \mapsto (Bu, \det(B)z + h(u))$, with $B \in \mathrm{GL}_2(k)$ and $h \in \mathrm{Hom}_k(k^2,k)$. For $B = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right)$, the corresponding binary lift with $h = 0$ is

$$(X,V,z) \longmapsto (aX + bV, cX + dV, \det(B)z + acX^2 + bdV^2).$$

The kernel is $(k^2,+)$, giving order $4^2|\,\mathrm{GL}_2(4)| = 2880$, as asserted in Remark 3.21.

A.2 Pure quadratic parameters and an intrinsic action rank

Pure quadratic parameters form a readable all-class subfamily of the canonical extremizers. Their ordinary polynomial degree refers to the fixed ring coefficients, and does not increase the binary degree of the associated map.

Proposition A.2. *Let $s = c - 1 \geq 3$. Among the canonical models with $p \in \Lambda^2 k^s$, there are exactly $\lfloor s/2 \rfloor + 1$ binary-linear conjugacy classes, represented by*

$$p_0 = 0, \qquad p_r = t_1 t_2 + t_3 t_4 + \cdots + t_{2r-1} t_{2r} \quad (1 \leq r \leq \lfloor s/2 \rfloor).$$

For $r \geq 1$, multiplication by p_r on the intrinsic derived module J has k-rank

$$\rho_{s,r} = 2^{s-1} - 2^{s-r-1} - 1. \tag{85}$$

The corresponding binary rank is $2\rho_{s,r}$.

Proof. Linear substitutions preserve the homogeneous degree. On $\Lambda^2 k^s$, the standard alternating-bivector normal form is determined by its even rank $2r$, giving the displayed representatives. Common nonzero scalar multiplication creates no further identifications between different ranks. The canonical orbit correspondence therefore gives the stated complete classification of this subfamily.

For the rank calculation, regard multiplication by p_r as a square-zero differential on R. Each paired factor $k[a,b]/(a^2, b^2)$, with differential given by multiplication by ab, is the direct sum of the acyclic block $k1 \to k(ab)$ and the zero differential on $ka \oplus kb$. Tensoring the r paired factors gives homology dimension 2^r. The remaining $s - 2r$ square-zero variables have zero differential and multiply that dimension by 2^{s-2r}. Thus

$$\dim_k H(R, p_r \cdot) = 2^{s-r}.$$

Since a square-zero operator has homology dimension equal to its ambient dimension minus twice its rank, multiplication by p_r on R has rank $2^{s-1} - 2^{s-r-1}$. Moreover $p_r R = k p_r \oplus p_r J$: the first summand is homogeneous of degree two, whereas the second has degree at least three. Subtracting this one dimension proves (85). □

In the canonical model, kv is the intrinsic kernel of the quotient action modulo the square of the radical of its generated operator algebra. Consequently the rank of $P(v)$, up to rescaling v, is an intrinsic invariant. Formula (85) makes the alternating rank of a pure quadratic parameter visible directly in the extremal map. For example, at $s = 4$, parameters of ranks two and four give binary action ranks six and ten.

B The unrestricted metabelian equality classification

We prove Theorem 3.6. Throughout, $k = \mathbb{F}_4$, $c \geq 5$, $d = c - 1$, $m = d - 1$,

$$U = ke_0 \oplus H, \quad H = \operatorname{span}_k\{e_1, \ldots, e_m\}, \quad R = k[t_0, \ldots, t_m]/(t_0^2, \ldots, t_m^2), \quad K = (t_1, \ldots, t_m).$$

Put $z = t_1 \cdots t_m$, $I = (z) = \operatorname{span}_k\{z, t_0 z\}$, and $\tau = t_0 z$. In particular I is two-dimensional, whereas $\dim_k K = 2^d - 2$. All cohomology below is ordinary Chevalley–Eilenberg cohomology of the abelian Lie algebra U, acting by $e_i w = t_i w$. The Koszul complex, Frobenius pairing, contraction homotopies and alternating-form reductions used here are standard linear-algebraic tools; see also [4, Theorem 1 and p. 279] for the classical Koszul background. We give the required calculations explicitly.

B.1 Equality forces a common derived module

Lemma B.1. *Every minimum-dimensional metabelian solution of class $c \geq 5$ has $d = c - 1$ quotient generators and derived R-module*

$$W \simeq D(R/I) \simeq K, \qquad D = \mathrm{Hom}_k(-, k).$$

Moreover $\mathrm{Ann}_R K = I$ *and* $\mathrm{End}_R K \simeq R/I$.

Proof. Equality in Theorem 3.4 gives $\dim_k W = 2^d - 2$ and $d = c - 1$. The quotient $W_d \to W$ retains a nonzero top component. Pull back a functional on W nonzero there to $\phi \in W_d^*$. By Lemma 3.3, a linear change of generators normalizes its highest homogeneous component to a top functional λ_0 with $\mathrm{Ann}_R \lambda_0 = I$. Write $\phi = \lambda_0 + \psi$, where the dual coefficient degrees of ψ are at most $d-2$. Since z has degree $d-1$, it annihilates ψ, and it annihilates λ_0 as well. Hence $I\phi = 0$, and $\dim_k R\phi \leq 2^d - 2$. The filtered contraction argument of the same lemma gives the reverse inequality. Thus $\mathrm{Ann}_R \phi = I$. As $R\phi \subseteq W^*$ already has dimension $\dim_k W$, $W^* = R\phi \simeq R/I$.

The pairing $(r, s) \mapsto [\tau](rs)$ is nondegenerate on R. It identifies $D(R/I)$ with $\mathrm{Ann}_R I = K$: a polynomial annihilates z precisely when its constant and t_0 coefficients vanish. The same observation gives $\mathrm{Ann}_R K = I$. Finally, duality identifies $\mathrm{End}_R K$ with the opposite of $\mathrm{End}_R(R/I) = R/I$, which is commutative. □

Lemma B.2 (One-dimensional derived-module defect)**.** *Let $d \geq 4$. Suppose a quotient W of the universal module W_d retains a nonzero top component and has $\dim_k W = 2^d - 1$. After a linear change of quotient generators, there is an exact sequence of R-modules*

$$0 \longrightarrow k \longrightarrow W \longrightarrow K \longrightarrow 0.$$

After this normalization, the number of possible R-module isomorphism types is at most $4^{\binom{d}{2}}$, and their ordinary cohomology satisfies $\dim_k H^2(U, W) \leq d^2$.

Proof. Pull back a functional nonzero on the top to $\phi \in W_d^*$, and normalize its highest component as in Lemma B.1. The same degree argument gives $I\phi = 0$, while the filtered contraction bound gives $\dim_k R\phi = 2^d - 2$. Thus $R\phi \simeq R/I$ is a codimension-one submodule of W^*. Its quotient is the trivial module k, since each t_i acts nilpotently. Dualizing gives the displayed sequence.

Such dual extensions are classified by $\mathrm{Ext}^1_R(k, R/I)$. Finite duality identifies this with $D\,\mathrm{Tor}^R_1(k, K)$. To compute the latter, put $S = k[t_1, \ldots, t_{d-1}]/(t_i^2)$ and $J_S = (t_1, \ldots, t_{d-1})$. Then $K = k[t_0]/t_0^2 \otimes_k J_S$. The tensor-periodic minimal resolution of k over S, as in (80), has free rank $\binom{d+j-2}{j}$ in degree j. Deleting its final copy of S gives a resolution of J_S; extending scalars gives a minimal R-resolution of K. Its degree-one free rank is $\binom{d}{2}$. Hence

$$\dim_k \mathrm{Ext}^1_R(k, R/I) = \binom{d}{2},$$

giving the asserted bound on module types. Finally the ordinary CE long exact sequence of $0 \to k \to W \to K \to 0$ and Lemma B.3 give

$$\dim_k H^2(U, W) \leq \dim_k H^2(U, k) + \dim_k H^2(U, K) = \binom{d}{2} + \left(d + \binom{d}{2}\right) = d^2.$$

Here module Ext_R and ordinary CE cohomology have different complexes; each is used with its stated meaning. □

B.2 Ordinary cocycles and intrinsic generation

Let $\tau_{ij} = \prod_{\ell \notin \{i,j\}} t_\ell$. Write $\beta_\eta(e_0, e_i) = t_i$, with all its other basis values zero. For $\kappa \in H^*$, let

$$\beta_\kappa(e_i, e_j) = t_0(\kappa_i t_j + \kappa_j t_i) \quad (i, j \geq 1), \qquad \beta_\kappa(e_0, e_i) = 0.$$

For $\lambda = (\lambda_{ij})$, set $\beta_\lambda(e_i, e_j) = \lambda_{ij}\tau_{ij}$. Thus $\beta_{\kappa,\lambda} = \beta_\eta + \beta_\kappa + \beta_\lambda$ is exactly (29).

Lemma B.3. *Every class in $H^2(U, K)$ has a unique expression*

$$\eta[\beta_\eta] + [\beta_\kappa] + [\beta_\lambda], \qquad \eta \in k, \quad \kappa \in H^*, \quad \lambda \in k^{\binom{d}{2}}.$$

Consequently $\dim_k H^2(U, K) = d + \binom{d}{2}$. *Also* $\dim_k H^1(U, K) = d + 1$. *For a closed cocycle β, the condition*

$$K = \mathfrak{m}K + \operatorname{span}_k\{\beta(x, y) : x, y \in U\}, \qquad \mathfrak{m} = (t_0, \ldots, t_m), \tag{$\dagger$}$$

holds if and only if $\eta \neq 0$. In that case the associated solution has class $d + 1$.

Proof. The ordinary CE complex for the R-module R is the tensor product of the one-variable complexes $k[t_i]/t_i^2 \xrightarrow{t_i} k[t_i]/t_i^2$. Their zeroth cohomology is kt_i, and their first cohomology has representative 1. Hence $H^n(U, R)$ has basis

$$\Big(\prod_{i \notin S} t_i\Big) e_S^*, \qquad |S| = n.$$

In particular its coefficient degree is $d - n$.

Put $A = R/K = k[t_0]/t_0^2$. The H-action on A is zero, so $H^1(U, A)$ has representatives

$$h(e_0) = 1,\ h(H) = 0; \qquad h(e_i) = \kappa_i t_0,\ h(e_0) = 0.$$

Its dimension is $1 + m = d$. Its elements have coefficient degree zero or one; the same is true of $H^2(U, A)$. Since $d \geq 4$, the maps $H^1(U, R) \to H^1(U, A)$ and $H^2(U, R) \to H^2(U, A)$ vanish by coefficient degree. The long exact sequence therefore gives

$$0 \longrightarrow H^1(U, A) \longrightarrow H^2(U, K) \longrightarrow H^2(U, R) \longrightarrow 0. \tag{86}$$

In degree zero, $H^0(U, R) = k\tau \to H^0(U, A) = kt_0$ is zero. Together with the vanishing map on H^1 above, this gives $0 \to k \to H^1(U, K) \to H^1(U, R) \to 0$, proving its dimension $d + 1$. The connecting map gives β_η and β_κ. The displayed $H^2(U, R)$-basis lifts to the closed K-valued cocycles β_λ, proving the basis claim. Equivalently, the coefficient of τ_{ij} in component (i, j) extracts λ_{ij}, because a coboundary in that component is divisible by t_i or t_j.

The quotient $K/\mathfrak{m}K$ has basis $t_1, \ldots, t_m$. The only degree-one closed cocycles are multiples of β_η; there are no degree-one coboundaries, since K has no constant component. Thus the images of the cocycle values in this quotient are precisely those of $\eta\beta_\eta$. They span it exactly when $\eta \neq 0$. This proves ($\dagger$); one may also lift the spanning statement by Nakayama's lemma. Finally, $\mathfrak{m}^{d-1}K = k\tau \neq 0$ and $\mathfrak{m}^d K = 0$. Equation (16) gives class $d + 1$. □

Multiplication by a nonzero scalar normalizes η to one. Lemmas B.1 and B.3, followed by Theorem 2.6, prove the asserted coverage by $F_{\kappa,\lambda}$.

B.3 All morphisms and the normalized parabolic

Lemma B.4. *On the slice $\eta = 1$, auxiliary-Lie isomorphism is exactly* (30).

Proof. A Lie isomorphism preserves the intrinsic derived module K. Let $B \in \mathrm{GL}(U)$ be its induced quotient map and let C be its restriction to K. Intertwining the actions gives $C(rw) = \sigma_B(r)C(w)$, and hence $\sigma_B(I) = I$. The degree-m line of I is kz, and $\{u \in U : uz = 0\} = H$. Thus $B(H) = H$, so

$$B = \begin{pmatrix} a & 0 \\ v & M \end{pmatrix}.$$

Conversely every such B preserves I and K. By Lemma B.1, all intertwining C have the form $u\sigma_B$, with $u \in (R/I)^\times$. Changes of section add exactly ordinary CE coboundaries.

Positive-degree multiplication is null-homotopic on this CE complex. Indeed, if ι_i is contraction with e_i, then $\delta\iota_i + \iota_i\delta = t_i$. Writing a positive-degree element as $\sum_i t_i r_i$ gives the homotopy $\sum_i r_i\iota_i$. Consequently the unit u acts on cohomology by its constant term u_0 alone. The star transforms as $\eta' = u_0\eta/a$, as is seen either in degree one or on its lift $h(e_0) = 1$ in A. The normalization $\eta = \eta' = 1$ therefore forces $u_0 = a$. This proves necessity of (30). For sufficiency, use $C = a\sigma_B$; equality of cohomology classes supplies the required change of section.

There are no further quotient substitutions: B is the actual linear map induced on U. For the original quadratic solutions, Theorem 2.6 identifies these Lie isomorphisms with binary conjugacies, including their uniquely determined off-diagonal binary-linear parts. In particular their diagonal blocks are k-linear, not Frobenius-semilinear. □

B.4 The graded action when $d \geq 5$

The coefficient degrees of the three parts of Lemma B.3 are $1, 2, d-2$. For $d \geq 5$ these are distinct, so the displayed splitting is equivariant. On A, the substitution is $t_0 \mapsto at_0$. Pulling back the lift $h(e_i) = \kappa_i t_0$ gives $\kappa' = a^2 M^{-T}\kappa$; its extra e_0-component is a multiple of $t_0 e_0^*$, which is a boundary on A. For the other part, the complementary-minor identity gives

$$[\tau_{ij}]\sigma_B(\tau_{kl}) = \det(B)(\Lambda^2 B^{-T})_{ij,kl}. \tag{‡}$$

The pullback on the two cochain inputs contributes the same minor. Extracting the τ_{ij}-coefficient in component (i,j) therefore proves (32). For example, take $B = \mathrm{diag}(b_0, \ldots, b_m)$, so $a = b_0$. The complementary monomial contributes

$$\sigma_B(\tau_{ij}) = \frac{\det B}{b_i b_j}\tau_{ij},$$

and the two pulled-back inputs contribute a further $(b_i b_j)^{-1}$. Consequently

$$\lambda'_{ij} = a\det(B)(b_i b_j)^{-2}\lambda_{ij}.$$

The same inverse minor occurs once from each source, explaining the entrywise square in (32). Identity (‡) also proves the action on the quotient $H^2(U, R)$ when $d = 4$; only its splitting fails at that endpoint.

Put $\Omega = \lambda^{[2]}$. This is a bijective reparametrization of coefficients, not an additional Galois identification of models. Since squaring twice is the identity on k, and $a^4 = a$,

$$\Omega' = \chi\Lambda^2 B^{-T}\Omega, \qquad \chi = a(\det M)^2.$$

Writing $\Omega = e_0^* \wedge \ell + b$, with $\ell \in H^*$, $b \in \Lambda^2 H^*$, and $w = M^{-1}v$, gives the complete action

$$\begin{aligned} \kappa' &= a^2 M^{-T}\kappa, \\ b' &= \chi M^{-T} b M^{-1}, \\ \ell' &= (\det M)^2 M^{-T}(\ell + \iota_w b). \end{aligned} \tag{87}$$

B.5 Complete rank and radical-pair reduction

Fix $\operatorname{rank} b = 2r$, put $Z = \operatorname{rad} b$, $n = \dim Z = m - 2r$, and choose $H = S \oplus Z$, with b standard symplectic on S. The parameter v removes exactly $\operatorname{im} b$ from ℓ. Its residual is thus $\ell_Z = \ell|_Z$. Write $\kappa = (\kappa_S, \kappa_Z)$. For $r > 0$, the stabilizer of the chosen b has

$$M = \begin{pmatrix} A & 0 \\ C & D \end{pmatrix}, \qquad A^T b A = \rho b, \quad D \in \mathrm{GL}(Z), \quad C \text{ arbitrary.}$$

The Pfaffian identity gives $\det A = \rho^r$. The condition $\chi = \rho$ is equivalent to

$$a = \rho^{1-2r} (\det D)^{-2}. \tag{88}$$

Every $\rho \in k^\times$ and every $D \in \mathrm{GL}(Z)$ occur. When $r = 0$, the same formulas hold with S absent and the freely selectable parameter $\rho = a(\det D)^2$. Because $x^3 = 1$ for $x \in k^\times$, the radical-pair action is

$$\kappa'_Z = \rho^{2-r} (\det D)^{-1} D^{-T} \kappa_Z, \qquad \ell'_Z = \rho^{-r} (\det D)^{-1} D^{-T} \ell_Z. \tag{89}$$

If $\kappa_Z \neq 0$, the free block C removes κ_S. If $\kappa_Z = 0$, it cannot change κ_S, and the symplectic group is transitive on its nonzero values. Thus there is exactly one zero/nonzero flag when $r > 0$ and $\kappa_Z = 0$, and no such flag otherwise.

The radical pair has five possible incidence types:

$$(0,0), \quad (0,\neq 0), \quad (\neq 0, 0), \quad \text{nonzero proportional}, \quad \text{independent}. \tag{$*$}$$

The last requires $n \geq 2$. In the proportional case the nonzero ratio is multiplied by ρ, so gives no further invariant. For $n \geq 2$, $\mathrm{SL}_n(k)$ is transitive on nonzero covectors. For $n \geq 3$, it is also transitive on ordered independent two-frames: adjust the determinant of a frame extension on an unused coordinate. These observations prove transitivity for all types in $(*)$, except possibly the last at $n = 2$. For $n = 1$, putting $D = (d_0)$ in (89) gives a freely selectable factor d_0 on a nonzero covector, so the same conclusion holds.

For $n = 2$ and an independent pair, let $\Delta = \det[\kappa_Z, \ell_Z]$. Equation (89) gives

$$\Delta' = \rho^{2-2r} \Delta, \tag{90}$$

since $(\det D)^3 = 1$. If $r \not\equiv 1 \pmod 3$, this character covers $k^\times$. If $r \equiv 1 \pmod 3$, the three nonzero values of Δ are separate invariants. These are complete: choose ρ using (90) when permitted, and then use $\mathrm{SL}_2(k)$, which is transitive on ordered bases of a fixed determinant.

This proves the following exhaustive count at each rank:

Rank condition	Radical condition	Number of orbits
$r = 0$	$n = m \geq 4$	5
$r > 0$	$n = 0$	2
$r > 0$	$n = 1$	6
$r > 0$	$n \geq 2$, except the next row	7
$r > 0,\ r \equiv 1 \pmod 3$	$n = 2$	9

Indeed, when $n \geq 1$ the first two types in $(*)$ each acquire the extra κ_S-flag for $r > 0$; the other types do not. For $n = 0$ only the first type remains, with its two flags. For $r = 0$ all five types occur and there is no flag. Summation gives

$$E_c = \begin{cases} 7h + 2\mathbf{1}_{h \equiv 2 \ (3)}, & m = 2h \geq 4, \\ 7h + 4, & m = 2h + 1 \geq 5. \end{cases}$$

Since $m = c - 2$, these are exactly (31) for $c \geq 6$.

B.6 The scalar invariant at $6 \mid c$

The three-valued case has a coordinate-free relative-invariant description. Suppose $m = 2r+2$ and $\operatorname{rank} b = 2r$. Let $\gamma_r(b)$ be the divided exterior power: its coefficient on a $2r$-subset is the Pfaffian of the corresponding submatrix. This polynomial definition uses no factorial division. It is natural under linear maps and homogeneous of degree r. Since $\gamma_r(b)$ spans the top exterior power of $\operatorname{im} b$, $(\iota_w b) \wedge \gamma_r(b) = 0$. Consequently the coefficient J of

$$\kappa \wedge \ell \wedge \gamma_r(b) \in \Lambda^m H^*$$

in a fixed volume form satisfies

$$J' = a^{r+2} (\det M)^{2r+1} J. \tag{91}$$

For $r \equiv 1 \pmod 3$ this character is trivial on $k^\times$; equivalently $c = 2r + 4$ is divisible by six. For example,

$$b = \sum_{j=1}^{r} e^*_{2j-1} \wedge e^*_{2j}, \quad \kappa = e^*_{2r+1}, \quad \ell = t e^*_{2r+2}, \qquad t \in \{1, \omega, \omega^2\},$$

gives three inequivalent parameters with identical rank and incidence data. The divided power is essential in characteristic two: ordinary b^r is zero for $r \geq 2$.

B.7 The nonsplit endpoint $d = 4$

Now $m = 3$. Define a symmetric matrix S and vector q by

$$S = \begin{pmatrix} \lambda_{23} & \kappa_3 & \kappa_2 \\ \kappa_3 & \lambda_{13} & \kappa_1 \\ \kappa_2 & \kappa_1 & \lambda_{12} \end{pmatrix}, \qquad q = \begin{pmatrix} \lambda_{01} \\ \lambda_{02} \\ \lambda_{03} \end{pmatrix}, \qquad D(S) = \begin{pmatrix} 0 & S_{33} & S_{22} \\ S_{33} & 0 & S_{11} \\ S_{22} & S_{11} & 0 \end{pmatrix}.$$

For $B = \left(\begin{smallmatrix} a & 0 \\ v & M \end{smallmatrix}\right)$ and $w = M^{-1}v$, the full action is

$$S' = a^2 (\det M)^{-1} M S M^T, \qquad q' = \det M (M^{-T})^{[2]} \big(q + D(S) w^{[2]}\big). \tag{92}$$

Here is a direct derivation. The coefficients of t_0 in the H-pair cocycles, with columns ordered as $(23), (13), (12)$, form a linear map $H^* \to H$; Jacobi is exactly symmetry of its matrix S. These coefficients cannot be altered by a K-valued degree-one coboundary. The change of volume on H, the factor $t_0 \mapsto a t_0$, and the normalizing scalar a give the first formula. Factoring $B = \operatorname{diag}(a, M) \left(\begin{smallmatrix} 1 & 0 \\ w & I \end{smallmatrix}\right)$, the unipotent factor leaves these t_0-coefficients unchanged. On q, the complementary-minor calculation gives exactly $D(S) w^{[2]}$; the diagonal factor gives $\det M (M^{-T})^{[2]}$. Thus no splitting of the colliding degrees has been assumed.

A single shear shows the collision explicitly. Take $a = 1$, $v = 0$, and $M = I + E_{21}$, so $e_1 \mapsto e_1 + e_2$. Start with $\lambda_{23} = 1$ and every other λ- and κ-coefficient zero. Then $S = E_{11}$, and

$$M S M^T = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus $\lambda'_{23} = \lambda'_{13} = \kappa'_3 = 1$, with all other coefficients zero. A pure λ-class has acquired a κ-component, so the split action for $d \geq 5$ cannot be used at this endpoint.

In fact the degree-two part of (86) is nonsplit already on the Levi $\mathrm{GL}_3(k)$. For $a = 1, v = 0$, the S-space is $\det(H)^{-1} \otimes \Gamma^2 H$, the symmetric tensors. Its alternating, or zero-diagonal, subspace is the κ-space; its diagonal quotient is the corresponding Frobenius-natural twist.

A diagonal-torus-equivariant splitting would have to lift each quotient basis vector to its diagonal symmetric matrix, since the weights x_i^2 and x_ix_j are distinct characters of $(k^\times)^3$, also after the common determinant twist. But a transvection $e_i \mapsto e_i + e_j$ sends that diagonal matrix to the two diagonal matrices plus a nonzero off-diagonal symmetric matrix. This contradicts equivariance. Projecting to S shows that the separate q-summand cannot split the full sequence either.

It remains to classify (92). Over the perfect field k, symmetric 3×3 matrices have five congruence types: zero, alternating rank two, and nonalternating ranks one, two and three. One may obtain this list by Gram–Schmidt; if an alternating plane remains after a nonsingular line, the identity $\langle 1\rangle \perp \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) \simeq I_3$ diagonalizes the final nonalternating case. These are also the S-orbits in (92): any ordinary congruence M is realized by choosing $a^2 = \det M$. For each type the remaining complete invariants are

S	Remaining invariant	Orbits
0	$q = 0$ or $q \neq 0$	2
Alternating, rank 2	$q = 0;\ q \neq 0,\ Sq^{[2]} = 0;\ Sq^{[2]} \neq 0$	3
Nonalternating, rank $1, 2$, or 3	$\operatorname{diag}(S)^T q = 0$ or $\neq 0$	2 each

For $S = 0$, $\mathrm{SL}_3(k)$ is transitive on nonzero q. For alternating $S = J_2 \oplus 0$, its stabilizer has

$$M = \begin{pmatrix} A & b \\ 0 & a^2 \end{pmatrix}, \qquad A \in \mathrm{GL}_2(k), \quad b \in k^2.$$

If the first two coordinates of q are nonzero as a vector, the free block b removes the third coordinate and $\mathrm{SL}_2(k)$ is transitive on the remaining vector. If they vanish, $\det A$ scales the third coordinate freely. This proves the three cases in the table. Finally, if $u = \operatorname{diag} S \neq 0$, then $\operatorname{im} D(S) = \ker u^T$, so translations identify exactly the q with equal $u^T q$. Formula (92) gives $u'^T q' = a^2 u^T q$. The simultaneous choice $M = a^{-1} I_3$ fixes every S and multiplies q by a^2. All nonzero residues are therefore equivalent, proving the two cases for each nonalternating rank. There are exactly $2 + 3 + 2 + 2 + 2 = 11$ classes. This is E_5, and completes the proof of Theorem 3.6.

C Fixed surplus over finite characteristic-two fields

We prove the upper bound in Theorem 5.3. Throughout this appendix, k is a finite field of characteristic two, $|k| = q \geq 4$, and all Lie isomorphisms are k-linear. By Lemma 5.2, the metabelian 3-Engel identity is equivalent to

$$(\operatorname{ad} x)^2 L' = 0 \qquad (x \in L).$$

Put

$$d = c - 1 \geq 4, \qquad C_d = \binom{d}{2}, \qquad w_0 = 2^d - 2,$$

and use the ring and modules

$$R = k[t_0, \ldots, t_{d-1}]/(t_i^2), \quad \mathfrak{m} = (t_0, \ldots, t_{d-1}), \quad K = (t_1, \ldots, t_{d-1}), \quad I = (t_1 \cdots t_{d-1}). \tag{93}$$

The ideal I has dimension two. We write $D = \mathrm{Hom}_k(-, k)$, and ordinary cohomology is the Chevalley–Eilenberg cohomology of the indicated abelian quotient. Module Ext_R is used separately.

Lemma C.1 (The minimum and its common module over k)**.** *Among metabelian 3-Engel k-Lie algebras of class $c \geq 5$, the minimum k-dimension is*

$$\ell(c) = 2^{c-1} + c - 3 = d + w_0.$$

Every equality case has quotient dimension d. After a linear change of quotient generators, its derived R-module is K. For the core quotient $U_0 = k^d$, acting by $e_i w = t_i w$,

$$\begin{aligned} \operatorname{Ann}_R K &= I, & \operatorname{End}_R K &\simeq R/I, \\ \dim_k H^1(U_0, K) &= d+1, & \dim_k H^2(U_0, K) &= d + C_d. \end{aligned}$$

The degree-one free rank in a minimal R-resolution of K is C_d, and

$$\dim_k \operatorname{Ext}^1_R(k, K) = 1.$$

In every minimum core the degree-one cocycle can be normalized to

$$\beta(e_0, e_i) \equiv t_i \pmod{\mathfrak{m}K} \quad (i \geq 1), \qquad \beta(e_i, e_j) \in \mathfrak{m}K \quad (i, j \geq 1). \tag{94}$$

In particular, the number of minimum-dimensional k-Lie isomorphism classes is at most q^{d+C_d}.

Proof. The universal commutator module has the same square-free presentation and triangle relations as in Section 3, with k in place of $\mathbb{F}_4$. The multigrading calculation in Proposition 3.1, the repeated-terminal argument in Lemma 3.2, and the contraction calculation in Lemma 3.3 use only characteristic two. The top module is $\det(U_0) \otimes U_0$, and $\mathrm{SL}_d(k)$ is transitive on its nonzero dual vectors. A normalized top functional has annihilator I and contraction dimension $2^d - 2$. Thus every class-c object has quotient dimension at least d and derived dimension at least w_0. The normalized contraction quotient attains both bounds.

For equality, the proof of Lemma B.1 applies over k: a top functional generates the whole dual derived module as R/I. The top-coefficient Frobenius pairing on R gives $D(R/I) \simeq K$, $\operatorname{Ann}_R K = I$, and $\operatorname{End}_R K \simeq R/I$. The ordinary Koszul calculation in Lemma B.3 likewise has no field-size dependence. It gives the displayed H^1 and H^2 dimensions. For clarity, its degree-one normalization can also be read directly from Jacobi. The degree-one coefficients lie in $\operatorname{span}_k\{t_1, \ldots, t_{d-1}\}$. Jacobi at $(0, i, j)$ first makes the degree-one part of $\beta(e_i, e_j)$ zero: its t_0-multiple cannot cancel against the other two terms. Comparing the remaining monomials $t_i t_l$, using $d - 1 \geq 3$, then gives $\beta(e_0, e_i)_{[1]} = \eta t_i$ with one common scalar η. Generation of $K/\mathfrak{m}K$ forces $\eta \neq 0$, and rescaling K normalizes it to one. Two-coboundaries have coefficients in $\mathfrak{m}K$, so no degree-one term is lost in this comparison. This proves (94). For a fixed derived module, cocycles modulo changes of section form $H^2(U_0, K)$; these classes cover all minimum-dimensional objects. This proves the last count.

For the module-resolution assertions, set $S = k[t_1, \ldots, t_{d-1}]/(t_i^2)$ and $J_S = (t_1, \ldots, t_{d-1})$. The tensor-periodic minimal resolution of k over S, as in (80), has degree-j free rank $\binom{d+j-2}{j}$. Its shifted tail resolves J_S, and extension of scalars resolves $K = R \otimes_S J_S$. Its degree-one free rank is therefore C_d. Finally finite duality gives

$$\operatorname{Ext}^1_R(k, K) \simeq D \operatorname{Tor}^R_1(k, R/I).$$

The cyclic presentation of R/I has the principal minimal relation ideal I, so $\operatorname{Tor}^R_1(k, R/I) \simeq I/\mathfrak{m}I$ is one-dimensional. □

Lemma C.2 (Bounded derived defect)**.** *Let L be a metabelian 3-Engel k-Lie algebra of class c, with derived dimension $w_0 + h$ and quotient dimension at least d. There is a class-c core with d quotient generators. After choosing normalized core quotient coordinates, the full derived*

ideal $W = L'$, as an R-module, has a filtration with one factor K and h factors k. The possible normalized R-module types are covered by at most

$$M_h(d) = (h+1)q^{hC_d+dh^2+h} \tag{95}$$

representatives. Every such W satisfies

$$\dim_k \mathrm{End}_R(W) \le w_0 + hd + h^2, \qquad \dim_k \mathrm{Soc}_R(W) \le h+1, \tag{96}$$

$$\dim_k H^1(U_0, W) \le (h+1)d + 1, \qquad \dim_k H^2(U_0, W) \le d + (h+1)C_d. \tag{97}$$

Here the core quotient U_0 acts through R.

Proof. Lemma 3.2 supplies the core. Its generators may be chosen as linear lifts of a basis of a d-dimensional quotient subspace. Changing a lift inside W leaves a terminal commutator unchanged, since $\mathfrak{m}^{c-1}W = 0$. Thus the chosen core H still has class c, and its abelianization is that quotient subspace. Write $W_0 = H'$. Its dimension is $w_0 + t$ for some $0 \le t \le h$, and it is a quotient of the universal module W_d retaining its top component.

Choose a functional on W_0 nonzero on that top component, and pull it back to $\phi \in W_d^*$. Normalize its highest component to have annihilator I. Write this component as λ, so the remaining dual degrees of $\phi - \lambda$ are at most $d-2$. Multiplication by $t_1 \cdots t_{d-1}$ kills those lower components and kills λ. Therefore $I\phi = 0$. The filtered contraction bound and this annihilator bound give $\dim_k R\phi = w_0$, hence $R\phi \simeq R/I$. Its quotient in W_0^* has dimension t. Dualizing gives

$$0 \longrightarrow F_t \longrightarrow W_0 \longrightarrow K \longrightarrow 0, \qquad 0 \longrightarrow W_0 \longrightarrow W \longrightarrow N_a \longrightarrow 0, \qquad a = h - t, \tag{98}$$

where $\dim_k F_t = t$ and $\dim_k N_a = a$. The local algebra R has residue field k; refining the two small modules by composition series gives the claimed filtration. In particular the full W is handled through these two extensions; only W_0 is required to be a quotient of W_d.

A j-dimensional R-module can be specified by d matrices of size j, so there are at most q^{dj^2} possible module types. For fixed F_t, the degree-one free rank in Lemma C.1 gives

$$\dim_k \mathrm{Ext}^1_R(K, F_t) \le tC_d.$$

The first extension in (98) is thus covered by $q^{dt^2+tC_d}$ choices. Since $\dim_k \mathrm{Ext}^1_R(k,k) = d$, composition series and the long exact sequence give

$$\dim_k \mathrm{Ext}^1_R(k, F_t) \le td, \qquad \dim_k \mathrm{Ext}^1_R(k, W_0) \le td + 1.$$

Applying the long exact sequence in its first argument to a composition series of N_a then gives

$$\dim_k \mathrm{Ext}^1_R(N_a, W_0) \le a(td+1).$$

Thus the second extension costs at most $q^{da^2+a(td+1)}$ choices. Summing over $t + a = h$, the total is bounded by

$$\sum_{t=0}^{h} q^{d(t^2+a^2)+tC_d+a(td+1)} \le (h+1)q^{hC_d+dh^2+h}.$$

These extension representatives cover all normalized module types. Every unmarked Lie isomorphism class admits a normalized presentation, so choices of normalization introduce no additional factor in this upper bound.

For the remaining bounds, dimensions of Hom_R are subadditive across a short exact sequence in either argument. The filtration factors are one K and h copies of k, and

$$\dim_k \mathrm{Hom}_R(K, k) = d - 1, \quad \dim_k \mathrm{Hom}_R(k, K) = 1, \quad \dim_k \mathrm{End}_R(K) = w_0.$$

Applying subadditivity in both arguments proves

$$\dim_k \mathrm{End}_R(W) \le w_0 + h(d-1) + h + h^2 = w_0 + hd + h^2.$$

The same filtration gives $\dim_k \mathrm{Soc}_R(W) = \dim_k \mathrm{Hom}_R(k, W) \le h + 1$. Ordinary CE cohomology is dimension-subadditive across its long exact sequence. Combining Lemma C.1 with $\dim H^1(U_0, k) = d$ and $\dim H^2(U_0, k) = C_d$ proves (97). □

Proposition C.3 (The fixed-surplus upper bound)**.** *Let $B_{c,r}(q)$ count metabelian 3-Engel k-Lie isomorphism classes of class c and dimension $\ell(c) + r$. For $r \ge 1$, put*

$$Q_r(d) = (2r+1)C_d + (r^2 + r + 1)d + r^3 + r + \binom{r}{2}(r+1).$$

Then

$$B_{c,r}(q) \le (r+1)^2 q^{rw_0 + Q_r(d)}. \tag{99}$$

More precisely, the branch with derived dimension $w_0 + h$, where $0 \le h \le r$ and $s = r - h$, contributes at most

$$(h+1)q^{sw_0 + (2h+1)C_d + d(h^2+2sh+s+1) + sh^2 + r + \binom{s}{2}(h+1)}. \tag{100}$$

In particular, for fixed q, r,

$$\log_q B_{c,r}(q) \le r2^{c-1} + O_{q,r}(c^2).$$

Proof. The separate minimum bounds in Lemma C.1 force quotient and derived dimensions

$$(d + s, w_0 + h), \qquad s + h = r, \quad s, h \ge 0.$$

Choose and normalize a d-generator core, then complete its quotient basis by $v_1, \dots, v_s$. For the fixed core R-module W, write $P_i = t_i \cdot$ and $T_a = \mathrm{ad}\, v_a|_W$. Metabelianity makes every T_a commute with all P_i, so $T_a \in \mathrm{End}_R(W)$. The action tuple therefore has at most

$$q^{s(w_0 + hd + h^2)}$$

choices. The commuting and square-zero conditions select a subset of these tuples.

The core bracket cocycle $\beta_{ij} = [e_i, e_j]$ is closed for the ordinary CE complex on U_0. Changing the core section by vectors of W adds a two-coboundary. By (97), the core cocycle can consequently be represented by at most $q^{d+(h+1)C_d}$ choices. Such section changes leave W, the P_i, and the T_a unchanged, so the module and action census continues to cover the resulting representatives.

Next put $A_{ia} = [e_i, v_a]$. The core/core/extra Jacobi equations are

$$P_i A_{ja} + P_j A_{ia} = T_a \beta_{ij}.$$

For fixed actions and core cocycle, two solutions for a given a differ by an ordinary one-cocycle. Replacing v_a by $v_a + w_a$, $w_a \in W$, adds the one-coboundary $(P_i w_a)_i$. Thus all mixed tuples can be represented by at most

$$q^{s((h+1)d+1)}$$

choices. These changes may also replace $[v_a, v_b]$ by $[v_a, v_b] + T_a w_b + T_b w_a$; those remaining brackets are counted after the mixed tuples have been fixed.

Write $Z_{ab} = [v_a, v_b]$. The core/extra/extra Jacobi equations give

$$P_i Z_{ab} = T_a A_{ib} + T_b A_{ia}.$$

Once the preceding data are fixed, two solutions for Z_{ab} differ by a vector in $\bigcap_i \ker P_i = \mathrm{Soc}_R(W)$. Therefore all extra pairs together have at most $q^{\binom{s}{2}(h+1)}$ choices. The remaining Jacobi equations and the generation and class conditions select a subset of the displayed data.

Multiplying these counts by (95) gives exponent

$$\begin{aligned} &hC_d + dh^2 + h + s(w_0 + hd + h^2) \\ &\qquad + d + (h+1)C_d + s((h+1)d + 1) + \binom{s}{2}(h+1) \\ &= sw_0 + (2h+1)C_d + d(h^2 + 2sh + s + 1) + sh^2 + r + \binom{s}{2}(h+1), \end{aligned}$$

which proves (100). Every Lie algebra under consideration admits one of these presentations; multiple choices of a core only enlarge this upper count. Finally

$$h^2 + 2sh + s + 1 \le r^2 + r + 1, \quad sh^2 \le r^3, \quad \binom{s}{2}(h+1) \le \binom{r}{2}(r+1),$$

and $\sum_{h=0}^{r}(h+1) \le (r+1)^2$. Summing the branches proves (99) and its logarithmic consequence. □

C.1 Proof of the transition scale

Write $n = 2^d$, $C_d = \binom{d}{2}$, $w_0 = n - 2$, and $D_{\mathrm{par}} = n - d - 3$. We use B_h for the branch with derived dimension $w_0 + h$, so $B_0 = B^{\min}_{d+1,r}(q)$ and $B_{d+1,r}(q) = \sum_{h=0}^{r} B_h$.

A uniform frame estimate. For $1 \le r \le D_{\mathrm{par}}$, Theorem 5.3 gives

$$\log_q B_{d+1,r}(q) \ge rn - d^2 - 2dr - r^2 - 3r - 1. \tag{101}$$

Indeed, after extracting $q^{rD_{\mathrm{par}}}$ from the frame numerator, its remaining product is at least

$$\prod_{j\ge 1}(1 - q^{-j}) \ge 1 - \sum_{j\ge 1} q^{-j} \ge \frac{2}{3}.$$

The denominator is at most $q^{dr+d^2+r^2}$, proving the estimate.

The minimum-derived branch below the transition scale. For $h \ge 1$, put $s = r - h$. The exponent in (100), apart from sw_0, is at most

$$h\big(3C_d + 3dr + 2r^2 + r\big).$$

Here $2h + 1 \le 3h$, $h^2 + 2sh + s + 1 \le 3rh$, $sh^2 \le r^2 h$, and $\binom{s}{2}(h+1) \le r^2 h$. Since $\log_q(h+1) \le h/2$, comparison with (101) gives, with

$$T = 3C_d + d^2 + 5dr + 3r^2 + 2r + 2,$$

the bound

$$\frac{B_h}{B_{d+1,r}(q)} \le q^{-h(w_0 - T)}. \tag{102}$$

When $a = w_0 - T > 0$, summing a geometric series yields

$$1 - \pi_{d,r}(q) \le \frac{q^{-a}}{1 - q^{-a}}. \tag{103}$$

If $r = o(\sqrt{n})$, then eventually $r \le D_{\mathrm{par}}$ and $T = o(n)$, so this tends to zero. Moreover the coarse upper bound (99) has an error $O(rd^2 + r^2 d + r^3)$. Together with (101), it gives $\log_q B_{d+1,r}(q) \sim rn$ in the same range.

A cubic lower family and a small characteristic quotient. Fix one minimum class-$d+1$ algebra L_0 from Lemma C.1. Its derived module is K, and

$$\dim_k\big(L_0/\gamma_3(L_0)\big) = d + \dim_k(K/\mathfrak{m}K) = 2d-1.$$

For $u = \lfloor 2r/3 \rfloor$ and $w = r-u$, each alternating map $b : \Lambda^2 k^u \to k^w$ defines a two-step algebra N_b on $k^u \oplus k^w$, with central second summand. No surjectivity condition is needed. All $q^{w\binom{u}{2}}$ labelled brackets give algebras $L_b = L_0 \oplus N_b$ counted by $B_{d+1,r}(q)$. Since $\gamma_3(N_b) = 0$,

$$L_b/\gamma_3(L_b) \simeq \big(L_0/\gamma_3(L_0)\big) \oplus N_b$$

has dimension $m = r+2d-1$. Distinct tensors give distinct brackets on this fixed labelled quotient space. Each quotient isomorphism class has at most $|\operatorname{GL}_m(q)|$ such labelled structures. Isomorphic L_b's have isomorphic characteristic quotients. Consequently,

$$B_{d+1,r}(q) \geq \frac{q^{w\binom{u}{2}}}{|\operatorname{GL}_{r+2d-1}(q)|}, \qquad \log_q B_{d+1,r}(q) \geq \frac{2}{27}r^3 - O\big((r+d)^2\big). \tag{104}$$

This argument uses the classical alternating-tensor count for two-step Lie brackets; see also the coordinate-explicit enumeration in [18]. It neither transfers a group-counting theorem to characteristic two nor assumes cancellation of direct summands.

The minimum-derived branch above the transition scale. Taking $h = 0$ in (100) gives, for every $r \geq 1$,

$$B_0 \leq q^{rw_0 + C_d + d(r+1) + r + \binom{r}{2}}. \tag{105}$$

If $r/\sqrt{n} \to \infty$, then $r \gg d$ and $n = o(r^2)$. Dividing (105) by (104) gives

$$\log_q \pi_{d,r}(q) \leq rn - \frac{2}{27}r^3 + O(r^2 + rd + d^2) = -\big(2/27 - o(1)\big)r^3.$$

This proves (62).

The uniform order of growth. The explicit upper bound in Proposition C.3 is $O(rn+r^3)$ on the logarithmic scale, since $d^2 = O(n)$ and $r^2 d \leq (r^3 + rd^2)/2$. For $r \leq \sqrt{n}$, (101) is at least a fixed positive multiple of rn, uniformly in r, once d is large. For $r \geq \sqrt{n}$, (104) is at least a fixed positive multiple of r^3, again uniformly in r. The larger of rn and r^3 is at least half their sum. This proves (60) and completes the proof of Theorem 6.1.

D Exact principal orbits and multiplicative enumeration

Let $k = \mathbb{F}_q$, $q \geq 4$, have characteristic two, and use $d, C_d, D_{\mathrm{par}}, w_0, R, K, I$ as in Section 5.3. Put $\tau = t_0 \cdots t_{d-1}$, $\tau_j = \prod_{i \neq j} t_i$, and $V_r = \langle v_1, \ldots, v_r \rangle_k$. The binary-map interpretation is used only at $q = 4$.

D.1 All ordinary cocycles

For each $p_a \in \mathcal{P}_d$, choose $a_{0a} = 0$, $a_{ia} \in \mathfrak{m}$, with

$$p_a = \sum_{i=1}^{d-1} t_i a_{ia}. \tag{106}$$

Assigning each monomial to its smallest positive index fixes this choice. Let $b = \eta\beta_\eta + \beta_\kappa + \beta_\lambda$ be the core representative from Lemma B.3. Use the core one-cocycles

$$\zeta_*(e_0) = 0, \quad \zeta_*(e_i) = t_0 t_i \ (i \geq 1), \qquad \zeta_j(e_i) = \mathbf{1}_{i=j}\tau_j \quad (0 \leq j < d).$$

Their classes form a basis of $H^1(U_0, K)$, where $U_0 = k^d$.

Lemma D.1. *For any tuple* $\mathbf{p} \in \mathcal{P}_d^r$,

$$\begin{aligned} H^2(U_0 \oplus V_r, K) &\simeq H^2(U_0, K) \oplus H^1(U_0, K) \otimes V_r^* \oplus H^0(U_0, K) \otimes \Lambda^2 V_r^*, \\ \dim H^2(U_0 \oplus V_r, K) &= d + C_d + r(d+1) + \binom{r}{2}, \\ \dim H^1(U_0 \oplus V_r, K) &= d + 1 + r. \end{aligned}$$

Explicit representatives are, with $\zeta^a \in \operatorname{span}\{\zeta_*, \zeta_0, \ldots, \zeta_{d-1}\}$ *and* $\theta_{ab} \in k$,

$$\begin{aligned} \beta(e_i, e_j) &= b(e_i, e_j), \\ \beta(e_i, v_a) &= \sum_j a_{ja} b(e_i, e_j) + \zeta^a(e_i), \\ \beta(v_a, v_b) &= \sum_{i<j} (a_{ia}a_{jb} + a_{ib}a_{ja}) b(e_i, e_j) + \sum_i (a_{ia}\zeta^b(e_i) + a_{ib}\zeta^a(e_i)) + \theta_{ab}\tau. \end{aligned} \tag{107}$$

These representatives are unique modulo coboundaries. Intrinsic generation is equivalent to $\eta \neq 0$.

Proof. In the ordinary Koszul cochain complex, replace e_i^* by $e_i^* + \sum_a a_{ia} v_a^*$, using (106). This triangular invertible R-linear change converts the differential $\sum_i t_i e_i^* \wedge + \sum_a p_a v_a^* \wedge$ to the core differential, with zero differential on $\Lambda(V_r^*)$. It is an operation on cochains, not a k-linear change of Lie quotient generators. Expanding the resulting representatives gives (107). The core cohomology bases and $H^0(U_0, K) = k\tau$ prove the dimensions and uniqueness. Every mixed coefficient has degree at least two, so modulo $\mathfrak{m}K$ only the core star remains. It spans $K/\mathfrak{m}K$ exactly when $\eta \neq 0$; apply Nakayama. □

Scalar multiplication of K normalizes $\eta = 1$. The minimum derived-module result covers every object with $h = 0$. Each extra action lies in $\operatorname{End}_R K = R/I$; square zero forces its constant term to vanish, and subtracting its linear part from the corresponding quotient generator gives a unique tuple in $\mathcal{P}_d^r$. Thus the normalized fibers cover the whole minimum-derived stratum.

D.2 The complete effective action

We use ordinary abelian-extension equivalence and section changes as in [3, Theorems 1.1–1.2]. The intrinsic kernel of $U \to \operatorname{rad}(R/I)/\operatorname{rad}(R/I)^2$ is V_r, whether or not the tuple is independent. Every isomorphism has quotient matrix $A_g = \left(\begin{smallmatrix} B & 0 \\ \alpha & \Delta \end{smallmatrix}\right)$, and action intertwining gives (56). The nilpotent implicit-function principle [1, Sections 1.2–1.3] also applies: iteration from zero increases the radical valuation of a difference, hence gives a unique target tuple. Its associated substitution has invertible linear part B.

Preserving $I = (z)$, $z = t_1 \cdots t_{d-1}$, in degree $d-1$ forces $B(H) = H$. Conversely, this condition suffices. The substitution σ_g is an automorphism of R and preserves K, since all p_a' lie in $\mathfrak{m}^2 \subset K$. In $\sigma_g(z)$, terms with at least two additional parameter factors have degree at least $d+1$ and vanish. The remaining terms give

$$\sigma_g(z) = (\det M)z + \xi\tau,$$

so $\sigma_g(I) = I$. No product between distinct parameters has been discarded where its degree could survive.

All module intertwiners are $u\sigma_g|_K$, $u \in (R/I)^\times$. Positive-degree multiplication is null-homotopic, by $\delta\iota_i + \iota_i\delta = t_i$. Consequently only the constant term u_0 acts on ordinary cohomology. The degree-one star transforms by u_0/a; its normalization forces $u_0 = a$. Section changes add precisely two-coboundaries. This proves that (57) is necessary and sufficient.

The formulas give a genuine group action. Under composition, two canonical R-lifts may differ by terms in I. Their maps on K intertwine the same automorphism of R/I, so their ratio is a module unit. Their leading maps agree, making its constant term one; its cohomological action is trivial. Thus the induced fiber maps compose as the actual quotient matrices, without assuming that the two R-lifts agree pointwise on K. This proves Theorem 5.7.

D.3 A summed fixed-point bound

Write

$$F_{d,r} = \prod_{j=0}^{r-1} (1 - q^{j - D_{\mathrm{par}}}),$$

which tends to one for fixed q, r. The frame base has size $q^{rD_{\mathrm{par}}} F_{d,r}$. All its cohomology fibers have the same cardinality, so a fixed-base bound also bounds the fixed proportion in $\mathcal{Y}^{\mathrm{fr}}_{d,r}$.

Suppose $(B, \Delta) \neq (I, I)$. The fixed equation is triangular by coefficient degree: in degree j its linear part is $\Lambda^j B \otimes I_r - I \otimes \Delta^T$, and all nonlinear terms depend on lower degrees. On the full exterior algebra tensored with k^r, put $T = \Lambda B \otimes I_r - I \otimes \Delta^T$. Then

$$T(L_v \otimes I_r) - (L_{Bv} \otimes I_r)T = L_{Bv - v} \otimes \Delta^T.$$

If $B \neq I$, this gives $\operatorname{rank} T \geq r2^{d-2}$. If $B = I$ but $\Delta \neq I$, then $\operatorname{rank} T \geq 2^d$. Removing degrees zero and one and quotienting by I loses at most $r(d+3)$. The common lower bound $2^{d-2} - r(d+3)$ therefore suffices. Since $|G_{d,r}| < q^{d^2 + rd + r^2 + 1}$, the sum of these fixed proportions is at most

$$F_{d,r}^{-1} q^{d^2 + rd + r^2 + 1 - 2^{d-2} + r(d+3)}. \tag{108}$$

It remains to control pure shears, where $B = I$, $\Delta = I$, and $\alpha \neq 0$. Let $k_0 = \operatorname{rank} \alpha \leq r$. For counting, enlarge $\mathcal{P}^r_d$ to $(\mathfrak{m}^2 R)^r$, adding $2r$ dimensions. Invertible linear changes of ring variables and parameter components put the displacement map into the form

$$(x_1, \ldots, x_{k_0}, y) \longmapsto (x_1 + p_1, \ldots, x_{k_0} + p_{k_0}, y).$$

These changes need not preserve I; it becomes a two-dimensional space I'. Set $S = k[y_1, \ldots, y_m]/(y_i^2)$, $m = d - k_0$, and expand

$$p_a = \sum_{T \subseteq \{1, \ldots, k_0\}} x_T b_{a,T}, \qquad a_a = b_{a,\varnothing} \in \mathfrak{m}_S^2.$$

A necessary condition for a fixed parameter is obtained by setting all active variables x_i to zero:

$$\sum_{T \neq \varnothing} \left(\prod_{i \in T} a_i \right) b_{a,T} \in \mathrm{pr}_S(I') \qquad (1 \leq a \leq r). \tag{109}$$

This is an exact specialization; it retains every product involving different parameters. The target space has dimension at most two.

Condition on all a_a. If some a_i, $1 \leq i \leq k_0$, has a nonzero quadratic coefficient, multiplication by a_i on S has rank at least 2^{m-2}: restrict inputs to variables outside a nonzero

quadratic monomial and project outputs to that monomial's full coefficient. The resulting map multiplies by a unit of the complementary ring. Restricting inputs to $b_{a,\{i\}} \in \mathfrak{m}_S$ loses at most one rank. Fixing all other coefficients in one equation of (109) therefore imposes codimension at least $2^{m-2} - 3$. The chosen a_i is held fixed independently of this coefficient block, so this is an affine-linear count with no independence assumption between the different fixed equations.

If all of $a_1, \dots, a_{k_0}$ have zero quadratic part, the enlarged parameter space has codimension $k_0\binom{m}{2}$. For $d \geq r + 2$, use $m \geq d - r$ and sum over fewer than q^{rd} nonzero matrices α. The total pure-shear fixed proportion is at most

$$F_{d,r}^{-1} q^{rd+2r} \left(q^{3-2^{d-r-2}} + q^{-\binom{d-r}{2}} \right). \tag{110}$$

For fixed q, r, both (108) and (110) tend to zero. They bound the sum of all nonidentity fixed proportions, not only a probability of nonfree points. Burnside's lemma gives

$$|\mathcal{Y}_{d,r}^{\mathrm{fr}}/G_{d,r}| = \frac{q^{rD_{\mathrm{par}}} F_{d,r}\, q^{f_{d,r}}}{|G_{d,r}|}(1 + o(1)).$$

Nonfree parameter and orbit proportions also tend to zero, by the same stabilizer-sum argument as in Corollary 3.23.

D.4 The whole layer and its typical automorphism order

Rank-deficient tuples number at most $q^{(r-1)D_{\mathrm{par}}+r(r-1)}$; multiplying by the entire fiber size costs only $q^{O_r(d^2)}$ beyond this exponent. The branches with $h \geq 1$ likewise contribute at most $q^{(r-1)2^d+O_{q,r}(d^2)}$, by Proposition C.3. Both are negligible compared with the displayed orbit main term. Substituting the group order proves (58) for all objects in the layer.

An automorphism inducing the identity on U has a principal-unit module block, with q^{w_0-1} choices. Every such unit lifts, because it acts trivially on H^2. Its section blocks form a torsor under $Z^1(U, K)$, whose dimension is

$$(w_0 - 1) + (d + 1 + r) = w_0 + d + r.$$

Thus each faithful minimum-derived object satisfies

$$|\operatorname{Aut}_k(L_y)| = q^{2w_0+d+r-1} |\operatorname{Stab}_{G_{d,r}}(y)|.$$

Generic freeness and the negligible complementary strata prove (59). At $q = 4$, apply the isomorphism correspondence of Theorem 2.6.

E The critical constant and coupled derived defects

The action-space normalization uses the standard Gaussian enumeration of subspaces over finite fields; see, for example, [18, Section 2]. Ordinary extension equivalence is used in the form of a module intertwiner together with cocycles modulo coboundaries [3, Theorems 1.1–1.2]. The large principal-unit kernel follows from the usual contraction identity in the Chevalley–Eilenberg complex: if T_i is the action of e_i, then $\delta\iota_{e_i} + \iota_{e_i}\delta = T_i$. The uniform comparison here combines these tools with the intrinsic near-minimum module and a coupled one-dimensional central-defect family. In particular, the critical constant comes from comparing complete isomorphism-type counts.

E.1 Action subspaces and the lower side of the critical constant

Put $n = 2^d$, $w_0 = n-2$, $C_d = \binom{d}{2}$, and $D_{\mathrm{par}} = n-d-3$. Let B_h count the class-$d+1$ objects of total dimension $d + w_0 + r$ with derived dimension $w_0 + h$, and write $s = r - h$. All counts are unlabelled k-linear isomorphism counts, where $k = \mathbb{F}_q$ has characteristic two and $q \geq 4$.

Lemma E.1 (Normalizing the action subspace)**.** *Fix a normalized core R-module W in the h-branch, with core actions $P_0, \dots, P_{d-1}$. These actions are linearly independent. Set*

$$E = \mathrm{End}_R(W), \qquad S_0 = \mathrm{span}_k\{P_0, \dots, P_{d-1}\}.$$

The possible tuples of s additional actions, up to changes of the additional quotient basis and addition of core quotient directions, have a covering set of size at most

$$\sum_{j=0}^{\min(s,L)} \begin{bmatrix} L \\ j \end{bmatrix}_q, \qquad L = w_0 + hd + h^2 - d.$$

If $2s \leq L$, this is at most $\frac{3}{2}(s+1)q^{s(L-s)}$.

Proof. The filtration in Lemma C.2 has one factor K. A linear relation among the P_i on W would hold on that factor, contradicting the absence of a nonzero linear form in $\mathrm{Ann}_R K = I$. Thus $\dim S_0 = d$, and $\dim E/S_0 \leq L$.

For each subspace $F \subset E/S_0$, choose once and for all an ordered basis and lifts to E. Given an actual tuple $(T_1, \dots, T_s)$, let F be the span of its images and put $j = \dim F$. A change in $\mathrm{GL}_s(k)$ makes those images the chosen basis followed by zeros. Adding suitable core directions to the additional quotient generators then makes their actual actions equal to the chosen lifts followed by zeros. This is an invertible block change of quotient coordinates fixing the core. It keeps the R-module W and all core actions unchanged.

The brackets transform along with these coordinates. Once the actions have been normalized, the same core, mixed, and extra-pair cocycle covering used in Proposition C.3 applies. Some preselected lifts may give no admissible Lie algebra; including them only enlarges an upper bound. The argument chooses presentations and never divides a parameter count by a group with unknown stabilizers.

There are $\begin{bmatrix} L \\ j \end{bmatrix}_q$ subspaces of dimension j in an L-dimensional space. The usual product formula and $\prod_{i \geq 1}(1 - q^{-i}) \geq 2/3$ give $\begin{bmatrix} L \\ j \end{bmatrix}_q \leq \frac{3}{2} q^{j(L-j)}$. If $2s \leq L$, the exponent increases for $0 \leq j \leq s$, proving the last assertion. □

In particular, replacing the raw action-tuple census in Proposition C.3 by this covering gives

$$B_h \leq \tfrac{3}{2}(h+1)(s+1)q^{\,sw_0 + (2h+1)C_d + d(h^2+2sh+1) + sh^2 + r + \binom{s}{2}(h+1) - s^2}, \qquad 2s \leq w_0 + hd + h^2 - d. \tag{111}$$

The core section changes leave the actions fixed, and the later changes of additional lifts affect only the remaining brackets, as in that proposition. Thus the new normalization does not invalidate the successive ordinary-cohomology quotients used there.

Lemma E.2 (The minimum-derived count at growing surplus)**.** *Uniformly for $1 \leq r \leq D_{\mathrm{par}}/2$,*

$$\log_q B_0 = rn - \tfrac{1}{2}r^2 + O(rd + d^2). \tag{112}$$

More explicitly,

$$\log_q B_0 \geq rn - \tfrac{1}{2}r^2 - dr - C_d - \tfrac{5}{2}r - 3,$$
$$\log_q B_0 \leq rn - \tfrac{1}{2}r^2 + C_d + d - \tfrac{5}{2}r - 1 + \log_q\big(\tfrac{3}{2}(r+1)\big).$$

Proof. For $W = K$, subtracting the linear parts of the additional actions places them in the D_{par}-dimensional space $\mathcal{P}_d$. The standard Koszul elimination of redundant generators gives

$$H^2(U_0 \oplus k^r, K) \simeq H^2(U_0, K) \oplus \big(H^1(U_0, K) \otimes (k^r)^*\big) \oplus \big(\mathrm{Soc}_R(K) \otimes \Lambda^2(k^r)^*\big).$$

Indeed, write each $p_a = \sum_i t_i a_{ia}$ with $a_{ia} \in \mathfrak{m}$, and change cochain generators by $e_i^* \mapsto e_i^* + \sum_a a_{ia} v_a^*$. All mixed coefficients have degree at least two, so intrinsic generation is still exactly the nonzero core-star condition. After normalizing that coefficient to one, every action tuple has q^{f_r} cocycle classes, where

$$f_r = d + C_d + r(d+1) + \binom{r}{2} - 1.$$

Apply the action-subspace normalization directly in $\mathcal{P}_d$:

$$B_0 \le q^{f_r} \sum_{j=0}^{r} \begin{bmatrix} D_{\mathrm{par}} \\ j \end{bmatrix}_q \le \tfrac{3}{2}(r+1) q^{f_r + r(D_{\mathrm{par}} - r)}.$$

This is the stated upper bound.

For the lower bound take every independent ordered r-frame in $\mathcal{P}_d$, together with all its normalized cocycle classes. The vector implicit-recovery equation in Theorem 5.3 makes the target frame unique for each fixed quotient matrix. There are at most $q^{dr} |\mathrm{GL}_d(q)||\mathrm{GL}_r(q)|$ such matrices. For a fixed quotient matrix, any two module intertwiners differ by a unit of R/I. Principal units act trivially on ordinary cohomology, by the core contraction homotopies. Thus there are at most $q-1$ different effects on the target cocycle class. This is a fiber bound; it requires neither a free action nor an explicit full orbit classification. Consequently

$$B_0 \ge \frac{q^{f_r} \prod_{j=0}^{r-1} (q^{D_{\mathrm{par}}} - q^j)}{(q-1) q^{dr} |\mathrm{GL}_d(q)||\mathrm{GL}_r(q)|}.$$

The frame product is at least $\frac{2}{3} q^{r D_{\mathrm{par}}}$, while the denominator is at most $q^{dr + d^2 + r^2 + 1}$. Taking logarithms and using $\log_q(2/3) \ge -1$ gives the stated lower bound. □

Proposition E.3 (Minimum-derived dominance below the constant)**.** *Put*

$$Z(d, r) = 2d^2 + 3dr + 2r + 6.$$

For $1 \le r \le D_{\mathrm{par}}/2$ and $1 \le h \le r$,

$$\frac{B_h}{B_0} \le q^{\,h[-n + \frac{1}{2}r^2 - \frac{1}{2}h^2 + Z(d,r)]}. \tag{113}$$

In particular, if $r/\sqrt{n} \to \alpha < \sqrt{2}$ as $d \to \infty$, then $B_0 / \sum_h B_h \to 1$.

Proof. The condition $r \le D_{\mathrm{par}}/2$ implies the condition on s in (111) for every h. Subtract the lower exponent in Lemma E.2 from the exponent there, and substitute $s = r - h$. After removing $h(-n + r^2/2 - h^2/2)$, the remainder is

$$d^2 h + d^2 - dh^2 + 2dhr - dh + dr + \tfrac{1}{2}hr + \tfrac{5}{2}h + r + 3.$$

For $h \ge 1$, dropping the negative terms bounds this by $h(2d^2 + 3dr + 3r/2 + 11/2)$. Also $\log_q[\frac{3}{2}(h+1)(s+1)] \le (r+1)/2$, since $\log_q(t+1) \le t/2$ for nonnegative integers t. Absorbing this into $hZ(d, r)$ proves (113).

If $a = n - r^2/2 - Z(d, r) > 0$, a geometric sum gives

$$1 - \frac{B_0}{\sum_h B_h} \le \sum_{h \ge 1} \frac{B_h}{B_0} \le \frac{q^{-a}}{1 - q^{-a}}.$$

For the stated sequence, $a/n \to 1 - \alpha^2/2 > 0$. No monotonicity assumption on the population proportions is used. □

E.2 A coupled central defect and the upper side of the constant

Retain the notation of Appendix E.1: $n = 2^d$, $w_0 = n-2$, $C_d = \binom{d}{2}$, and $D_{\mathrm{par}} = n-d-3$, with $d \geq 4$ and $k = \mathbb{F}_q$ of characteristic two, $q \geq 4$. Here B_h counts the class-$d+1$ objects of total dimension $d + w_0 + r$ and derived dimension $w_0 + h$. Put $A = R/I$, and write $\tau = t_0 \cdots t_{d-1}$. All cohomology below is ordinary Chevalley–Eilenberg cohomology of an abelian quotient, with its specified module action.

Proposition E.4 (One coupled derived defect)**.** *For $s = r-1$ with $2 \leq s \leq D_{\mathrm{par}}$, put*

$$f_s = d + C_d + s(d+1) + \binom{s}{2} - 1.$$

Then

$$B_1 \geq \frac{q^{f_s} \prod_{j=0}^{s-1} (q^{D_{\mathrm{par}}} - q^j)\big(q^{\binom{s}{2}} - 1\big)}{q^{ds+d+2} |\operatorname{GL}_d(q)|\,|\operatorname{GL}_s(q)|}. \tag{114}$$

Every algebra in the family used for this lower bound is indecomposable. In particular,

$$\log_q B_1 \geq sn - ds - C_d - d - 3s - 4. \tag{115}$$

For $3 \leq r \leq D_{\mathrm{par}}/2$, it follows that

$$\frac{B_0}{B_1} \leq q^{\,n - r^2/2 + dr + d^2 + r/2 + \log_q(\frac{3}{2}(r+1))}. \tag{116}$$

Proof. Apply Proposition F.1 with $h = 1$, restricting its central cocycle to $\Lambda^2 V_s$ and setting all other central coefficients to zero. A nonzero form on that subspace gives exactly the earlier coupled slice. Its complete isomorphism fibre is bounded by $(q-1)^2 q^{ds+d} |\operatorname{GL}_d(q)||\operatorname{GL}_s(q)|$, which is at most the denominator in (114). The indecomposability proof in that proposition applies to this subfamily. Extracting the frame product and the nonzero-form factor, using bounds 2/3 and 3/4, gives (115). Subtracting it from the upper bound in Lemma E.2 gives (116). The general construction retains further central cocycles; its stronger bound is used in Corollary E.7. □

Remark E.5 (Indecomposability of the binary solutions). At $q = 4$, the corresponding binary quadratic solutions are indecomposable under binary-linear conjugacy. Indeed, suppose a solution is conjugate to $f_1 \oplus f_2$ on two nonzero binary subspaces. Its polar Lie algebra splits accordingly, so both factors are metabelian. The intrinsic derived ideal, quotient, and their scalar actions also split. In the proof of Theorem 2.6, choose the bases and normalized-section lifts separately in the two factors. Scalar-weight projection, the actions P_i, and the cocycle refinement Q then preserve these blocks. The auxiliary bracket (15) is consequently the direct sum of the two auxiliary Lie algebras, contradicting Proposition E.4. A nonzero linear factor gives a nonzero abelian auxiliary factor and is excluded by the same argument. This statement concerns direct sums of solutions. It does not assert indecomposability of the original polar Lie algebra, nor require an extension of the correspondence to arbitrary homomorphisms.

E.3 Principal orbits when the surplus grows

Fix the field $k = \mathbb{F}_q$, $q \geq 4$, of characteristic two. Retain $d, n = 2^d, D_{\mathrm{par}} = n-d-3, w_0 = n-2$, and write $P = \operatorname{rad}(R/I)^2$, a vector space of dimension D_{par}. Let B_0 count the minimum-derived stratum in dimension $d + w_0 + r$. The exact orbit model of Theorem 5.7 remains valid for every r.

Lemma E.6 (Pure shears through invariant subspaces)**.** *Let $\mathcal{F}_{d,r}$ be the ordered independent r-frames in P, and let N be the group of automorphisms of R/I whose linear part is the identity. For $\sigma \in N$, set*

$$U_\sigma = \operatorname{span}_k\{\sigma(t_i) - t_i : 0 \le i < d\}, \qquad F_\sigma = \ker(\sigma - 1 : P \to P).$$

Then the summed fixed proportion of all nonidentity pure shears is

$$\sum_{\alpha \neq 0} \frac{|\operatorname{Fix}_{\mathcal{F}_{d,r}}(\alpha)|}{|\mathcal{F}_{d,r}|} = \sum_{\substack{\sigma \in N \\ \sigma \neq 1}} \frac{|\{Q \le P : \dim Q = r,\ U_\sigma \le Q \le F_\sigma\}|}{\begin{bmatrix} D_{\mathrm{par}} \\ r \end{bmatrix}_q}. \tag{117}$$

Proof. Every choice $u_i \in P$ gives an automorphism $t_i \mapsto t_i + u_i$ of R/I. To see the lifting assertion, use the canonical representatives in $\mathfrak{m}^2 R$. Their substitution has identity linear part, hence is an automorphism of R. It preserves K; the image of $z = t_1 \cdots t_{d-1}$ lies in $kz + k\tau = I$, since terms with two positive-degree increments have degree at least $d+1$. Thus it preserves I, and the induced map is determined by the u_i's. Conversely every member of N has these increments.

For a fixed ordered frame $\mathbf{p}$, the coefficients of $u_i = \sum_a \alpha_{ai} p_a$ are unique. The pure-shear base equation is exactly $\sigma(p_a) = p_a$ for every a. Hence fixed pairs $(\alpha, \mathbf{p})$ correspond to pairs $(\sigma, \mathbf{p})$ with $U_\sigma \le \operatorname{span}(\mathbf{p}) \le F_\sigma$. The number of ordered bases is the same for every r-subspace. Dividing the pair count gives (117). □

For $2 \le j \le d-1$, put

$$a_j = \dim(P/\operatorname{rad}(R/I)^j), \qquad \rho_j = \max\{0, 2^{d-j-1} - d - 3\}.$$

If j is the least order of a nonzero increment of σ, then

$$\operatorname{rank}(\sigma - 1 : P \to P) \ge \rho_j. \tag{118}$$

Indeed lift σ to R, write $T = \sigma - 1$, and use

$$TL_{t_i} - L_{\sigma(t_i)}T = L_{\sigma(t_i) - t_i}.$$

An increment of order j has multiplication rank at least 2^{d-j}: restrict inputs to the variables complementary to one nonzero degree-j monomial and project to its full coefficient; the resulting multiplier has nonzero constant term. The commutator therefore gives $\operatorname{rank} T \ge 2^{d-j-1}$. Restricting to $\mathfrak{m}^2 R$ loses at most $d+1$ rank, and passage to the quotient by I loses at most two more. This proves (118).

If $k_0 = \dim U_\sigma$, the number of possible increment tuples of order at least j and rank k_0 is at most

$$\tfrac{3}{2} q^{k_0(D_{\mathrm{par}} - a_j + d - k_0)}.$$

For each actual tuple the Gaussian ratio in (117) is zero unless $U_\sigma \le F_\sigma$; otherwise it is at most

$$\tfrac{3}{2} q^{-k_0(D_{\mathrm{par}} - r) - (r - k_0)\rho_j}.$$

Thus the contribution from order j, rank k_0, is at most

$$\tfrac{9}{4} q^{k_0(r - a_j + d - k_0) - (r - k_0)\rho_j}. \tag{119}$$

This estimates the sum of stabilizers, rather than only the probability of a nontrivial stabilizer.

Let $j_0 = \lfloor d/2 \rfloor$, $a_0 = a_{j_0}$, and $\rho_0 = \rho_{j_0}$. Symmetry and the central-binomial estimate give $a_0 \sim n/2$, while $\rho_0 \geq \sqrt{n}/2 - d - 3$. For $r \geq 2d$, summing (119) separately over $j < j_0$ and $j \geq j_0$ gives the uniform bound

$$S_{\text{pure}} \leq \tfrac{9}{4} d^2 \left(q^{d(r+d)-(r-d)\rho_0} + q^{r+d-a_0} \right). \tag{120}$$

For the first group, drop $-k_0 a_j - k_0^2$ and use $k_0 \leq d$, $\rho_j \geq \rho_0$. For the second group use $a_j \geq a_0$, drop the nonpositive rank term, and use $k_0(r - a_0 + d - k_0) \leq r + d - a_0$ once $a_0 > r + d$. In particular (120) tends to zero uniformly on

$$2d \leq r \leq D_{\text{par}}/(4d). \tag{121}$$

For completeness the nonpure linear parts admit a bound on the same range. Put $F_{d,r} = \prod_{i<r}(1 - q^{i-D_{\text{par}}})$, which tends to one uniformly there. For $B \neq I$, retain the stronger exterior rank $r2^{d-2}$ in Appendix D. The resulting sum is at most

$$F_{d,r}^{-1} q^{d^2+rd+r^2+1-r2^{d-2}+r(d+3)}. \tag{122}$$

For $B = I$, put $t = \operatorname{rank}(\Delta - I)$. In every retained graded monomial the diagonal linear block is $\Delta^T - I$, so the total rank is $D_{\text{par}} t$. There are at most $\frac{3}{2} q^{t(2r-t)}$ such matrices Δ. Including all q^{rd} shears and summing over $t \geq 1$ gives

$$S_\Delta \leq \tfrac{3}{2} r F_{d,r}^{-1} q^{rd+2r-1-D_{\text{par}}}. \tag{123}$$

Both displays tend to zero uniformly in (121).

Proof of Theorem 6.4. Every normalized cohomology fiber has size $q^{f_{d,r}}$, so each fixed-base proportion also bounds the corresponding fixed proportion in the whole frame-fiber space. Equations (120)–(123) show that the summed nonidentity fixed proportion is $o(1)$, uniformly. Burnside therefore gives the frame count divided by $|G_{d,r}|$ with multiplicative error $1 + o(1)$, and makes nonfree orbit proportions negligible by the same stabilizer-sum argument.

It remains to cover rank-deficient action tuples. Normalizing their action subspace, while retaining all r quotient directions and all $q^{f_{d,r}}$ cocycles, bounds their class count by

$$q^{f_{d,r}} \sum_{j<r} \begin{bmatrix} D_{\text{par}} \\ j \end{bmatrix}_q \leq \tfrac{3}{2} r q^{f_{d,r}+(r-1)(D_{\text{par}}-r+1)}.$$

Its ratio to the faithful orbit main term is at most

$$\tfrac{3}{2} r F_{d,r}^{-1} q^{-D_{\text{par}}+rd+d^2+2r} = o(1),$$

uniformly in (121). This proves (65). The exact full-automorphism kernel calculation in Appendix D is valid for every faithful frame, so generic effective freeness gives the stated order. □

The independent subcritical estimate of Proposition E.3 transfers (65) to the entire dimension layer on the subrange $2d \leq r \leq \sqrt{2n} - 3d - 3$. No such whole-layer conclusion is made above the critical constant.

Corollary E.7 (A sharper upper-side localization)**.** *If $r \geq \sqrt{2n} - d + 1$ eventually, then $\pi_{d,r}(q) \to 0$. In the portion of* (121) *with $r \geq 3$, put*

$$T = \binom{d+r-1}{2} - (d-1).$$

The full-central-cocycle coupled family gives

$$\pi_{d,r}(q) \leq (1 + o(1)) \frac{(q^{D_{\text{par}}} - q^{r-1})(q-1)q^2(q^d-1)}{(q^r-1)(q^T-1)}. \tag{124}$$

Proof. Divide (65) by the independent coupled lower bound (125), with $h = 1$ and $s = r - 1$. Its full central-cocycle count is $q^{d-1}(q^T - 1)$; its effective module-block bound is $(q-1)^2 q^d$. Use $|\operatorname{GL}_m|/|\operatorname{GL}_{m-1}| = q^{m-1}(q^m - 1)$. This gives (124). The logarithm of its right-hand side is

$$n - \tfrac{1}{2}(r+d)^2 + \tfrac{1}{2}(r+d) + 2d - 3 + \log_q(q-1) + o(1).$$

The quadratic expression decreases for $r+d \geq 1$. At $r = \sqrt{2n} - d + 1$ it equals $-\sqrt{2n}/2 + 2d - 3$, which tends to minus infinity. This proves the assertion on the indicated range. On all remaining indices with $r > D_{\mathrm{par}}/(4d)$, one has $r/\sqrt{n} \to \infty$; invoke (62). □

F Coupled central defects and the postcritical population

Fix $k = \mathbb{F}_q$ of characteristic two, $q \geq 4$, and retain $R, K, I, A = R/I$ from Appendix E. Put $d = c - 1 \geq 4$, $n = 2^d$, $w_0 = n - 2$, $C_d = \binom{d}{2}$, $D_{\mathrm{par}} = n - d - 3$, and $\tau = t_0 \cdots t_{d-1}$. Write B_h for the branch of $B_{d+1,r}(q)$ with derived dimension $w_0 + h$, and put $s = r - h$. Ordinary cohomology always means the Chevalley–Eilenberg cohomology for the specified abelian quotient action. No restricted cohomology is used.

F.1 A family with arbitrary central defect

Proposition F.1 (The coupled central-defect family)**.** *Suppose* $1 \leq s \leq D_{\mathrm{par}}$, $r = s + h$, *and*

$$T_{d,s} = \binom{d+s}{2} - (d-1), \qquad 1 \leq h \leq T_{d,s}.$$

Set

$$f_s = d + C_d + s(d+1) + \binom{s}{2} - 1.$$

There is an indecomposable family in the branch B_h *giving*

$$B_h \geq \frac{q^{f_s + h(d-1)} \prod_{j=0}^{s-1} (q^{D_{\mathrm{par}}} - q^j) \prod_{j=0}^{h-1} \left(q^{T_{d,s}} - q^j\right)}{(q-1) q^{dr} |\operatorname{GL}_d(q)| |\operatorname{GL}_s(q)| |\operatorname{GL}_h(q)|}. \tag{125}$$

Consequently the following estimate is uniform in this range:

$$\log_q B_h \geq sn + h\binom{d+s}{2} - \tfrac{1}{2}s^2 - dr - C_d - h^2 - \tfrac{5}{2}s - 3. \tag{126}$$

At $q = 4$, *the corresponding binary quadratic solutions are indecomposable under binary-linear conjugacy.*

Proof. Let $Z = k^h$, $W = K \oplus Z$, and $U = U_0 \oplus V_s$, where $U_0 = k^d$ has basis $e_0, \ldots, e_{d-1}$. Choose an ordered independent s-frame $\mathbf{p} = (p_1, \ldots, p_s)$ in $\mathcal{P}_d$. On K, the core and added quotient generators act by t_i and p_a, respectively; they kill Z. Take every K-valued ordinary two-cohomology class with normalized core-star coefficient one. The complete Koszul decomposition of Lemma D.1 shows that there are exactly q^{f_s} such classes, retaining all core, mixed, and extra-pair coefficients.

Write $\mathcal{N} \subset \Lambda^2 U$ for the span of all basis pairs except the $d-1$ star pairs $e_0 \wedge e_i$, $1 \leq i < d$. Thus $\dim \mathcal{N} = T_{d,s}$. Independently choose an alternating map

$$\gamma : \Lambda^2 U \longrightarrow Z$$

whose restriction to $\mathcal{N}$ is surjective; its star values are arbitrary. It is a Z-valued cocycle because the quotient acts trivially on Z. The combined cocycle defines an abelian Lie extension of U by W.

To check its derived ideal, the specified action already gives $[U, W] = \mathfrak{m}K \subset L'$, as a pure submodule, without first assuming that $W = L'$. Every nonstar K-component in the complete cocycle formula (107) lies in $\mathfrak{m}K$: the nonstar core terms have degree at least two, including the λ-terms of degree $d - 2 \geq 2$; mixed terms have degree at least two; extra-pair terms have degree at least three, apart from their degree-d socle term. Subtracting these pure $\mathfrak{m}K$-components from nonstar brackets gives all values of $\gamma|_{\mathcal{N}}$ as pure central vectors. Its surjectivity supplies all of Z. The central star components may then be subtracted. Their K-components q_i satisfy $q_i \equiv t_i \pmod{\mathfrak{m}K}$, and hence generate all of K by Nakayama. More generally, for any central map γ in this slice,

$$\dim_k L' = w_0 + \operatorname{rank}(\gamma|_{\mathcal{N}}).$$

Indeed, modulo $[U, W] = \mathfrak{m}K$, the star brackets have independent $K/\mathfrak{m}K$-components, whereas every nonstar K-component is zero. The bracket image therefore has dimension $d - 1 + \operatorname{rank}(\gamma|_{\mathcal{N}})$ in that quotient. Thus surjectivity on $\mathcal{N}$ is precisely the condition for $L' = W$, independently of the star values. For our surjective choice, from the third term onwards the lower central series is $\gamma_j(L) = \mathfrak{m}^{j-2}K$, so its class is $d + 1$. The commuting quotient actions square to zero, and the extension is metabelian 3-Engel. Its dimension is $d + s + w_0 + h = d + w_0 + r$.

We bound complete isomorphism fibres rather than assume a free parameter action. The action algebra on W is faithfully A, and the quotient action itself is injective: the degree-one terms detect U_0, and independence of the p_a's detects V_s. The kernel of $U \to \operatorname{rad}(A)/\operatorname{rad}(A)^2$ is intrinsically V_s. Thus an isomorphism between two members has a quotient matrix

$$\begin{pmatrix} B & 0 \\ \alpha & \Delta \end{pmatrix}, \qquad B \in \mathrm{GL}_d(k), \quad \Delta \in \mathrm{GL}_s(k).$$

There are at most $q^{ds} |\mathrm{GL}_d(q)||\mathrm{GL}_s(q)|$ such matrices. For each fixed matrix, the vector implicit-recovery equation (53) fixes at most one target frame. It uses only the faithful algebra A and the radical degree of the added actions, so adding the trivial summand Z does not alter its conclusion.

The full module automorphism group has blocks

$$\operatorname{Aut}_R(K \oplus Z) = \left\{ \begin{pmatrix} u & \tau v \\ \lambda & C \end{pmatrix} : u \in A^\times,\ v \in k^{1\times h},\ \lambda \in \operatorname{Hom}_R(K, Z),\ C \in \mathrm{GL}_h(k) \right\}.$$

Indeed, $\operatorname{Hom}_R(Z, K)$ consists of socle-valued maps, and $\dim \operatorname{Hom}_R(K, Z) = h(d-1)$. Since $\tau \in \mathfrak{m}K$, one has $\lambda(\tau) = 0$. Every automorphism must have $u_0 \neq 0$: on $W/\mathfrak{m}W = (K/\mathfrak{m}K) \oplus Z$, its upper-right block is zero and its upper-left block is $u_0 I_{d-1}$. For a unit u, the Schur complement is just C, since $u^{-1}\tau = u_0^{-1}\tau$ and $\lambda(\tau) = 0$. This proves the displayed block description and gives

$$|\operatorname{Aut}_R(W)| = (q-1)|\mathrm{GL}_h(q)|q^{w_0-1+hd}.$$

The subgroup $\operatorname{diag}(1 + a, I_h)$, $a \in \operatorname{rad}(A)$, has order q^{w_0-1} and acts trivially on all of $H^2(U, W)$. To see this, lift $a = \sum_i t_i b_i$ in $\mathfrak{m}R$ and use the ordinary contraction identity

$$\delta\Big(\sum_i b_i \iota_{e_i}\Big) + \Big(\sum_i b_i \iota_{e_i}\Big)\delta = a.$$

On W, positive-degree multiplication is the map $\operatorname{diag}(a, 0)$. Hence each displayed principal unit differs from the identity by a null-homotopic map. The effective module effects on cohomology number at most $(q-1)|\mathrm{GL}_h(q)|q^{hd}$.

For a fixed quotient matrix and target frame, all intertwiners differ by an element of $\mathrm{Aut}_R(W)$, and section changes are two-coboundaries. Each effective module effect gives at most one target cohomology class. The selected K-valued classes and Z-valued surjections inject into $H^2(U,K) \oplus H^2(U,Z)$, so each such class represents at most one member of the chosen family. An effect need not preserve our slice: an image outside it contributes no target. This bounds every complete isomorphism fibre by the denominator in (125). The number of surjections $k^{T_{d,s}} \to k^h$ is the second product in its numerator; the first product counts independent action frames. The arbitrary star values contribute $q^{h(d-1)}$. This proves the exact lower bound.

Every family member is indecomposable. The endomorphism ring $\mathrm{End}_R K = A$ is local, so K is indecomposable and is not isomorphic to the trivial module k. If $L = L_1 \oplus L_2$, then $W = L_1' \oplus L_2'$ as R-modules. By finite-length Krull–Schmidt, at most one derived summand contains the indecomposable factor K; any other nonzero derived summand is a sum of trivial modules. All core and polynomial extra actions kill that summand. Its Lie factor also kills the other derived summand by the direct-sum condition. Faithfulness of L/W then puts the whole factor inside the abelian ideal W, contradicting its nonzero derived ideal. If one derived summand is zero, its Lie factor kills W; faithfulness puts it inside W, which lies in the other factor, so it is zero. At $q = 4$, suppose the corresponding quadratic solution were conjugate to $f_1 \oplus f_2$ on two nonzero binary subspaces. Its derived ideal, quotient, and their intrinsic scalar actions split accordingly. Choose the bases and normalized-section lifts block by block in Theorem 2.6. The scalar-weight projections, operators P_i, and quadratic refinement Q preserve these blocks, so the resulting auxiliary Lie algebra is the direct sum of their two nonzero auxiliary algebras. This contradicts the Lie indecomposability just proved. A nonzero linear factor likewise gives a nonzero abelian auxiliary factor and is excluded. This proves indecomposability of the quadratic solution; it makes no assertion about arbitrary decompositions of its original polar Lie algebra.

Finally, both products after extracting their leading powers of q are at least $\prod_{j\ge1}(1 - q^{-j}) \ge 2/3$. Use $(q-1) < q$ and $|\mathrm{GL}_j(q)| \le q^{j^2}$. Substituting $D_{\mathrm{par}} = n - d - 3$ and f_s gives (126), with the constant -3 covering the extracted product factors. For $h = 1$, $s = r - 1$, the resulting bound is $\log_q B_1 \ge sn - 3s - d - 4$. Relative to the extra-pair-only family, the larger central cocycle space contributes the factor q^{ds+C_d} at leading order. □

G Socle-visible actions and the exact leading population law

The residual action entropy in the branch upper bound can be coupled to a loss of ordinary two-cohomology. This coupling closes the leading-order comparison of Appendix F for all bounded values of $r/2^{d/2}$, including the supercritical regime. Fix $k = \mathbb{F}_q$ of characteristic two, $q \ge 4$, and retain $R, K, d, n = 2^d, w_0 = n - 2$ and $C_j = \binom{j}{2}$. Write B_h for the branch with derived dimension $w_0 + h$ and quotient dimension $d + s$, where $h + s = r$. All cohomology in this appendix is ordinary Chevalley–Eilenberg cohomology of the indicated abelian quotient. The one-direction cohomology calculation is the classical codimension-one exact sequence of [7, Proposition 1].

G.1 Restriction to the core socle

Lemma G.1 (The socle-restriction kernel)**.** *Let W be any normalized bounded-defect core module supplied by Lemma C.2. Put*

$$S = \mathrm{Soc}_R(W), \qquad m = \dim_k S, \qquad t = h + 1 - m.$$

Then $1 \le m \le h+1$, and the restriction map $\mathrm{End}_R(W) \to \mathrm{End}_k(S)$ has kernel of dimension at most

$$e_{h,m} = w_0 + h(d-1) + tm. \tag{127}$$

Proof. Use the two exact sequences in (98), written here as

$$0 \longrightarrow F \longrightarrow W_0 \longrightarrow K \longrightarrow 0, \qquad 0 \longrightarrow W_0 \longrightarrow W \longrightarrow N \longrightarrow 0, \qquad \dim F + \dim N = h.$$

Let $a = \dim(S \cap F)$, $b = \dim(S \cap W_0)$, and $\epsilon = b - a$. The image of $S \cap W_0$ in K is annihilated by the maximal ideal of R, hence lies in $\mathrm{Soc}_R K = k\tau$. Thus $\epsilon \in \{0, 1\}$. The quotient by S has the more precise filtration

$$0 \longrightarrow F/(S \cap F) \longrightarrow W_0/(S \cap W_0) \longrightarrow K/k^\epsilon \longrightarrow 0, \qquad W/(W_0 + S) \simeq N/\operatorname{im}(S \to N).$$

Here k^ϵ is either zero or the socle $k\tau$ of K. The two small factors have combined length

$$(\dim F - a) + (\dim N - m + b) = h - m + \epsilon = t - 1 + \epsilon.$$

Refine them by composition series. Subadditivity of $\mathrm{Hom}_R(-, W)$, and precomposition with the quotient map $K \longrightarrow K/k^\epsilon$, give

$$\dim \mathrm{Hom}_R(W/S, W) \le \dim \mathrm{Hom}_R(K, W) + (t - 1 + \epsilon)m.$$

Target subadditivity for the original two extensions gives

$$\dim \mathrm{Hom}_R(K, W) \le \dim \mathrm{End}_R(K) + h \dim \mathrm{Hom}_R(K, k) = w_0 + h(d-1).$$

The kernel of restriction is precisely $\mathrm{Hom}_R(W/S, W)$. Since $\epsilon \le 1$, the result follows. This argument uses K as a subquotient; it does not assume that K embeds in W. □

Lemma G.2 (The action–cohomology loss)**.** *Let the extra quotient $V = k^s$ act on W through commuting square-zero R-endomorphisms $T_1, \dots, T_s$. Denote their restrictions to S by Q_a, and put*

$$\rho = \max_{v \in k^s} \operatorname{rank}\Big(\sum_a v_a Q_a\Big), \qquad A_{d,h,s} = d + C_d + s(d+1) + C_s + hC_{d+s}.$$

Then $0 \le \rho \le \lfloor m/2 \rfloor$, and

$$\dim_k H^2(U_0 \oplus V, W) \le A_{d,h,s} - (t + \rho)C_s. \tag{128}$$

Proof. Take core cohomology first in the ordinary CE double complex. Its total degree-two term bounds the resulting two-cohomology by

$$\dim H^0(V, H^2(U_0, W)) + \dim H^1(V, H^1(U_0, W)) + \dim H^2(V, S).$$

Lemma C.2 bounds the first two terms by $d + C_d + hC_d$ and $s(d + 1 + hd)$, respectively.

If $s \ge 1$, choose an actual k-rational quotient direction attaining rank ρ on S. On this one-dimensional abelian quotient both degree-zero and degree-one cohomology have dimension $m - \rho$. Taking its cohomology first, then that of a complementary $(s-1)$-space, gives

$$\dim H^2(V, S) \le \big(C_{s-1} + s - 1\big)(m - \rho) = C_s(m - \rho).$$

For $s = 0$ this term is zero directly. Summing gives (128). Neither spectral sequence is assumed to collapse. All linear combinations of the Q_a's are square zero, so the stated range of ρ follows. Its definition uses the finite base field, not an algebraic closure. □

G.2 Rank-sensitive action coverings

Lemma G.3 (A common-flag rank covering)**.** *The number of ordered commuting square-zero s-tuples of m-square matrices whose span has maximum rank ρ over k is at most*

$$K_{m,s,\rho} := [m]_q!\, 2^{ms} q^{s\rho(m-\rho)}, \qquad [m]_q! = \prod_{j=1}^{m} \frac{q^j - 1}{q - 1}. \tag{129}$$

For $\rho = 0$ one may replace $K_{m,s,0}$ by one. Moreover

$$\log_q K_{m,s,\rho} \le s\rho(m-\rho) + \tfrac{1}{2}m(m-1) + ms \log_q 2 + m \log_q (1 - q^{-1})^{-1}. \tag{130}$$

Proof. Let $a_{m,j}$ count strictly upper triangular square-zero matrices of rank j. The classical rank recurrence is

$$a_{m,j} = q^j a_{m-1,j} + (q^{m-j} - q^{j-1}) a_{m-1,j-1}.$$

Indeed a last column v appended to A must lie in $\ker A$. If $\operatorname{rank} A = j$, the rank remains j for the q^j choices in $\operatorname{im} A$; if $\operatorname{rank} A = j-1$, it increases for the $q^{m-j} - q^{j-1}$ choices outside $\operatorname{im} A$. Induction gives $a_{m,j} \le \binom{m}{j} q^{j(m-j)}$, hence

$$\sum_{j=0}^{\rho} a_{m,j} \le 2^m q^{\rho(m-\rho)}.$$

Commuting nilpotent operators have a common complete flag over k: their common kernel contains a rational line, after which one argues inductively on the quotient. They are strictly triangular relative to that flag. There are $[m]_q!$ complete flags. For each flag, ignoring mutual commutation and counting each generator independently gives (129). This is a covering argument and assumes no free flag action. The estimate $[m]_q! \le q^{m(m-1)/2}(1 - q^{-1})^{-m}$ gives (130). □

Classical input. The triangular rank recurrence is Kirillov's, as recorded in [9, p. 1, equation *(Sasha)*]. The common-flag covering and the displayed inequality are elementary consequences. The population bound below couples this classical matrix count to the socle-restriction kernel and the rank-sensitive ordinary-cohomology loss.

Proposition G.4 (A full branch covering)**.** *Put*

$$S_e(s) = \sum_{j=0}^{\min(s,e)} \begin{bmatrix} e \\ j \end{bmatrix}_q, \qquad M_h(d) = (h+1) q^{hC_d + dh^2 + h}.$$

For every $h, s \ge 0$, the full branch satisfies

$$B_h \le M_h(d) q^{A_{d,h,s}} \sum_{m=1}^{h+1} S_{e_{h,m}}(s) \sum_{\rho=0}^{\lfloor m/2 \rfloor} K_{m,s,\rho}\, q^{-(h+1-m+\rho)C_s}. \tag{131}$$

Terms $\rho > 0$ may be omitted when $s = 0$.

Proof. The module census in Lemma C.2 supplies at most $M_h(d)$ representatives. Fix one of them, of socle dimension m, and let E_0 be the kernel of $E = \operatorname{End}_R(W) \to \operatorname{End}_k(S)$. Choose a vector-space section of $E \to E/E_0$. Each action then has a kernel coordinate and a visible socle coordinate.

Normalize the kernel-coordinate map $k^s \to E_0$ by $\mathrm{GL}_s(k)$: for its image choose a prescribed basis, followed by zeros. This supplies $S_{\dim E_0}(s)$ choices, at most $S_{e_{h,m}}(s)$ by Lemma G.1.

The same quotient-coordinate change transforms the visible tuple; allowing all such tuples still covers every algebra. For a fixed socle tuple the visible coordinate has a unique lift to the chosen complement of E_0, so no additional factor is required. Lemma G.3 supplies its count.

For each action tuple, ordinary two-cohomology covers every abelian Lie extension up to section changes. Lemma G.2 bounds the number of these classes. The class and derived-generation conditions impose further restrictions and may be dropped in an upper covering. Every unlabelled Lie type admits a normalized presentation, so multiplying these coverings, then summing over m and ρ, proves the result. No orbit size is divided out and no module splitting is assumed. □

G.3 The branch envelope

Define the continuous function

$$\Psi(\beta,\gamma)=\begin{cases}\gamma+\frac{1}{2}\beta\gamma^2, & \beta\le\gamma/2,\\ \gamma+\frac{1}{3}\beta^2\gamma+\frac{1}{6}\beta\gamma^2+\frac{1}{12}\gamma^3, & \beta\ge\gamma/2.\end{cases}\tag{132}$$

Proposition G.5 (Uniform leading branch bound)**.** *For each fixed $A_*>0$, uniformly for integers $h,s\ge 0$ with $h+s\le A_*\sqrt{n}$,*

$$\log_q B_h\le n^{3/2}\Psi(h/\sqrt{n},s/\sqrt{n})+O_{q,A_*}(dn).\tag{133}$$

Proof. In (131), put $\beta=h/\sqrt{n}$, $\gamma=s/\sqrt{n}$, $\zeta=m/\sqrt{n}$, and $\xi=\rho/\sqrt{n}$. The module census costs $O_{A_*}(dn)$. Moreover

$$e_{h,m}=n+(h+1-m)m+O_{A_*}(d\sqrt{n}),$$

so eventually $2s\le e_{h,m}$ uniformly. The Gaussian estimate of Appendix E gives $S_e(s)\le\frac{3}{2}(s+1)q^{s(e-s)}$. The flag and triangular prefactors in (130) cost $O_{q,A_*}(n)$. All remaining terms involving d, the difference $C_s-s^2/2$, and the number of summands contribute $O_{q,A_*}(dn)$. The resulting leading expression is

$$F(\beta,\gamma,\zeta,\xi)=\gamma+\tfrac{1}{2}\beta\gamma^2+\gamma(\beta-\zeta)\zeta-\tfrac{1}{2}(\beta-\zeta+\xi)\gamma^2+\gamma\xi(\zeta-\xi).$$

The true range $0\le\zeta\le\beta+n^{-1/2}$ can be replaced by $0\le\zeta\le\beta$ at a cost $O_{A_*}(n)$, since F has uniformly bounded first derivatives on the relevant compact set. The range of ξ is $0\le\xi\le\zeta/2$.

For $\gamma>0$, complete squares in

$$F/\gamma=1+\zeta(\beta+\gamma/2)-\zeta^2+\xi(\zeta-\gamma/2)-\xi^2.$$

The maximizing ξ is $(\zeta-\gamma/2)_+/2$. If $\beta\le\gamma/2$, the maximizing ζ is β; if $\beta\ge\gamma/2$, it is $2\beta/3+\gamma/6$. The resulting values are exactly (132). For $\gamma=0$ all leading expressions are zero. This proves the uniform estimate. □

G.4 The exact leading law and the typical allocation

Put

$$\varphi(\alpha)=\begin{cases}\alpha, & 0\le\alpha\le\sqrt{2},\\ \dfrac{\alpha^3+9\alpha+(\alpha^2+6)^{3/2}}{27}, & \alpha\ge\sqrt{2},\end{cases}\tag{134}$$

and

$$\beta_*(\alpha)=\begin{cases}0, & 0\le\alpha\le\sqrt{2},\\ \dfrac{2\alpha-\sqrt{\alpha^2+6}}{3}, & \alpha>\sqrt{2}.\end{cases}\tag{135}$$

Theorem G.6 (The full leading population law)**.** *For every fixed $A_* > 0$, uniformly for integers $1 \le r \le A_*\sqrt{n}$,*

$$\log_q B_{d+1,r}(q) = n^{3/2}\varphi(r/\sqrt{n}) + O_{q,A_*}(dn). \tag{136}$$

In particular, if $r/\sqrt{n} \to \alpha > 0$, then

$$\frac{\log_q B_{d+1,r}(q)}{n^{3/2}} \longrightarrow \varphi(\alpha).$$

For each $\epsilon > 0$, there is $\delta > 0$ such that along this sequence the proportion of types satisfying

$$\left|\frac{\dim_k L' - w_0}{\sqrt{n}} - \beta_*(\alpha)\right| \ge \epsilon$$

is at most $q^{-\delta n^{3/2}}$ for all sufficiently large d. Thus the allocation of surplus to the derived ideal has a unique limiting value. This assertion concerns scaled defects and makes no claim of exact minimum-derived dominance in a shrinking window.

Proof. Fix $\alpha \ge 0$ and write $\gamma = \alpha - \beta$. On the second branch of Ψ, equivalently $0 \le \gamma \le 2\alpha/3$, its value is

$$E_\alpha(\gamma) = \gamma + \alpha^2\gamma/3 - \alpha\gamma^2/2 + \gamma^3/4.$$

It is strictly increasing because

$$E'_\alpha(\gamma) = 1 + \tfrac{3}{4}(\gamma - 2\alpha/3)^2 > 0.$$

Hence its maximum occurs at the boundary with the first branch. On the first branch the value is

$$L_\alpha(\gamma) = \gamma + \tfrac{1}{2}\alpha\gamma^2 - \tfrac{1}{2}\gamma^3, \qquad 2\alpha/3 \le \gamma \le \alpha.$$

Its unique maximum is at $\gamma = \alpha$ when $\alpha \le \sqrt{2}$, and otherwise at $\gamma_* = (\alpha + \sqrt{\alpha^2 + 6})/3$. For $\alpha > \sqrt{2}$ this point is strictly greater than $2\alpha/3$. Substitution gives $\varphi(\alpha)$, and the unique complementary value is $\beta_*(\alpha)$.

Apply Proposition G.5 and sum the $r + 1$ branches. This gives the upper bound in (136) with its stated uniform error.

For the lower bound put $\alpha = r/\sqrt{n}$ and choose $h = \lfloor \beta_*(\alpha)\sqrt{n} \rfloor$. Then $0 \le h \le r$. If $h = 0$, Lemma E.2 gives $\log_q B_0 \ge rn - O_{A_*}(dn)$. When this rounding occurs with $\alpha > \sqrt{2}$, replacing the real optimizer by zero changes the leading expression by at most $O_{A_*}(n)$, by the uniform Lipschitz bound for L_α on $0 \le \alpha \le A_*$.

If $h \ge 1$, put $s = r - h$. Uniformly in this case, $s \le A_*\sqrt{n}$, and the supercritical optimizer has $s \ge 2r/3 - O(1)$. For all sufficiently large d, $1 \le s \le D_{\mathrm{par}}$ and $h \le C_{d+s} - (d-1)$. Thus Proposition F.1 applies and gives

$$\log_q B_h \ge sn + \tfrac{1}{2}hs^2 - O_{q,A_*}(dn).$$

Changing the maximizing real h by at most one costs $O_{A_*}(n)$. This is the required lower bound, uniformly through $\alpha = \sqrt{2}$.

Finally, the maximizer of the continuous compact-domain envelope is unique. For fixed $\alpha > 0$ and $\epsilon > 0$, its value on the closed subset $|\beta - \beta_*(\alpha)| \ge \epsilon$ has a strictly positive gap from $\varphi(\alpha)$, unless that subset is empty. Continuity preserves a smaller fixed gap when $r/\sqrt{n}$ is sufficiently close to α. The uniform branch upper, the total lower, and $\log_q(r+1) = o(n^{3/2})$ then prove the exponential concentration estimate. □

What the argument establishes. For $\alpha > \sqrt{2}$, the typical derived surplus is $\beta_*(\alpha)\sqrt{n} + o(\sqrt{n})$; its leading value is positive. The matching lower is furnished by the full-central family, whereas the upper covers every bounded-defect derived module, including nonsplit modules. Equality of the leading exponents does not assert that the full-central family has density one, or that all typical algebras are indecomposable. These finer properties require information below the $n^{3/2}$ scale.

H The discrete critical transition

Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$. Retain R, K, I from (93), and put

$$n = 2^d, \quad w_0 = n - 2, \quad D_{\mathrm{par}} = n - d - 3, \quad C_j = \binom{j}{2}, \quad A = R/I, \quad \tau = t_0 \cdots t_{d-1}, \quad d \geq 4.$$

The branch B_h has derived dimension $w_0 + h$, quotient dimension $d + s$, and $h + s = r$. All cohomology is ordinary Chevalley–Eilenberg cohomology. Write

$$T = \sqrt{2n}, \quad R_d^* = T - d + \tfrac{1}{2}, \quad x_{d,r} = r - R_d^*, \quad E_{d,r} = -n + C_{d+r}, \quad z'(L) = \dim_k(Z(L) \cap L').$$

H.1 Maximum core socle and polynomial rigidity

Proposition H.1 (The modules at maximum core socle)**.** *Every normalized core module W with $\dim W = w_0 + h$ and $\dim \mathrm{Soc}_R W = h + 1$ is, after a linear core-coordinate change, a module $E_Q = K \oplus Z$, $Z = k^h$, as follows. The core kills Z and acts by usual monomial multiplication on K, except that*

$$t_i y_i = q_i \in Z \quad (1 \leq i < d), \qquad t_i y_j + t_j y_i = 0 \quad (i \neq j),$$

where y_i lifts t_i. The map $Q : k^{d-1} \to Z$, $Q(e_i) = q_i$, is classified by its rank $0 \leq u \leq \min(h, d-1)$. Moreover $\mathrm{Ann}_R E_Q = I$.

Proof. In the two extensions (98), $0 \to F \to W_0 \to K \to 0$ and $0 \to W_0 \to W \to N \to 0$, one has

$$\dim \mathrm{Soc}\, W \leq \dim \mathrm{Soc}\, W_0 + \dim \mathrm{Soc}\, N \leq \dim \mathrm{Soc}\, F + 1 + \dim \mathrm{Soc}\, N \leq h + 1.$$

Equality forces F, N to be trivial and $\mathrm{Soc}\, W \to N$ to be surjective. A section into the socle splits the second extension, leaving $0 \to Z \to W \to K \to 0$. The minimal presentation of K has relations $t_i y_i$ and $t_i y_j + t_j y_i$; the t_0-factor is free. Thus this extension has diagonal values q_i and mixed values q_{ij}. The core-star connecting class is $\sum_{i<j} q_{ij} e_0^* \wedge e_i^* \wedge e_j^*$; all other core representatives have coefficient degree at least two and map to zero. The nonzero generating star therefore forces $q_{ij} = 0$. A positive core change M transforms the diagonal map by $M^{[2]}$; Frobenius is bijective over k, so row and column changes leave only its rank. Distinct ranks have distinct top dimensions $d - 1 + h - u$. Finally $E_Q/Z = K$ forces its annihilator into I, and the displayed monomial action shows that I kills E_Q. □

Lemma H.2 (Low core cohomology)**.** *The connecting maps from $H^j(U_0, K)$ to $H^{j+1}(U_0, Z)$ vanish for $j = 0, 1, 2$. Consequently*

$$\dim H^0(U_0, E_Q) = h + 1, \quad \dim H^1(U_0, E_Q) = d + 1 + hd, \quad \dim H^2(U_0, E_Q) = d + C_d + hC_d,$$

and $H^2(U_0, Z) \to H^2(U_0, E_Q)$ is injective.

Proof. The socle vector τ lifts. Representatives for $H^1(U_0, K)$ are $\zeta_*(e_i) = t_0 t_i$ for $i > 0$, $\zeta_*(e_0) = 0$, and $\zeta_j(e_i) = \mathbf{1}_{i=j}\tau/t_j$. Their coefficients have degree at least two, as do the κ- and λ-representatives in Lemma B.3. These lift unchanged, while the star lifts because the mixed relations vanish. The long exact sequence proves the assertions; no higher-degree connecting map is being used. □

Proposition H.3 (Polynomial actions and the normalized fiber)**.** *An R-endomorphism of E_Q that kills its socle and satisfies $T_*[\beta] = 0$ for a generating core cocycle is multiplication by a unique element of* $\operatorname{rad} A$*. Subtracting core linear parts therefore puts every extra action in $\mathcal{P}_d = \operatorname{rad}(A)^2$, of dimension D_{par}. For s such actions, put*

$$A_{d,h,s} = d + C_d + s(d+1) + C_s + hC_{d+s}, \qquad F_{d,h,s} = A_{d,h,s} - 1 - h(d-1).$$

The full ordinary two-cohomology has dimension $A_{d,h,s}$, and normalized generating presentations are covered, up to module isomorphism, by

$$q^{F_{d,h,s}} \tag{137}$$

fiber representatives.

Proof. Since T kills Z, it induces multiplication by $p \in A$ on K; killing τ forces $p_0 = 0$. Subtract that actual polynomial endomorphism. The difference is $N_\Lambda(y_i) = \Lambda_i \in Z$, killing Z and nonsingleton monomials. Positive multiplication is null-homotopic on the core complex, whereas

$$(N_\Lambda)_*[\beta] = \sum_i \Lambda_i e_0^* \wedge e_i^*.$$

These classes are independent by Lemma H.2, so compatibility forces $N_\Lambda = 0$. Faithfulness on the quotient K gives uniqueness.

Write $p_a = \sum_i t_i a_{ia}$, $a_{ia} \in \mathfrak{m}R$. The triangular R-linear change of exterior cochain generators eliminates the polynomial actions, giving

$$H^2(U_0 \oplus k^s, E_Q) \simeq H^2(U_0, E_Q) \oplus \big(H^1(U_0, E_Q) \otimes (k^s)^*\big) \oplus \big(\operatorname{Soc}_R(E_Q) \otimes \Lambda^2 (k^s)^*\big).$$

This gives $A_{d,h,s}$. Scalar multiplication normalizes the nonzero star coefficient to one. The commuting automorphisms $1 + N_\Lambda$ then translate its $h(d-1)$ central star coordinates independently. Their action on this value-one fiber is free, since restriction to core cohomology detects every nonzero Λ. Its orbit representatives have the cardinality (137). No further stabilizer is divided out. □

The principal finite-product formula gives

$$\log_q B_0 = rn - 3r - C_{d+r} + d - 1 - \log_q(q-1) + O_q(1). \tag{138}$$

This is Theorem 6.4, specialized to the compact positive square-root range used below.

H.2 Uniform control of the other module and action strata

For a normalized core write $m = \dim \operatorname{Soc}_R W$, $t = h + 1 - m$, and let ρ be the maximum rank of its visible extra action on that socle. Since

$$Z(L) \cap L' = \operatorname{Soc}_R W \cap \bigcap_a \ker Q_a,$$

the condition $z'(L) < h + 1$ is equivalent to $t > 0$ or $\rho > 0$, independently of the chosen normalized core. Denote these types by B_h^{bad}. Put

$$U = hC_d + dh^2 + h + A_{d,h,s} + s\big(w_0 + h(d-1) - s\big).$$

For $2s \le w_0$, $h \le s/4$, and $s \ge 40$, the full branch covering (131) implies

$$\log_q B_h \le U + O_q(\log(r+2)), \tag{139}$$

$$\log_q B_h^{\mathrm{bad}} \le U - s^2/16 + O_q(\log(r+2)). \tag{140}$$

Indeed use $S_e(s) \le \frac{3}{2}(s+1)q^{s(e-s)}$ with the upper ambient dimension $e = e_{h,m} \ge w_0$. Relative to $m = h+1, \rho = 0$, the exponent changes by

$$t(sm - C_s) + \rho\big(s(m-\rho) - C_s\big).$$

For $t \ge 1$, the first term is at most $-t(s^2/4 - 3s/2)$. For $\rho = 0$ there is no flag factor. For $\rho \ge 1$, the second term is at most $-s^2/4 + s/2$, and the common-flag prefactor has logarithm at most

$$\tfrac{1}{2}m(m-1) + \tfrac{1}{2}ms + \tfrac{1}{4}m \le 5s^2/32 + 11s/16 + 1/4.$$

Here $\log_q 2 \le 1/2$ and $\log_q(1-q^{-1})^{-1} \le 1/4$. The combined loss is at least $s^2/16$ for $s \ge 40$. All census and summation prefactors cost only $O_q(\log(r+2))$, proving both estimates.

Subtracting (138) gives

$$\log_q(B_h/B_0) \le \Delta_{d,n}(r,h) + O_q(\log(r+2)), \tag{141}$$

where the exact polynomial is

$$\begin{aligned}\Delta_{d,n}(r,h) = {} & h^3/2 - (r+d-1)h^2 \\ & + h\big[-n + r^2/2 + (2d-1/2)r + d^2 - 2d + 5/2\big] \\ & + r(2d+1) + d^2 - d + 1.\end{aligned}$$

The complete multiplicative branch census is proved in Appendix I. The full critical partition and integer selection follow in Appendices L–M.

I Complete defect fibers and uniform multiplicative counts

We retain the modules and ordinary cohomology of Appendix H. Throughout this appendix $m = d-1$, $D = n-d-3$, $Z = k^h$, $u = \operatorname{rank} Q$, and $Z_1 = \operatorname{im} Q$. In particular, m here denotes the core hyperplane dimension, not the variable socle dimension in the all-module covering. Write $Z = Z_1 \oplus Z_0$.

I.1 All module maps and all coordinate changes

Lemma I.1 (The complete endomorphism algebra)**.** *For $d \ge 4$, every R-endomorphism of E_Q has a unique expression*

$$M_f + N_\Lambda + \Theta_\ell + D_J, \qquad f \in A, \quad \Lambda \in \operatorname{Hom}(k^m, Z), \quad \ell \in Z^*, \quad JQ = 0.$$

Here M_f is actual multiplication, $N_\Lambda(y_i) = \Lambda_i$ and kills Z and nonsingleton monomials, while

$$\Theta_\ell(z) = \ell(z)\tau, \qquad \Theta_\ell(y_i) = \ell(q_i)\tau/t_i, \qquad D_J(z) = Jz.$$

The other values of Θ_ℓ, D_J are zero. If $c = f_0$ and $C = c\,\mathrm{id}_Z + J$, invertibility is equivalent to $c \ne 0$ and invertibility of C on Z_0. Thus

$$\begin{aligned}\dim \operatorname{End}_R(E_Q) &= n - 2 + hm + h + h(h-u), \\ |\operatorname{Aut}_R(E_Q)| &= (q-1)q^{n-3+hm+h+u(h-u)}|\operatorname{GL}_{h-u}(k)|.\end{aligned}$$

Proof. The displayed maps preserve the diagonal and mixed relations; for Θ the only nonzero diagonal check is $t_i\Theta_\ell(y_i) = \ell(q_i)\tau$. For completeness, take any endomorphism ϕ. Since $\operatorname{Soc} E_Q = k\tau \oplus Z$, write $\phi(z) = Cz + \ell(z)\tau$, without assuming that Z is preserved. In the quadratic monomial component of the mixed relations, a third positive index kills every off-diagonal coefficient of the linear part of $\phi(y_i)$. Comparison of t_it_j makes the diagonal coefficients equal to one scalar c. This uses $m \geq 3$. The diagonal relations then give $CQ = cQ$. Subtract Θ_ℓ. The projected images of the y_i in K now satisfy $t_ia_i = 0$ and $t_ia_j + t_ja_i = 0$, hence define a map $K \to K$. The monomial endomorphism calculation $\operatorname{End}_R K = A$ gives its unique multiplier f, with $f_0 = c$. Subtracting actual M_f leaves exactly $N_\Lambda + D_J$, with $JQ = 0$. These successive components also prove uniqueness. On the head $k^m \oplus Z_0$, the matrix is lower triangular, with diagonal $c\,\mathrm{id}_m, C_{00}$. Nakayama's lemma gives the criterion. There are $u(h-u)$ free cross-block entries in C, $hm + h$ in Λ, ℓ, and $n-3$ positive multiplier coefficients, proving the order. □

For independent polynomial frames $p = (p_a) \in \mathcal{P}_d^s$, the quotient acts faithfully through A. The algebra generated by its actions, its radical filtration, and $V_s = \ker(U \to \operatorname{rad} A/(\operatorname{rad} A)^2)$ are intrinsic. The lowest homogeneous part of I recovers the positive hyperplane. The coordinate recovery of Appendix D therefore gives every quotient change in the form

$$g = \begin{pmatrix} B & 0 \\ \alpha & \Delta \end{pmatrix}, \qquad B = \begin{pmatrix} a & 0 \\ b & M \end{pmatrix}, \qquad \Delta^T p' = p(Bt + \alpha^T p'). \tag{142}$$

The last equation has its unique nilpotent implicit solution. In particular all changes of core, including b and α, are retained.

Lemma I.2 (Necessary and sufficient semilinear lifting)**.** *A matrix in* (142) *lifts to a normalized isomorphism for the rank-u module if and only if*

$$M \in H_{m,u} := \{M \in \mathrm{GL}_m(k) : M^{[2]} \ker Q = \ker Q\}.$$

The complete base group has order

$$|G_u| = (q-1)q^{ds+m}|H_{m,u}||\,\mathrm{GL}_s(k)|, \quad |H_{m,u}| = q^{u(m-u)}|\,\mathrm{GL}_u(k)||\,\mathrm{GL}_{m-u}(k)|. \tag{143}$$

Proof. The line $k\tau = (\operatorname{rad} A)^{d-1}E_Q$ is intrinsic, so a semilinear map has $\phi|Z = C + \tau\ell$, with C invertible. After changing to inverse M-coordinates, the same quadratic mixed-relation comparison as above forces $\phi(y_i)_{[1]} = c\sum_j M_{ji}y_j$. Reading the diagonal central values gives $CQ = cQM^{[2]}$. The nonzero core-star coefficient changes by c/a; its normalization forces $c = a$. This proves necessity, including preservation of the kernel.

Conversely the implicit substitution $\sigma(t) = Bt + \alpha^T p'$ preserves K, I. For the fixed monomial lift of an element of K, the action table gives $v\widetilde{w} + w\widetilde{v} = 0$ and $v\widetilde{v} = Q(v_{[1]}^{[2]})$. Choose invertible C with $CQ = aQM^{[2]}$, first on Z_1 and then on a complement. The assignments

$$\phi(y_i) = a\widetilde{\sigma(t_i)}, \qquad \phi(z) = Cz$$

preserve every relation and normalize the star. There is no additional factor a on the already chosen C. The t_0-factor is free, so its shear introduces no relation. This proves sufficiency. Frobenius is bijective; the stabilizer is a kernel parabolic of the stated order. Counting a, b, α, Δ gives (143). Different u cannot collide, since the intrinsic module head has dimension $m + h - u$. □

I.2 The residual action on the whole cohomology fiber

Put

$$A_h = d + C_d + s(d+1) + C_s + hC_{d+s}, \qquad F = A_h - 1 - hm.$$

Ordinary section changes are the two-coboundaries. Scalar normalization of the star, followed by the hm independent N_Λ translations, leaves q^F cohomology classes. Positive multipliers act trivially on cohomology: for each core variable t_i, the identity $\delta\iota_i + \iota_i\delta = t_i$ gives a contraction, and linear combinations give the assertion for every positive polynomial.

Proposition I.3 (Complete residual fiber)**.** *After these normalizations the complete effective vertical group is*

$$\mathcal{R}_u = \{(C,\ell) : C \in \mathrm{GL}(Z),\ C|Z_1 = \mathrm{id},\ \ell \in Z^*\}, \qquad |\mathcal{R}_u| = q^{h+u(h-u)}|\,\mathrm{GL}_{h-u}(k)|.$$

On the extra-pair socle coordinate $\theta\tau + \Gamma$, with $\Gamma : \Lambda^2 V_s \to Z$, it acts by

$$(\theta, \Gamma) \longmapsto (\theta + \ell\Gamma, C\Gamma). \tag{144}$$

It acts by C on the remaining central coordinates. Uniformly in the frame, its generating fiber orbit count b_u satisfies

$$\left| \frac{b_u}{q^F/|\mathcal{R}_u|} - 1 \right| \le \frac{q^{h^2+2h-C_s}}{q-1}. \tag{145}$$

For $u = h$ the exact value is $b_h = q^{F-h}[1 + (q^h - 1)q^{-C_s}]$.

Proof. The preceding complete endomorphism calculation leaves exactly the stated C, ℓ after the star translations and multipliers. To check the whole fiber, use the low core representatives in Lemma H.2. The image of the star under Θ_ℓ is a boundary, obtained on positive index i by the one-cochain value $\ell(q_i)\tau/(t_0 t_i)$. All other core two-representatives have degree at least two and are killed. On one-cohomology, any remaining τe_i^* is $\delta(\tau/t_i)$; thus Θ_ℓ is also zero there. It is nonzero on zero-cohomology, sending z to $\ell(z)\tau$. The triangular elimination of polynomial extra actions is R-linear and commutes with these maps. The direct-sum cohomology decomposition in Proposition H.3 therefore gives (144), including all mixed and core parameters.

If Γ is surjective this action is free and the derived module is generated: the core supplies the K-head and Γ supplies Z/Z_1. More generally generation is surjectivity of all nonstar central values onto Z/Z_1, a condition independent of the polynomial frame. Hence both the generating set and its orbit count are frame-independent. The proportion of nonsurjective Γ is at most $e = q^{h-C_s}/(q-1)$, by a union over nonzero dual lines. Count only the free open set for a lower bound; count its entire complement point by point for an upper bound. Since $|\mathcal{R}_u| \le q^{h^2+h}$, the relative error is at most $|\mathcal{R}_u|e$, proving (145) even when the complement has nontrivial stabilizers. If $u = h$, generation already follows from the core. Each nonzero ℓ fixes the q^{F-C_s} classes satisfying $\ell\Gamma = 0$; Burnside gives the exact formula. □

First divide by all vertical module maps and section changes, and denote the resulting fiber by X_p. Any normalized semilinear lift of $g \in G_u$ induces a map $X_p \to X_{p'}$. Two choices differ by a vertical isomorphism, already divided out; the same holds for the difference between a composite and a chosen lift of its product. Thus G_u acts genuinely on $\coprod_p X_p$. Every Lie isomorphism has quotient, module and section parts of these forms, so its orbits are exactly the unlabelled types. This argument does not require an invariant choice of normalized cohomology slice.

I.3 All ranks and all exceptional modules

Theorem I.4 (Uniform complete branch count)**.** *Fix q and $0 < a < b < \infty$. Uniformly for integers*

$$0 \le h \le H = \lfloor \sqrt{n}/d^2 \rfloor, \qquad a\sqrt{n} \le r \le b\sqrt{n}, \qquad s = r - h,$$

there is $\kappa > 0$, depending only on the fixed parameters, such that $B_h = M_h(1 + O_{q,a,b}(q^{-\kappa n}))$, where

$$M_h = \frac{N_s q^{C_d+s+C_s+hC_{d+s}-h}}{(q-1)|\operatorname{GL}_m(k)||\operatorname{GL}_h(k)||\operatorname{GL}_s(k)|}, \qquad N_s = \prod_{j<s}(q^D - q^j), \tag{146}$$

$$= \frac{q^{V_h+m-h(h+1)}\mathcal{F}_s}{(q-1)P_mP_hP_s}, \qquad V_h = sn - 3s + (h-1)C_{d+s}, \tag{147}$$

with $P_j = \prod_{i=1}^{j}(1-q^{-i})$, $P_0 = 1$, and $\mathcal{F}_s = \prod_{j<s}(1-q^{j-D})$. On this branch the intrinsic rank u, outside an exceptional set of probability $O_{q,a,b}(q^{-\kappa n})$, differs in total variation by that order from the rank of a uniform $h \times m$ matrix over k.

Proof. For faithful frames the complete rank-u main term is

$$L_{h,u} = \frac{N_s q^{F-ds-m-h}}{(q-1)|\operatorname{GL}_s(k)||H_{m,u}|q^{u(h-u)}|\operatorname{GL}_{h-u}(k)|}. \tag{148}$$

Indeed (145) is exponentially small uniformly: $h^2 \le n/d^4$, $h = o(s)$, and $C_s = \Omega_a(n)$. The nonidentity base stabilizer sum is bounded term by term by the full-group sum in (120)–(123), with r there replaced by s. The pure exponents are at most

$$d(s+d) - (s-d)(\sqrt{n}/2 - d - 3), \qquad s + d - a_0, \quad a_0 \sim n/2;$$

both are $-\Omega_a(n)$. The other two exponents are $d^2 + sd + s^2 + 1 - sn/4 + s(d+3)$ and $sd + 2s - 1 - D + O_q(\log s)$, also at most $-\kappa n$. Each fixed fiber contributes at most its frame-independent size b_u. Burnside therefore proves the count $L_{h,u}(1 + O_{q,a,b}(q^{-\kappa n}))$, uniformly in h, u. There is no group-index loss on restricting the stabilizer sum.

Independently, choose the kernel, image and quotient isomorphism of a rank-u matrix. Its classical count is

$$R_{h,m}(u) = \begin{bmatrix} m \\ u \end{bmatrix}_q \begin{bmatrix} h \\ u \end{bmatrix}_q |\operatorname{GL}_u(k)| = \frac{|\operatorname{GL}_m(k)||\operatorname{GL}_h(k)|}{|H_{m,u}|q^{u(h-u)}|\operatorname{GL}_{h-u}(k)|}. \tag{149}$$

Thus $L_{h,u}$ is one common factor times $R_{h,m}(u)$, and $\sum_u R_{h,m}(u) = q^{hm}$. Summing proves the stated main term for faithful polynomial families, including $h > m$.

It remains to cover the whole branch. Equations (139)–(140) apply uniformly here. Their exponent satisfies exactly

$$U - V_h = (h+2)C_d + dh^2 + h + d + s(hm + 2d + 1) = O_b(n/d + d\sqrt{n}) = o(n).$$

Meanwhile $\log_q M_h = V_h + m - h(h+1) + O_q(1)$, uniformly, since all finite product factors stay between fixed positive bounds. Hence the loss $s^2/16$ makes all bad socle and visible nonpolynomial-action types exponentially negligible relative to M_h. For deficient polynomial frames use, for each module rank, the raw cover $q^F \sum_{j<s} \begin{bmatrix} D \\ j \end{bmatrix}_q$, without dividing by an unproved orbit size. After division by M_h, and summation over at most $m+1$ ranks, its logarithm is at most

$$-D + ds + 2s + m + h + m^2 + h^2 - hm + O_q(\log(s+1) + \log(m+1)) = -n + o(n).$$

The maximum-socle classification and polynomial rigidity in Appendix H exhaust the remainder.

Finally every rank main term has the same relative error. Positive summation preserves this bound; normalization changes its vector of probabilities in total variation by $O(q^{-\kappa n})$. Add the exceptional type mass just bounded. The rank is intrinsic because $\dim E_Q/(\operatorname{rad} A)E_Q = m + h - u$. This proves the claim for the uniform type measure, rather than merely a labelled model. □

The finite products in (147) are material to its error: replacing P_m, P_s by P_∞ gives only relative $O_{q,a,b}(q^{-m})$. The factor P_h must be retained. For fixed h, full row rank has probability $1 - O_{q,h}(q^{-d})$; near $h = m$ the deficient ranks have positive limiting mass, and beyond m only full column rank is possible. These are consequences of the same complete rank census.

J Common head normalization and all-module coverings

Retain the normalized core ring of (93). Throughout this appendix $\sigma = \dim \operatorname{Soc}_R W$, $t = h + 1 - \sigma$; it is distinct from the matrix width $d - 1$.

Lemma J.1 (A socle-defect module census)**.** *The normalized module types with socle defect t are covered by at most*

$$\mathcal{N}_{d,h,t} = 2(t+1)q^{(h+1)(C_d+1)+dt(h+2)}.$$

For $Y = W/\operatorname{Soc}_R W$, $\beta_1^R(Y) \le C_d + 1 + dt$ and $\dim Y/\operatorname{rad} Y \le d - 1 + t$.

Proof. The filtration in the proof of Lemma G.1 has large factor K/k^ε, $\varepsilon \in \{0, 1\}$, and two small factors of combined length $t - 1 + \varepsilon$. Here $k^1 = k\tau$. The minimal resolutions give

$$\beta_1^R(K/k^\varepsilon) \le C_d + 1, \qquad \dim \operatorname{Ext}_R^1(k, K/k^\varepsilon) \le 1 + C_{d+1}.$$

Indeed apply the module long exact sequences to $0 \to k\tau \to K \to K/k\tau \to 0$, using $\beta_1^R(K) = C_d$, $\dim \operatorname{Ext}_R^1(k, K) = 1$ and $\dim \operatorname{Ext}_R^2(k, k) = C_{d+1}$. This last dimension is the degree-two rank in the tensor-periodic module resolution, not the ordinary CE dimension C_d.

If the two small lengths are a, b, fixed small modules cost at most $q^{d(a^2+b^2)}$. The first extension costs $q^{a(C_d+1)}$; the second costs at most $q^{b(da+1+C_{d+1})}$. Summing over the two possible total lengths $t - 1, t$ bounds the types of Y by $2(t+1)q^{dt^2+t(C_{d+1}+1)}$. Subadditivity of the first module Betti number gives $\beta_1^R(Y) \le C_d + 1 + dt$. Recovering W from $0 \to k^\sigma \to W \to Y \to 0$ costs at most $q^{\sigma(C_d+1+dt)}$. The total exponent is

$$dt^2 + t(C_{d+1} + 1) + (h + 1 - t)(C_d + 1 + dt) = (h+1)(C_d+1) + dt(h+2).$$

Extension classes with fixed ends cover unmarked module types; dropping the condition that the kernel be the whole socle only enlarges this covering. The same filtration, now on heads, gives $\dim Y/\operatorname{rad} Y \le d - 1 + t$. □

Proof of Theorem 4.1. Write $S_0 = \operatorname{Soc}_R W$, and choose Z complementary to $S_0 \cap \operatorname{rad} W$ in S_0. Its injection into the head extends to a projection $W \to Z$ factoring through the head; this is R-linear. Its kernel M gives

$$W = M \oplus Z, \qquad \operatorname{rad} M = \operatorname{rad} W, \qquad \operatorname{Soc}_R M = S_0 \cap \operatorname{rad} W \subseteq \operatorname{rad} M.$$

Put $a = \dim \operatorname{Soc}_R M$, $z = \dim Z$ and $h_0 = \dim M - w_0 = a + t - 1$. The tensor connecting map $\operatorname{Tor}_1^R(k, Y) \to S_0$ has image $S_0 \cap \operatorname{rad} W$. The preceding lemma therefore gives

$$a \le C_d + 1 + dt, \quad h_0 \le C_d + (d+1)t, \quad p := \dim M/\operatorname{rad} M = \dim Y/\operatorname{rad} Y \le d - 1 + t.$$

Project the core cocycle to M and let D_M be the R-span of its values. Its core extension retains class $d+1$: a terminal commutator lies in $\operatorname{rad} W$, whose intersection with Z is zero. Applying the minimum and socle bounds to this actual core derived module gives $\dim(D_M/\operatorname{Soc}_R D_M) \geq w_0 - 1$. Since $\dim(M/\operatorname{Soc}_R M) = w_0 + t - 1$, the quotient $M/(D_M + \operatorname{Soc}_R M)$ has length at most t. Thus the head image J_η of the core bracket in $P = M/\operatorname{rad} M$ has codimension at most t.

Let $\eta : \Lambda^2 U_0 \to P$ be this head map. Every map $\psi : P \to Z$ defines an actual module shear $1 + \psi$ on $M \oplus Z$, fixing S_0 pointwise. Choose a fixed complement to $\operatorname{im}\eta^*$ in $\Lambda^2 U_0^*$. The shears put β_Z in its tensor product with Z. Conjugate the whole extra-action tuple at the same time.

Write an R-endomorphism as $(A, B; C, D)$ in this decomposition. Here $B : Z \to \operatorname{Soc}_R M$, $C : P \to Z$. If $T_*[\beta] = 0$, the Z-part of $T\beta$ is zero, because core CE boundaries have zero Z-part. Hence $C\eta + D\beta_Z = 0$. The two terms belong to the complementary row spaces just chosen, so $C|J_\eta = 0$. All liftable extra actions consequently belong to one linear space $\mathcal{D}_\eta$, even when $D \neq 0$.

Its restriction kernel on S_0 has $B = D = 0$, $A|\operatorname{Soc}_R M = 0$, and $C \in \operatorname{Hom}(P/J_\eta, Z)$. Its dimension is therefore

$$e_\eta = \dim\ker\big(\operatorname{End}_R M \to \operatorname{End}_k(\operatorname{Soc}_R M)\big) + z\dim(P/J_\eta).$$

Apply the contraction of Lemma C.2 first to the actual core derived module D_M, a quotient of the universal commutator module retaining its terminal. Choose a functional there nonzero on the terminal and pull it back to the universal module. A core linear change normalizes its highest dual component. The lower dual components of this pullback are annihilated by I, so its cyclic contraction is R/I. Writing $v = \dim D_M - w_0$, duality gives $0 \to F_v \to D_M \to K \to 0$. The second sequence $0 \to D_M \to M \to N_{h_0 - v} \to 0$ then supplies the required filtration of M in the same core coordinates. The coordinate change acts simultaneously on W, M, Z; it preserves their radical and socle decomposition. No assertion about the annihilator of an arbitrary functional extended to all of M is needed. The proof of Lemma G.1, which uses only these module filtrations, now gives

$$e_\eta \leq w_0 + h_0(d-1) + ta + tz = w_0 + h_0(d-1) + t\sigma.$$

Substitute the bound on h_0. No quotient of the full Lie algebra by Z has been taken. □

Proposition J.2 (The normalized covering)**.** *Put*

$$E_{d,t} = w_0 - d + (d-1)C_d + d^2 t + th.$$

For fixed h, s, t, ρ, all actual unlabelled types are covered by at most

$$\mathcal{N}_{d,h,t}\, q^{(d-1+t)C_d} S_{E_{d,t}}(s) K_{h+1-t,s,\rho} q^{A_{d,h,s} - (t+\rho)C_s}. \tag{150}$$

The notation $S_e(s), K_{\sigma,s,\rho}, A_{d,h,s}$ is as in Appendix G.

Proof. For each module choose one decomposition above. There are at most $q^{pC_d} \leq q^{(d-1+t)C_d}$ possible head maps η. Fix η and the linear space $\mathcal{D}_\eta$. The d independent core multipliers lie in its restriction kernel. Adding core directions to the extra quotient removes them; the remaining kernel dimension is at most $E_{d,t}$. Indeed the preceding bound minus d is at most $E_{d,t} - t^2$.

Choose a linear section of restriction and normalize the kernel-coordinate tuple by GL_s, exactly as in Proposition G.4. Allow all visible tuples in the common-flag covering. The linear container need not be an algebra: retaining nonactual tuples only enlarges a covering. For each retained tuple, the full ordinary H^2 contributes at most the last factor by Lemma G.2. It counts core, mixed and extra-pair data together; no additional core-cocycle factor is used. Every unlabelled type has at least one such normalized representative, proving the assertion. □

K The macroscopic full-layer partition

Proof of Theorem 6.11. First suppose $h \le s/4$ and $a\sqrt{n} \le s \le b\sqrt{n}$ for fixed positive a, b. Call a type bad if $z'(L) = \dim(Z(L) \cap L') < h+1$, equivalently $t + \rho \ge 1$. Apply (150) and $S_E(s) \le \frac{3}{2}(s+1)q^{s(E-s)}$, valid eventually uniformly since $E \ge 2s$. Write $L_{d,t} = (d-1)C_d + d^2 t$, $V_h = sn - 3s + (h-1)C_{d+s}$. The exact polynomial subtraction from $V_h + d - 1 - h(h+1)$ is

$$2C_d + 1 + s(d + 1 + L_{d,t} + th) + h(h+1) - (t+\rho)C_s. \tag{151}$$

The module and head factors must still be included.

For $\rho > 0$, the flag logarithm is at most

$$s\rho(\sigma - \rho) + \tfrac{1}{2}\sigma(\sigma-1) + \tfrac{1}{2}\sigma s + \tfrac{1}{4}\sigma, \qquad \sigma = h + 1 - t.$$

For $\rho = 0$, its whole factor can be one. In either case, $h \le s/4$ gives

$$t(sh - C_s) \le -ts^2/4 + ts/2, \qquad \rho\{s(\sigma-\rho) - C_s\} \le -\rho s^2/4 + 3\rho s/2.$$

The remaining quadratic terms are at most $7s^2/32 + O(s)$. For $k = t + \rho \ge 1$ their combination is at most $-ks^2/32 + O(ks)$. All other positive logarithms in (150), including the module census, head choices, and finite prefactors, are $O_q(kd^3 s)$. Since $d^3/s \to 0$, summing over all t, ρ yields

$$B_h^{\mathrm{bad}}/M_h \le q^{-s^2/128} \tag{152}$$

eventually, with a possible further fixed reduction of the exponent. Here $\log_q M_h = V_h + d - 1 - h(h+1) + O_q(1)$ uniformly; all finite products are retained.

For $t = \rho = 0$, the exact maximum-socle classification, polynomial rigidity and full group calculation in Appendices H–I apply without a small-h assumption. Their residual fiber error is at most $q^{h^2+2h-C_s}/(q-1)$, hence exponentially small in n here. The generating vertical orbit count is strictly frame-independent. On its quotient, all choices of semilinear lift give the same base action. Each nonidentity fixed fiber is bounded by its full size, so the nonidentity Burnside sum is bounded by the old summed frame stabilizers with r replaced by s, without a group-index factor. The four exponents in (120)–(123) are

$$d(s+d) - (s-d)(\sqrt{n}/2 - d - 3), \qquad s + d - a_0,$$
$$d^2 + sd + s^2 + 1 - sn/4 + s(d+3), \qquad sd + 2s - 1 - D + O_q(\log s),$$

where $a_0 \sim n/2$, $D = n - d - 3$. All are at most $-\kappa n$, uniformly. This proves the rank-u main term $L_{h,u}$ of (148) with relative error $O_q(q^{-\kappa n})$, for all u. Their exact matrix-rank sum is M_h, including $h > d - 1$.

Deficient polynomial frames are covered by their whole normalized fiber and $\sum_{j<s} \begin{bmatrix} D \\ j \end{bmatrix}_q$, with no unknown orbit division. After division by M_h, their summed logarithm is at most

$$-D + ds + 2s + (d-1) + h + (d-1)^2 + h^2 - h(d-1) + O_q(\log(s+1) + \log d).$$

On the stated macroscopic window $h \le s/4$, $h^2 \le 4n/25$, so this is at most $-4n/5$ eventually. Together with (152), this proves $B_h = M_h(1 + O_q(q^{-\kappa n}))$ uniformly for $h \le r/5$.

The unique allocation from Theorem G.6 satisfies $\beta_*(\alpha)/(\alpha - \beta_*(\alpha)) < 1/4$ with a uniform strict gap on $[3/2, 8/5]$. The all-module envelope therefore bounds the actual type proportion with $h > r/5$ by $q^{-\eta n^{3/2}}$. Sum the positive branch estimates and then normalize by their sum. This proves the full partition and the joint total variation bound, assigning all bad and deficient objects the exception symbol.

Finally $h \ge b_0\sqrt{n}$ outside a probability $q^{-\eta n^{3/2}}$, for a fixed $b_0 > 0$. A uniform $h \times (d-1)$ matrix fails full column rank with probability at most $q^{d-1-h}/(q-1)$, by a union over projective kernel lines. The joint comparison proves the last claim. □

L The full-layer tail, partition and integer rank law

We prove Theorem 6.6 in dependency order: the complete fibers above, an all-module tail, and normalization. Throughout q is fixed, $H = \lfloor \sqrt{n}/d^2 \rfloor$, $T = \sqrt{2n}$, and $|x| = |r-(T-d+1/2)| \le H/4$.

Lemma L.1 (Uniform all-module tail)**.** *For some $c_q > 0$, uniformly on the stated integer window,*

$$\sum_{h>H} B_h/B_0 \le q^{-c_q T H^2}.$$

Proof. The exact all-module polynomial (141), not a fixed-window asymptotic remainder, gives for $s = r-h$, $2s \le n-2$, $h \le s/4$, $s \ge 40$,

$$\begin{aligned}\Delta = T[-h^2 + (d+x)h + 2d + 1] + \tfrac{1}{2}h^3 - (x - \tfrac{1}{2})h^2 \\ + h[\tfrac{1}{2}x^2 + dx - \tfrac{1}{2}d^2 - d + \tfrac{19}{8}] + (2d+1)x - d^2 - d + \tfrac{3}{2}.\end{aligned}$$

For $H \le h \le r/5$, all these hypotheses hold uniformly eventually, and $h \le T/4$. Divide the positive costs by Th^2. Their sum is at most

$$\begin{aligned}\frac{d}{H} + \frac{1}{4} + \frac{2d+1}{H^2} + \frac{1}{8} + \frac{H/4 + 1/2}{T} + \frac{H^2/32 + dH/4 + 19/8}{TH} \\ + \frac{(2d+1)H/4 + 3/2}{TH^2} = \frac{3}{8} + o(1).\end{aligned}$$

Here $H/d \to \infty$, $H/T \to 0$, and every remainder is uniform in h, x. Thus $\Delta \le -Th^2/2$. Absorbing $O_q(\log(r+2))$ and summing gives $\sum_{H<h\le r/5} B_h/B_0 \le q^{-c_1 T H^2}$. This bounds every module in each branch.

For $h \ge r/5$, use the all-module envelope (133), with $\alpha = r/\sqrt{n} \to \sqrt{2}$ uniformly and $\beta = h/\sqrt{n} \ge \alpha/5$. At $\alpha = \sqrt{2}$, on its first branch the gap is

$$\alpha - \Psi(\beta, \alpha - \beta) = \beta^2(\alpha - \beta/2) \ge 9\alpha^3/250 \quad (\alpha/5 \le \beta \le \alpha/3).$$

On the second branch, the expression in the complementary variable $\gamma = \alpha - \beta$ has derivative $1 + 3(\gamma - 2\alpha/3)^2/4 > 0$; its maximum is at the junction. Compactness and continuity preserve a fixed positive gap throughout this macroscopic tail when $\alpha \to \sqrt{2}$. The envelope error is $O_q(dn) = o(n^{3/2})$, whereas $\log_q B_0 = rn - O_q(n + d\sqrt{n} + d^2)$. Hence $\sum_{h \ge r/5} B_h/B_0 \le q^{-c_2 n^{3/2}}$. The two estimates prove the lemma; $TH^2/n \to \infty$. □

Proof of Theorem 6.6. Here $r/\sqrt{n}$ stays in a fixed compact positive interval. Theorem I.4 applies to all $h \le H$, with one relative error. Its sum has the same relative error because the M_h are positive. Since $Z_{d,r} \ge M_0 \sim B_0$, Lemma L.1 is negligible even after normalization by this sum. This proves the full count.

For faithful polynomial rank strata the actual counts are $L_{h,u}(1 + \epsilon_{h,u})$, with $\sup|\epsilon_{h,u}| = O_q(q^{-\kappa n})$. Their normalized vector differs from the normalized main terms by that order in total variation. The complement, consisting of bad modules, deficient frames, and $h > H$, has probability of the same order by the two preceding results. Equation (149) identifies the normalized vector with the stated joint mixture. Thus every comparison uses actual unlabelled type counts. □

L.1 Exact ratios and concentration on adjacent branches

Direct division of the finite products gives, for $h < H$,

$$R_h := \frac{M_{h+1}}{M_h} = \frac{q^{e_h}}{1-q^{-h-1}} \frac{1-q^{-s}}{1-q^{s-1-D}}, \qquad e_h = -n + C_{d+s} - h(d+s) + 1 - h. \tag{153}$$

There is no factor $1-q^{-(m-h)}$: summing all matrix ranks has removed it, including at $h = m$. Two useful exact identities are

$$e_{h+1} - e_h = -2(d+r-h) + h + 1, \tag{154}$$

$$e_h = T\delta + \delta^2/2 + 7/8 - h(h+3)/2, \qquad \delta = x - 2h. \tag{155}$$

Each of the three correction factors in (153) is nonincreasing in h. The exponent increment is $-2T - 2x + 3h \le -2T + 7H/2$, so eventually $R_{h+1}/R_h \le q^{-T}$, uniformly.

Let i maximize $M_0, \dots, M_H$. It cannot equal H: the all-module estimate at $h = H$, compared with Theorem I.4, gives $M_H/M_0 \ll 1$. If $i = 0$, choose the pair $\{0, 1\}$. Otherwise choose its larger neighbor and i, obtaining $\{j, j+1\}$. The two neighbor-to-mode ratios have product at most q^{-T}; the smaller is at most $q^{-T/2}$. Farther terms decrease geometrically by the ratio bound. At the endpoint the same conclusion follows from $R_0 \le 1$. Thus the mass outside this pair is $O_q(q^{-T/2})$. The two masses are $(1, R_j)/(1+R_j)$, with relative error $O_q(q^{-T/2})$. This proves Corollary 6.7.

For fixed $|x| \le A$, summing (155) over fixed h gives $\log_q(M_h/M_0) = T[hx - h(h-1)] + O_{q,h,A}(1)$. Choose a fixed integer $J > 2A+4$. The exact exponent is decreasing and at J is at most $-TJ$, eventually; the accumulated score at J is strictly negative of order T. The ratio bound discards every $h \ge J$. Taking the largest of the finitely many remaining scores proves Corollary 6.9, with its stated (weaker) $O_{q,A}(d)$ error.

L.2 True integer layers and the classical matrix limit

For fixed $\ell \in \mathbb{Z}$, let $r = \lfloor T \rfloor + d - 2 + 2\ell$ and $j = m + \ell$. These integers eventually lie in the theorem's window, with $0 \le j < H$. Writing $f = \{T\}$ gives $x = 2j - 1/2 - f$. Thus $\delta_j \le -1/2$ and $\delta_{j-1} \ge 1/2$. In (155) the other terms are $O_\ell(d^2) = o(T)$ at these two indices, and the finite-factor logarithms are bounded. Consequently $R_{j-1} \ge q^{T/3}$, $R_j \le q^{-T/3}$. Strong log concavity and full normalization give $\Pr(h = j) = 1 - O_{q,\ell}(q^{-cT})$.

For a uniform $h \times m$ matrix, set $v = \min(h, m)$, $k_0 = |h - m|$, and $J = v - u$. Its finite probability, by (149), is

$$q^{-t(k_0+t)} \frac{P_h P_m}{P_{v-t} P_t P_{k_0+t}}, \qquad 0 \le t \le v.$$

The classical limit is

$$p_{k_0}(t) = q^{-t(k_0+t)} \frac{P_\infty}{P_t P_{k_0+t}}, \quad t \ge 0. \tag{156}$$

Fulman–Goldstein [11, formulas (1)–(2), Theorem 1.1] give total variation error at most $3q^{-\max(h,m)-1}$. Apply this with $h = j$, add the full-layer error and the probability $h \ne j$, to obtain Corollary 6.8. The matrix limit and rank enumeration are classical; the conclusion here is their realization in the complete uniform Lie type layer. The matrix-limit error has order $q^{-\max(h,m)}$, separately from both the exponentially smaller Lie-to-finite-matrix error and the $O(q^{-m})$ finite-product replacement error.

M Integer selection from a dyadic approximation bound

Put $\delta_d = \operatorname{dist}(T, \mathbb{Z}+1/2)$. Ridout's theorem, in the dyadic form stated in [19, Theorem 1 and (2.5), p. 57], implies that, for every $\eta > 0$, eventually

$$\delta_d \geq T^{-\eta}. \tag{157}$$

Indeed for odd d, T is integral and the distance is $1/2$. For $d = 2v$, the nearest half-integer gives an odd numerator p and reduced denominator 2^{v+1} approximating $\sqrt{2}$. With $S_1 = \varnothing, S_2 = \{2\}$, the finite exception statement of Ridout reads

$$|\sqrt{2} - p/2^{v+1}| \geq 2^{-(v+1)(1+\varepsilon)}$$

eventually, uniformly in such numerators. Multiplying by 2^v and taking $\varepsilon < \eta$ proves (157). The denominator factor in Ridout is essential here. The resulting cutoff is ineffective; no assumption on digit frequencies is used.

For fixed $|x| \leq A$, choose the fixed cutoff J in the proof of Corollary 6.9. Actual integer surpluses have $x \notin 2\mathbb{Z}$. For $j = \max(0, \lceil x/2 \rceil)$, every competing score differs from the maximum by at least δ_d. The fixed-branch logarithms and complete tail give, for a fixed constant C,

$$\sum_{h \neq j} B_h/B_j \leq Jq^{-T\delta_d + Cd} + q^{-cT+Cd}.$$

The tail was first bounded relative to B_0; the lower $B_j/B_0 \geq q^{-Cd}$ justifies this change of denominator. Use (157) with $\eta = \epsilon/2$ and absorb the fixed factors to obtain the bounded-window rate in Corollary 6.10.

For the global assertion, use the shrinking outer localization $r \leq R_d^* - 4d/T \Rightarrow \pi_{d,r} \to 1$, $r \geq R_d^* + 4d/T \Rightarrow \pi_{d,r} \to 0$, proved below. Equation (157) shows that its intervening interval contains no integer eventually. Its integer separation is $\rho_d = \lfloor R_d^* \rfloor + 1$. Thus every integer sequence r_d has $\pi_{d,r_d} - \mathbf{1}_{r_d < \rho_d} \to 0$: apply the two localizations to their respective subsequences, completing each subsequence to a full one if needed; $r = 0$ has $\pi = 1$. If the claimed supremum did not tend to zero, choosing one surplus on a violating subsequence and completing it would contradict this sequential statement. The bounded-window exponential rate is not asserted uniformly for unbounded r.

The outer splice. In the strip $T - 3d - 3 \leq r \leq R_d^* + 1/4$, the exact ratio (153) and strong log concavity imply $\sum_{h \geq 2} B_h/B_0 \leq q^{-cT}$: its largest upper endpoint has $e_0 = T/4 + O(1)$, $e_1 = -7T/4 + O(1)$, so their product already has a negative exponent of order T; all subsequent ratios decrease. The errors from the whole-layer theorem are $q^{-\kappa n}$ times a partition of size at most $q^{T/4+O(1)}M_0$. The strip lies in its window eventually. In this strip $\pi = (1 + B_1/B_0)^{-1}(1 + O_q(q^{-cT}))$. At $r = R_d^* \pm 4d/T$, (155) for $h = 0$ has values $\pm 4d + O(1)$. Its increasing polynomial in r gives the left and right conclusions on the left part of the strip. For the remaining right indices with $r < T - d + 1$, one has $r = T - d + O(1)$; the uniform branch formula and $\pi \leq B_0/B_1$ give the same conclusion. For smaller r, the elementary lower-side localization in Section 6.1 applies; for $r \geq T - d + 1$, Corollary E.7 applies, including arbitrarily growing r. This proves the outer splice without assuming monotonicity of π.

N Compatible modules, complete fibers and conditional tails

All squares in this appendix have range in $W = L'$ and vanish on W. We count ordinary types after forgetting the square.

N.1 The module selected by the first socle defect

Proof of the module assertion in Theorem 7.1. If a core has maximum socle, its module is E_Q. Project to K and normalize the core star. For $b_0 = p(x_0)$, the square equation is $t_i b_0 = t_0 \beta_{0i}$. Its right side has nonzero degree-two term $t_0 t_i$. The left side cannot have that term, since the degree-one space of K has no t_0. Thus maximum core socle is impossible, and $t \geq 1$.

Suppose $t = 1$. Let D be the actual core derived module, $\dim D = w_0 + v$, and $S_0 = \operatorname{Soc}_R W$. The same projection argument gives $b_0 \notin D + S_0$. The length argument in the head-normalization proof gives $\dim W/(D + S_0) \leq 1$. Equality follows, and $\dim \operatorname{Soc}_R D = v + 1$. Only now apply the maximum-socle classification to D, obtaining E_Q.

Choose its module coordinates and core cohomology representatives so that $\beta_{0i} = y_i + \lambda_{0i}\tau_{0i}$. The coordinate changes on D are used by transporting the entire extension table on W, including the square values; one need not assert that every automorphism of D extends to an automorphism of the fixed W. Core section changes inside D preserve the generated core derived module and replace b_i by $b_i + t_i g_i$. Central star translations likewise transport the whole table. Thus all the square identities remain valid in these normalized coordinates.

Now $t_i^2 b_0 = 0$ gives $t_0 t_i \beta_{0i} = \lambda_{0i}\tau = 0$, so

$$t_i b_0 = t_0 y_i, \qquad t_0 b_0 = 0, \qquad W = D + S_0 + k b_0.$$

For $i > 0$ write $b_i = c_i b_0 + d_i + s_i$, with $d_i \in D$, $s_i \in S_0$. For a positive $j \neq i$, the projected right side of $t_j b_i = t_i \beta_{ij}$ has degree at least three. Comparing degree two first kills c_i, since D's positive head has no t_0. For each positive head index l, choose $j \neq i, l$; there are at least three positive indices. Comparison of $t_j t_l$ then kills every positive head coefficient of d_i. Central head terms can be absorbed into s_i. Thus $b_i \in \operatorname{rad} D + S_0$. But $t_0 \operatorname{rad} D$ has no component in the diagonal central space, whereas $t_0 b_i = t_i \beta_{0i} = q_i$. Hence every $q_i = 0$. Consequently $D = K \oplus k^v$, and $K + k b_0$ with the displayed actions is $J = \operatorname{rad} R$. The remaining socle directions split, giving $W = J \oplus k^{h-1}$. □

Lemma N.1 (Core compatibility before extra actions)**.** *For the core U_0 acting on $J \oplus Z$, the ordinary classes admitting square values in this module are exactly $\delta\ell + \Gamma$, where $\ell \in U_0^*$, Γ is any Z-valued alternating two-form, and $\delta\ell(i,j) = t_i \ell_j + t_j \ell_i$.*

Proof. The sequence $0 \to J \to R \to k \to 0$ and the tensor ordinary CE complex give

$$H^2(U_0, J) = \delta(U_0^*) \oplus \operatorname{span}\{\tau_{ij} e_i^* \wedge e_j^* : i < j\}.$$

Indeed the maps $H^1(U_0, R) \to H^1(U_0, k)$ and $H^2(U_0, R) \to H^2(U_0, k)$ vanish: their coefficient degrees are $d-1, d-2$, both positive. The connecting part admits square values $b_i = \ell_i t_i$, and central forms admit quadratic refinements. Subtract these parts. For a remaining coefficient λ_{ij}, $t_i b_j = t_j \beta_{ij} = \lambda_{ij}\tau / t_i$. The right monomial lacks t_i, whereas every monomial in $\operatorname{im} t_i$ contains it. Thus $\lambda_{ij} = 0$. □

Proof of polynomial rigidity in Theorem 7.1. For a core retaining class, the lemma gives $\ell \neq 0$. Choose core coordinates with $\ell = e_0^*$. Every module endomorphism of $J \oplus Z$ has four blocks: a multiplier in $R/(\tau)$, a head map $J \to Z$, a socle map $Z \to k\tau$, and an endomorphism of Z. This follows also from $D(R/(\tau)) = J$. An endomorphism T killing the whole core socle has the form $M_f + N_\Lambda$, with $f_0 = 0$. Positive multiplication is null-homotopic on ordinary core CE. The central star values of $N_\Lambda \beta$ are Λ_i, $i > 0$, independent in the direct central cohomology summand. Liftability forces all of these to vanish. The core square has $b_0 = t_0 + s_0$, s_0 in the socle. For an extra direction, $T b_0 = t_0 \beta_{0a}$ has zero Z-part on the right, so also $\Lambda_0 = 0$. Thus T is a positive multiplier. The action is faithful through $R/(\tau)$; removing core linear parts leaves $\operatorname{rad}^2(R/(\tau))$. This proof uses only the core lemma, not the full polynomial-action calculation that follows. □

N.2 The full compatible cohomology and its orbit fiber

Put $A_J = R/(\tau)$, $D' = n - d - 2$, $z = h - 1$, $W = J \oplus Z$, $\dim Z = z$, $N = d + s$. Choose an independent frame $p_a \in \operatorname{rad}^2 A_J$, with canonical representatives having zero top coefficient, and let $f_i = t_i$ on core directions and $f_a = p_a$ on extras.

Lemma N.2 (Compatible ordinary classes and generation)**.** *The full compatible ordinary $H^2(U, W)$ consists uniquely of*

$$\delta\ell + \theta\tau + \Gamma, \qquad \ell \in U^*, \quad \theta \in \Lambda^2 V_s^*, \quad \Gamma \in Z \otimes \Lambda^2 U^*.$$

Here $\delta\ell(x, y) = f_x\ell(y) + f_y\ell(x)$. Generation of the J-head is equivalent to $\ell|V_s \neq 0$. With $\kappa_\ell(x, y) = f_{x,[1]}\ell(y) + f_{y,[1]}\ell(x)$, full derived generation is then equivalent to surjectivity of $\Gamma|\ker\kappa_\ell$ onto Z.

Proof. Write $p_a = \sum_i t_i a_{ia}$, $a_{ia} \in \operatorname{rad} R$. The triangular change of exterior cochain generators eliminates the polynomial extra differentials. In the long exact sequence for $J \to R \to k$, augmentation still kills degrees one and two of R's cohomology: the extra corrections only increase coefficient degree. Thus $H^2(U, J)$ consists of the connecting U^*, core top pairs, mixed top pairs and extra top pairs. The core lemma eliminates the core top coefficients. After subtracting the connecting class, core square values are in the socle; extras kill them. The equations $p_a b_i = t_i \beta_{ia}$ therefore eliminate the mixed coefficients, whose right sides are their coefficients times τ. The remaining extra $\theta\tau$ and all central Γ admit square refinements.

The head bracket is κ_ℓ. If $\ell|V_s = 0$ its image has dimension at most $d - 1$; otherwise its image is the entire d-dimensional head of J. The actions already generate the pure submodule $\operatorname{rad} J$. Modulo it, the kernel of the head bracket supplies exactly $\Gamma|\ker\kappa_\ell$, proving the generation criterion. In particular it is independent of the polynomial frame. □

There are $q^d(q^s - 1)$ choices of ℓ with $\ell|V_s \neq 0$. The complete module effects are as follows. Positive units are trivial on H^2, by ordinary contraction. The scalar multiplier normalizes ℓ freely, a factor $q - 1$. All head maps $N_\Lambda : J \to Z$ translate Γ by $\Lambda \wedge \ell$, with zd independent parameters since $\ell|V_s \neq 0$. Their extra-pair values are zero. After division by them the full parameter count is

$$F_{\text{comp}} = \frac{q^d(q^s - 1)}{q - 1} q^{C_s + z(C_N - d)}.$$

The remaining effective group is $\mathrm{GL}(Z) \ltimes Z^*$, of order $q^z|\mathrm{GL}_z(k)|$. Its extra-pair action is

$$(\theta, \Gamma_{VV}) \longmapsto (\theta + \lambda\Gamma_{VV}, C\Gamma_{VV}).$$

On other central coordinates it acts by C. The core and mixed τ-values contributed by the socle map $\Theta_\lambda : Z \to k\tau$ are boundaries: use τ/t_i as one-cochain values. Every extra polynomial kills these values. The composite ΘN is a positive multiplier and is already ineffective. This checks the whole cohomology fiber, including the effect of taking the head-map quotient first.

When Γ_{VV} is surjective the residual action is free and generation holds. Its nonsurjective proportion is at most $q^{z - C_s}/(q - 1)$. Counting the free open set below and its whole complement pointwise above bounds the orbit count $b(p)$ of all generating fibers by

$$\left| \frac{b(p)}{F_{\text{comp}}/(q^z|\,\mathrm{GL}_z\,|)} - 1 \right| \leq \frac{q^{z^2 + 2z - C_s}}{q - 1}. \tag{158}$$

The generation criterion shows that $b(p)$ is strictly frame-independent, not merely asymptotically so.

N.3 All base changes and averaged stabilizers

Faithful frames recover the intrinsic algebra A_J, its radical filtration and V_s. All quotient changes have form $g = (B, 0; \alpha, \Delta)$, with $B \in \mathrm{GL}_d$, $\Delta \in \mathrm{GL}_s$. The target frame is the unique solution of

$$\Delta^T p' = p(Bt + \alpha^T p') \quad \text{in } A_J.$$

Existence follows by iteration in the nilpotent radical, and two solutions agreeing modulo rad^j agree modulo rad^{j+1}, since the frame has no linear terms. The substitution has invertible linear part in R, so preserves J and the top ideal (τ). It supplies a semilinear module lift. Other lifts differ by the full vertical group already divided out. Thus the actual base group, of order $q^{ds}|\,\mathrm{GL}_d\,||\,\mathrm{GL}_s\,|$, acts genuinely on the vertical quotient; no invariant choice of slice is needed.

For clarity the averaged fixed-frame bound can be taken in the form

$$\begin{aligned} S_{\mathrm{pure}} &\le \tfrac{9}{4} d^2 \{ q^{d(s+d)-(s-d)(\sqrt{n}/2-d-3)} + q^{s+d-a_0'} \}, \\ S_B &\le (\mathcal{F}_s')^{-1} q^{d^2+ds+s^2+1-sn/4+s(d+2)}, \\ S_\Delta &\le \tfrac{3}{2} s (\mathcal{F}_s')^{-1} q^{ds+2s-1-D'}, \end{aligned}$$

where $a_0' \sim n/2$ and $\mathcal{F}_s' = \prod_{j<s}(1-q^{j-D'})$. To obtain these bounds repeat the pair count (117) in A_J. Every high-order increment is now allowed because the top ideal is preserved. The commutator multiplication-rank bound loses at most $d+2$ dimensions after removing degrees below two and the top. For $B \neq I$, the full exterior rank is at least $s2^{d-2}$, with the same deletion cost; this argument uses all $B \in \mathrm{GL}_d$, not the core hyperplane parabolic. For $B = I$, the retained graded blocks have rank $D' \operatorname{rank}(\Delta - I)$. These are exactly the three bounds above, by the proof of (120)–(123). For $s \asymp \sqrt{n}$ all are $O_q(q^{-\kappa n})$. Each fixed vertical fiber has at most $b(p)$ fixed points. Burnside, with this constant fiber size, transfers the full averaged bound without a group-index factor. Together with (158), this gives $\widetilde{M}_h(1 + O_q(q^{-\kappa n}))$ for faithful polynomial types when $h \le s/10$.

N.4 An independent actual-family lower bound

Lemma N.3. *For $h \ge 1$, $s \ge 2$, $s \le D'$, $h - 1 \le C_s$, there are at least*

$$P_\infty(q)^2 q^{sD'+(h-1)C_N-N^2-E}, \qquad E = n - 1 + (h-1)(d+1) + (h-1)^2, \tag{159}$$

actual compatible ordinary types in branch h.

Proof. Take $W = J \oplus Z$, independent polynomial frames as above, and fix $\ell(x_0) = \ell(v_1) = 1$, with the other values zero. Allow all central forms Γ whose extra-pair restriction is surjective. The bracket is $\delta\ell + \Gamma$. An explicit square on $U \oplus J \oplus Z$ is

$$p(x, w, z) = (0, f_x w + f_x \ell(x), Q_\Gamma(x)),$$

where Q_Γ is the zero-diagonal quadratic refinement. Its polar is the bracket, its scalar homogeneity is Frobenius, and $\operatorname{ad} p(x, w, z) = \operatorname{ad}(x, w, z)^2$ follows from $f_x^2 = 0$. The actions give pure $\operatorname{rad} J$, extra brackets give pure Z, and the core stars with (x_0, v_1) give the full J-head. Thus $L' = W$, and $\gamma_j(L) = \mathrm{rad}^{j-2} J$ for $j \ge 3$, so class is $d + 1$.

The parameter number is $\prod_{a<s}(q^{D'} - q^a) q^{z(C_N - C_s)} \prod_{j<z}(q^{C_s} - q^j)$, at least $P_\infty^2 q^{sD'+zC_N}$. Different central forms give different ordinary cohomology classes. Each full isomorphism fiber has at most q^{N^2+E} parameters: there are at most q^{N^2} quotient matrices; intrinsic implicit recovery fixes their target frame; module intertwiners are a torsor under $\mathrm{Aut}_R(J \oplus Z)$, whose full endomorphism dimension is

$$(n-1) + zd + z + z^2 = E.$$

The four summands include both head and socle cross-maps. Sections have already been divided out in ordinary H^2. This proves the bound without a freeness assumption. □

N.5 Conditional tails and the rarity exponent

Proof of Theorem 7.2. For $h \le s/10$, the only compatible objects not yet counted have $t+\rho \ge 2$, by Theorem 7.1, or deficient polynomial frames. Put $D = n-d-3$. Direct division of the finite frame and general-linear products gives

$$\frac{\widetilde{M}_h}{M_h} = q^{-K_h}\frac{1-q^{-D-1}}{1-q^{s-1-D}}\frac{(1-q^{-s})(1-q^{-h})}{1-q^{-d}}, \quad K_h = C_s+2ds-s+h(d-2)+d^2-d-1. \tag{160}$$

In particular $\log_q(M_h/\widetilde{M}_h) = C_s + O(d\sqrt{n}+d^2)$ uniformly in the inner range. Relative to $\widetilde{M}_h$, the head covering has quadratic part

$$-(t+\rho-1)s^2/2 + (t+\rho)hs + 3h^2/2 + hs/2.$$

For $k = t+\rho \ge 2$, $h/s \le 1/10$, its coefficient after division by s^2 is at most $0.565-0.4k \le -0.1k$. All remaining positive terms are $O_q(kd^3s)$. This pays for all module and visible-action defects uniformly, including growing t, ρ.

For deficient frames use the compatible cohomology fiber, retaining all N quotient directions. The free head translations still divide out. The covering is at most $F_{\mathrm{comp}} \sum_{j<s} \begin{bmatrix} D' \\ j \end{bmatrix}_q$. Its ratio to $\widetilde{M}_h$ has logarithm at most $-D' + d^2 + ds + 2s + z^2 + z + O_q(\log s)$, at most $-4n/5$ eventually in this window. Thus $C_h = \widetilde{M}_h(1+O_q(q^{-\kappa n}))$ uniformly for $h \le r/11$.

To compare the remaining branches to the actual compatible denominator, use the independent family lemma. Its exponent differs from $V_h + d - 1 - h(h+1)$ by exactly $N^2 + n + s(d-1) + h(d-2) - 2$. Choose a maximizing ordinary M_h, which has $h \ge 1$ in this window. The family constraints hold eventually and the difference is less than $6n$, with room to absorb the finite factors and number of branches. Theorem 6.11 therefore gives $C/B \ge q^{-6n}$ eventually, independently of the present conditional tail argument. The ordinary optimizer has $\beta_*/\gamma_* < 1/10$ with a strict uniform gap on $[3/2, 8/5]$. Its all-module envelope bounds the ordinary mass with $h > r/11$ by $q^{-\eta n^{3/2}}$. Multiplication by $B/C \le q^{6n}$ still makes this a negligible compatible tail. Summing the inner branch estimates proves the full partition and the total variation comparison, with the stated typical structure.

Finally (160) gives $K_h/n \to \gamma_\alpha^2/2$ in every small neighborhood of the ordinary optimizer. The complement has ordinary mass $q^{-\eta n^{3/2}}$, negligible relative to C/B. Sandwich the positive sum between the minimum and maximum weights in such a neighborhood, then shrink it. This proves the rarity exponent. The same independent lower bound transfers the leading profile and scaled allocation without changing measure implicitly. □

O Square fibers and a change of leading allocation

The extension viewpoint for restricted Lie algebras is classical [13]. We give the particular square and section calculations needed here directly from the axioms.

O.1 Marking a square that vanishes on the derived ideal

Proof of Corollary 8.1. For a fixed action on a strongly abelian W, an extension with zero square on the quotient is represented by a closed ordinary cocycle β and basis square values $b_i \in W$ satisfying

$$P_i b_i = 0, \qquad P_i b_j = P_j \beta_{ij}.$$

The polar formula extends these values to a restricted square. A section $g : U \to W$ changes the pair by $(\beta, b_i) \mapsto (\beta + \delta g, b_i + P_i g_i)$.

On the polynomial $J \oplus Z$ family, $\dim Z = h - 1$, the raw squares above a fixed compatible ordinary cocycle form an affine space of size q^{Nh}. The sections preserving that cocycle are all of $Z^1(U, W)$. In the J-part its H^1 has one connecting class $g_i = f_i$, d core top classes, and s extra socle classes. The core top class at i has core value τ/t_i; extra corrections are $a_{ia}\tau/t_i$, from $p_a = \sum t_i a_{ia}$. The connecting class, extra socle classes, central part and one-boundaries have zero image under $g \mapsto (P_i g_i)_i$. The core top classes give exactly d independent square diagonals τ. Their extra diagonals vanish, since $a_{ia} \in \operatorname{rad} R$ and p_a has degree at least two. Thus the complete extension-level forget fiber has size

$$q^{Nh-d}. \tag{161}$$

This is not yet an assertion about the type fiber of an individual ordinary Lie algebra.

The full vertical action must also be checked on pairs. For the core multiplier t_i, take $g_j = \beta_{ij}$, $g_i = 0$. Jacobi gives $\delta g = t_i \beta$, and the square equations give $P_j g_j = t_i b_j$. For a general positive polynomial $\sum t_i a_i$, take the corresponding sum of these contractions. Hence all positive units remain ineffective on the pair quotient. The head translations are already free on the bracket coordinates. Socle maps send core and mixed central forms to actual pair boundaries; core τ-square values are removed by the top one-cocycles just listed. On extra square coordinates the socle maps retain the same translation as on extra alternating forms. The composite of a socle map and a head map is a positive multiplier and is ineffective on pairs as well.

Consequently the full effective residual group is still $\mathrm{GL}_z \ltimes Z^*$, now acting on extra quadratic data including its square diagonals. Surjectivity of the alternating Γ_{VV} makes this whole action free, and its complement has the same bound as in (158). It is the alternating bracket that must generate the ordinary derived ideal; square diagonals do not replace this condition. The base group and its complete semilinear lifts act on the pair quotient. Their summed fixed-frame estimate applies unchanged because the generating pair-fiber orbit count is frame-independent. Thus the faithful polynomial branch has main term $\widetilde{M}_h q^{Nh-d}$, with exponentially small relative error, after all isomorphisms.

For any compatible ordinary type, the raw squares are an affine $\mathrm{Hom}_{\mathrm{Frob}}(L/W, Z(L) \cap W)$, of size at most q^{Nh}. Apply this bound to the conditional bad and deficient branches from the preceding appendix, and compare with $\widetilde{M}_h q^{Nh-d}$; the loss is only q^d. For the far tail the raw bound is at most $q^{r(d+r)}$, while its compatible probability is $q^{-\eta n^{3/2}}$. Since $T^0_{d,r} \geq C$, this is still negligible relative to the actual restricted denominator. Positive summation proves the stated full partition.

Along the optimizer, (160) and the square factor give the n-scale cost $\gamma_\alpha^2/2 - \beta_*(\alpha)\gamma_\alpha$. The stationary equation for the ordinary leading profile is $1 + \alpha\gamma_\alpha - 3\gamma_\alpha^2/2 = 0$, so this cost equals one. Both conditional and marked tails have already been paid for; the same neighborhood sandwich used for Theorem 7.2 proves the limit. □

O.2 A relaxed square constraint

Lemma O.1 (Raw square parameters and full automorphisms)**.** *Let $S = Z(L) \cap W$, $z' = \dim S$, $D_{\mathrm{tot}} = \dim L$. The squares satisfying $p(L) \subseteq W$, $p(S) = 0$ exist exactly when L is square-compatible. For a fixed such ordinary algebra their parameter space is affine of size $q^{(D_{\mathrm{tot}} - z')z'}$. Its restricted type fiber is the orbit set under the complete $\mathrm{Aut}_k L$, and hence lies between this raw size divided by $|\mathrm{Aut}_k L|$ and the raw size. For the actual family in Lemma N.3,*

$$|\mathrm{Aut}_k L| \leq q^{N^2 + E + n - 1 + Nh}, \qquad E = n - 1 + (h-1)(d+1) + (h-1)^2.$$

Proof. Two restricted squares differ by an additive Frobenius-linear map into the center: subtract the polar, scalar and adjoint axioms. Conversely adding any such map preserves those axioms; this classical square-parameter method also underlies [14]. Since W is abelian, $p(W) \subseteq S$. Extend its Frobenius-linear restriction $W/S \to S$ to L/S, and subtract it. The resulting square vanishes on W and still has range in W. Conversely, starting with such a square p_0, all relaxed squares are exactly $p_0 + \mathrm{Hom}_{\mathrm{Frob}}(L/S, S)$. The constraints are characteristic, so the whole ordinary automorphism group acts on this affine space, and its orbits are precisely the restricted types. No freeness is assumed.

For the explicit $J \oplus Z$ family, each automorphism has quotient, module and section parts. Quotient matrices cost at most q^{N^2}. For fixed quotient matrix, if an intertwiner exists, all intertwiners form a torsor under $\mathrm{Aut}_R W$, with size at most q^E; the endomorphism dimension includes $\mathrm{End}\, J$, $\mathrm{Hom}(J, Z)$, $\mathrm{Hom}(Z, J)$, $\mathrm{End}\, Z$. For fixed quotient and module parts, sections, if nonempty, form an affine copy of the full ordinary $Z^1(U, W)$. The triangular CE change eliminates the extra polynomial differentials, giving $\dim H^1(U, J) = d + 1 + s = N + 1$. The trivial Z-part gives $(h-1)N$, and $\dim B^1(U, W) = n-2$. Therefore $\dim Z^1(U, W) = n-1+Nh$, proving the bound. For $h, s = O(\sqrt{n})$, its whole exponent is $O(n)$.

When $p_W \neq 0$, a section changes its square values by $P_i g_i + p_W(g_i)$. This action generally is not k-linear. The present full-orbit bounds avoid transferring (161) or the old ineffective kernel to that different action. □

Proof of Theorem 8.2. On the family of Lemma N.3, $\dim S = h$, so the raw square count is $q^{h(w_0+N)}$. Divide by the full automorphism bound and multiply the actual ordinary-family lower bound. This gives

$$\begin{aligned}\log_q A_{\mathrm{rel},h} &\geq s(n-d-2) + (h-1)C_N + hw_0 - 2N^2 - 2E - (n-1) + 2\log_q P_\infty \\ &\geq rn + hs^2/2 - O_q(n).\end{aligned}$$

The second inequality follows by expansion and discarding the favorable terms; all negative costs are uniformly $O(n)$. Choose the integer h nearest $r/3$, $s = r - h$. The family constraints $h - 1 \leq C_s$, $s \leq D'$ hold eventually, and $hs^2/2 = 2r^3/27 + O(r)$.

For the full upper bound use the original all-module cover (131), valid for every $h \leq r$. On its stratum (σ, ρ), the actual center satisfies $z' \leq \sigma - \rho$. Since $x(D_{\mathrm{tot}} - x)$ increases on $[0, h+1]$ eventually, multiply each cover term by

$$q^{(D_{\mathrm{tot}} - \sigma + \rho)(\sigma - \rho)}.$$

This upper bound may retain noncompatible ordinary presentations and overlaps; neither can remove any actual restricted type. In particular no old typical-allocation assertion is used here.

Set $\beta = h/\sqrt{n}$, $\gamma = s/\sqrt{n}$, $\zeta = \sigma/\sqrt{n}$, $\xi = \rho/\sqrt{n}$, $\alpha = \beta + \gamma$. The same Gaussian, flag and module estimates as the uniform envelope proof give the leading function

$$G = \gamma\{1 + \zeta(\beta + \gamma/2) - \zeta^2 + \xi(\zeta - \gamma/2) - \xi^2\} + \zeta - \xi,$$

with uniform error $O_q(dn)$ in the logarithm. Here $0 \leq \zeta \leq \beta$, $0 \leq \xi \leq \zeta/2$; the true upper endpoint $\beta + n^{-1/2}$ changes this by only $O(n)$, by bounded derivatives on the compact domain. All strata, including $s = 0$, are covered. On the specified window,

$$\begin{aligned}\partial_\zeta G &\geq 1 - \gamma(\beta - \gamma/2) \geq 43/75, \\ \partial_\xi G &\leq -1 + \gamma(\beta - \gamma/2) \leq -43/75,\end{aligned}$$

because $\gamma(\beta - \gamma/2) \leq \alpha^2/6 \leq 32/75$. Decrease ξ to zero, then increase ζ to β; both paths stay in the domain. Hence

$$G \leq \alpha + \frac{\beta(\alpha - \beta)^2}{2} \leq \alpha + \frac{2\alpha^3}{27}.$$

Summing the polynomial number of strata preserves the uniform error and matches the lower bound. Finally, with $\delta = \beta - \alpha/3$, the exact gap is

$$\alpha + 2\alpha^3/27 - \{\alpha + \beta(\alpha-\beta)^2/2\} = \delta^2(\alpha-\delta)/2 \geq \alpha\delta^2/6.$$

A fixed deviation of h/r or s/r thus loses $\eta n^{3/2}$ in the full upper exponent, while the actual total lower loses only $O(dn)$. This proves the claimed uniform probability estimate under restricted type measure. □

P Exact calculations and reproducibility

The general dimension, orbit-recovery and population bounds are proved in the text and preceding appendices. Exact finite controls accompany the equality classifications and coordinate actions. In particular the class-five noncompatible orbit calculation reconstructs the action matrices and exhausts all $4^9 = 262144$ parameter vectors, giving eleven orbits; compatible low-class orbit and mass calculations are checked separately. The package also reconstructs the universal modules, ordinary CE complexes, nonlinear coordinate recovery, automorphism kernels and fixed-surplus bookkeeping.

The verification groups for the critical constant, critical window, action cohomology and critical refinement give finite controls of the corresponding normalization and allocation interfaces. They do not substitute for the uniform estimates or asymptotic arguments. Each stored report states its model range and whether the calculation is exhaustive.

From this project directory run

```
python verification/run_replay.py --output-dir replay_core
```

The runner copies the supplied inputs into an isolated directory, collects failures and preserves the original reports. The `--no-write` option discards temporary output; `--extended` runs the additional listed controls. The selected entry point uses the Python standard library and SymPy 1.14.0 for the symbolic staircase identities. The additional group `full_partition/` controls the complete endomorphism and residual-action interfaces, nonbinary semilinear lifts, exact all-rank normalization and finite-product ratios, and the all-module tail polynomial. Run just this group with `--full-partition-only`. These are finite interface checks; the arbitrary-dimension fibers and uniform estimates are proved in Appendices I–M. Exact arithmetic controls are distinct from proof-assistant formalization.

The `population_constraints/` group gives direct finite controls of the common head gauge, the compatible and marked fibers, and the relaxed-square action and logarithmic budget. Run `--population-only` for this group and the two directly used complete-fiber interfaces. All general statements in Appendices J–O have proofs in the text; the finite controls state their ranges and do not certify uniform asymptotics or external mathematical priority.

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